
Quant Interview Questions

Probability & Statistics Reference

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1 Broken Stick Problem

Problem: A stick is broken into 3 pieces at two randomly chosen points. What is the probability that the three pieces form a triangle?

Triangle Inequality Condition

Three segments a, b, c form a triangle iff each is shorter than the sum of the other two. For pieces of a unit stick ($a + b + c = 1$) this reduces to:

$$a < \frac{1}{2}, \quad b < \frac{1}{2}, \quad c < \frac{1}{2}.$$

Solution: Two Independent Uniform Break Points

Choose break points $x, y \sim \text{Uniform}(0, 1)$ independently. The three piece lengths are

$$a = \min(x, y), \quad b = |x - y|, \quad c = 1 - \max(x, y).$$

The triangle conditions on the unit-square sample space become:

$$\begin{aligned} \min(x, y) &< \frac{1}{2}, \\ |x - y| &< \frac{1}{2}, \\ \max(x, y) &> \frac{1}{2}. \end{aligned}$$

The favorable region consists of exactly **two triangles** in the unit square (Figure 1):

- **Top-left:** $x \in (0, \frac{1}{2}), y \in (\frac{1}{2}, x + \frac{1}{2})$
- **Bottom-right:** $x \in (\frac{1}{2}, 1), y \in (x - \frac{1}{2}, \frac{1}{2})$

Each triangle has area $\frac{1}{8}$, giving total favorable area $\frac{1}{4}$.

$$P(\text{triangle}) = \frac{1}{4}$$

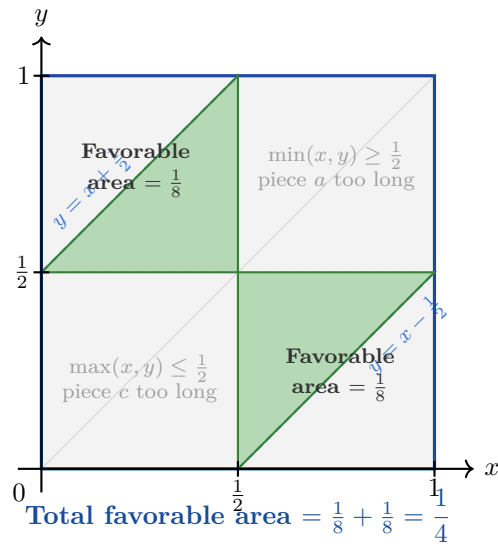


Figure 1: Geometric solution. The sample space is the unit square with break-point coordinates (x, y) . **Green triangles** satisfy all three triangle inequalities. The dashed blue lines are the boundaries $|x - y| = \frac{1}{2}$; outside them the middle piece $b = |x - y|$ exceeds $\frac{1}{2}$. The bottom-left and top-right white squares fail because an end piece exceeds $\frac{1}{2}$.

2 Geometric Distribution

Models the number of independent Bernoulli trials needed for the *first success*, each succeeding with probability $p \in (0, 1]$.

PMF and Key Properties

PMF (trials until first success, $k = 1, 2, 3, \dots$):

$$P(X = k) = (1 - p)^{k-1} p$$

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}, \quad M_X(t) = \frac{p e^t}{1 - (1 - p)e^t} \quad (t < -\ln(1 - p))$$

Property	Formula / Value
Notation	$\text{Geom}(p)$
Support	$k \in \{1, 2, 3, \dots\}$
PMF	$(1 - p)^{k-1} p$
Mean	$1/p$
Variance	$(1 - p)/p^2$
Mode	1
MGF	$p e^t / [1 - (1 - p)e^t], \quad t < -\ln(1 - p)$

Alternative: Failures Before First Success

Support $\{0, 1, 2, \dots\}$:

$$P(X = k) = (1 - p)^k p, \quad E[X] = \frac{1 - p}{p}.$$

Memoryless Property

The geometric distribution is the **only** memoryless discrete distribution:

$$P(X > m + n \mid X > m) = P(X > n).$$

Recursive Derivation of the Mean

$$E = p \cdot 1 + (1 - p)(1 + E) \implies pE = 1 \implies \boxed{E[X] = 1/p}.$$

Example: Rolling a Six-Sided Die

Gaming/Gambling: How many times do you need to roll a standard six-sided die before you roll a “2”?

Each roll succeeds with probability $p = \frac{1}{6}$, so $X \sim \text{Geom}(1/6)$.

$$P(X = k) = \left(\frac{5}{6}\right)^{k-1} \cdot \frac{1}{6}, \quad k = 1, 2, 3, \dots$$

Expected number of rolls:

$$E[X] = \frac{1}{p} = \frac{1}{1/6} = 6.$$

On average, you need to roll the die **6 times** to get your first “2”. For example, the probability it takes exactly 3 rolls is $P(X = 3) = (5/6)^2 \cdot (1/6) \approx 11.6\%$.

3 Binomial Distribution

$B(n, p)$ models the number of successes in n independent Bernoulli(p) trials.

PMF and Key Properties

PMF ($k = 0, 1, \dots, n$):

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p), \quad M_X(t) = ((1 - p) + pe^t)^n$$

Property	Formula / Value
Notation	$B(n, p)$
Support	$k \in \{0, 1, \dots, n\}$
PMF	$\binom{n}{k} p^k (1 - p)^{n-k}$
Mean	np
Variance	$np(1 - p)$
Skewness	$(1 - 2p) / \sqrt{np(1 - p)}$
Ex. kurtosis	$(1 - 6p(1 - p)) / (np(1 - p))$

Understanding the PMF

$$\binom{n}{k} p^k (1 - p)^{n-k} = \underbrace{\binom{n}{k}}_{\text{ways to choose } k \text{ successes}} \times \underbrace{p^k}_{\text{prob. of } k \text{ successes}} \times \underbrace{(1 - p)^{n-k}}_{\text{prob. of } (n-k) \text{ failures}}$$

Normal Approximation (CLT)

For large n : $B(n, p) \approx \mathcal{N}(np, np(1 - p))$. Reliable when $np \geq 5$ and $n(1 - p) \geq 5$.

Example: Multiple-Choice Quiz

Quiz scenario: You take a multiple-choice quiz with 5 questions. Each question has 4 choices and you guess randomly, so the probability of getting any one question correct is $p = \frac{1}{4}$. Let X = number of correct answers. Then $X \sim B(5, 1/4)$.

Key statistics:

$$E[X] = np = 5 \cdot \frac{1}{4} = 1.25, \quad \text{Var}(X) = np(1 - p) = 5 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{15}{16} \approx 0.94.$$

Probability of getting exactly 3 correct:

$$P(X = 3) = \binom{5}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 = 10 \cdot \frac{1}{64} \cdot \frac{9}{16} = \frac{90}{1024} \approx 8.8\%.$$

Probability of passing (3 or more correct):

$$P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) \approx 8.8\% + 1.5\% + 0.1\% \approx 10.4\%.$$

4 Conditional Probability

Definition

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication Rule

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A).$$

Bayes' Theorem

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}, \quad P(B) = \sum_i P(B | A_i) P(A_i).$$

Independence

A and B are independent iff $P(A | B) = P(A)$, equivalently $P(A \cap B) = P(A) P(B)$.

Pitfall: $P(A|B) \neq P(B|A)$ in general

$P(\text{positive test} \text{disease}) = 0.90$	High sensitivity
$P(\text{disease} \text{positive test}) = 0.15$	Disease is rare

Use Bayes' theorem to convert: $P(A|B) = P(B|A) \cdot P(A)/P(B)$.

Worked Example

Two fair dice. Given $D_1 + D_2 \leq 5$, find $P(D_1 = 2)$.

Numerator — outcomes where $D_1 = 2$ and $D_1 + D_2 \leq 5$:

$$D_1 = 2 \Rightarrow D_2 \leq 3, \quad \text{so } (2, 1), (2, 2), (2, 3) \Rightarrow 3 \text{ outcomes.}$$

Denominator — all outcomes where $D_1 + D_2 \leq 5$:

$$\begin{aligned}
 D_1 = 1 : & \quad (1, 1), (1, 2), (1, 3), (1, 4) \rightarrow 4 \text{ outcomes} \\
 D_1 = 2 : & \quad (2, 1), (2, 2), (2, 3) \rightarrow 3 \text{ outcomes} \\
 D_1 = 3 : & \quad (3, 1), (3, 2) \rightarrow 2 \text{ outcomes} \\
 D_1 = 4 : & \quad (4, 1) \rightarrow 1 \text{ outcome}
 \end{aligned}
 \qquad \text{Total: 10 outcomes.}$$

Result (the sample space has $6 \times 6 = 36$ equally likely outcomes):

$$P(D_1 = 2 \mid D_1 + D_2 \leq 5) = \frac{P(D_1 = 2 \text{ and } D_1 + D_2 \leq 5)}{P(D_1 + D_2 \leq 5)} = \frac{3/36}{10/36} = \frac{3}{10}.$$

Additional Problems (Bayesian Updating)

(a) Two-headed vs. fair coin. A pocket holds one fair coin and one two-headed coin. You pick one at random and flip it: heads. What is the probability the *other* side is tails (i.e. the coin is fair)?

$$P(F \mid H) = \frac{P(H \mid F)P(F)}{P(H)} = \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}} = \frac{1/4}{3/4} = \boxed{\frac{1}{3}}.$$

A head is evidence *for* the two-headed coin, so the fair coin drops from a $\frac{1}{2}$ prior to a $\frac{1}{3}$ posterior.

(b) Four coins, given at least one head. Flip four fair coins. Given that at least one is heads, what is the probability exactly one is heads?

$$P(\text{exactly one} \mid \geq 1) = \frac{\binom{4}{1}/16}{15/16} = \frac{4/16}{15/16} = \boxed{\frac{4}{15}}.$$

(c) Which coin is the double-headed one? Five coins: one double-headed, four fair. Pick one at random, flip it five times — all heads. Probability it is the double-headed coin?

$$P(D \mid 5H) = \frac{1 \cdot \frac{1}{5}}{1 \cdot \frac{1}{5} + \left(\frac{1}{2}\right)^5 \cdot \frac{4}{5}} = \frac{1/5}{1/5 + 1/40} = \frac{8/40}{9/40} = \boxed{\frac{8}{9}}.$$

Five straight heads is strong (but not conclusive) evidence for the certain-heads coin.

(d) $d4$ vs. $d6$. A fair 4-sided die and a fair 6-sided die; pick one at random and roll it, getting a 2. Probability it was the $d4$?

$$P(d4 \mid X=2) = \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2}} = \frac{1/4}{1/4 + 1/6} = \frac{3/12}{5/12} = \boxed{\frac{3}{5}}.$$

The $d4$ assigns higher likelihood to any low roll it can produce, so a 2 tilts the posterior toward it (60%).

Additional Problem — Rain on Both Days (Fréchet Bounds) (Q404)

Problem: There is a $p\%$ chance of rain on Saturday and a $q\%$ chance of rain on Sunday. What are the *maximum* and *minimum* possible probabilities that it rains on *both* days?

Key observation. We are told only the two *marginal* probabilities $P(A) = p/100$ (rain Saturday) and $P(B) = q/100$ (rain Sunday). Nothing is said about how the two days are related — they could be independent, strongly positively related (rainy weekends), or negatively related. So $P(A \cap B)$ is not a single number; we must find the *range* of values consistent with the marginals.

Step 1 — Upper bound. “Rain on both days” is a sub-event of “rain on Saturday” and also of “rain on Sunday”:

$$A \cap B \subseteq A \Rightarrow P(A \cap B) \leq P(A) = \frac{p}{100}, \quad A \cap B \subseteq B \Rightarrow P(A \cap B) \leq P(B) = \frac{q}{100}.$$

Hence $P(A \cap B) \leq \min(p, q)\%$. *Attained* when the less likely rainy day only happens together with the other one (e.g. with $p \leq q$: whenever it rains Saturday it also rains Sunday, $A \subseteq B$).

Step 2 — Lower bound. Use inclusion–exclusion,

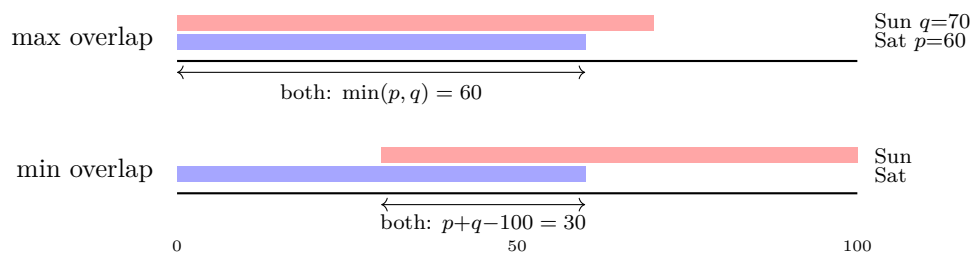
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \implies P(A \cap B) = P(A) + P(B) - P(A \cup B).$$

A probability can be at most 1, so $P(A \cup B) \leq 1$ and

$$P(A \cap B) \geq \frac{p}{100} + \frac{q}{100} - 1 = \frac{p + q - 100}{100}.$$

Also $P(A \cap B) \geq 0$. So $P(A \cap B) \geq \max(0, p + q - 100)\%$. *Attained* by making the two rainy events overlap as little as possible: if $p + q \leq 100$ they can be disjoint (overlap 0); if $p + q > 100$ they must overlap by at least the excess $p + q - 100$.

Picture. Think of the 100 equally likely “weather scenarios” laid out on a line; Saturday’s rain occupies p of them, Sunday’s q . Slide the two intervals to overlap as much or as little as possible:



$$\max(0, p + q - 100)\% \leq P(\text{rain both days}) \leq \min(p, q)\%.$$

Example: $p = 60$, $q = 70$ gives a range of 30% to 60%. (If the days were independent the answer would be $pq/100 = 42\%$, which always lies inside the range.)

Takeaway. These are the **Fréchet bounds**: marginals alone pin a joint probability only to an interval. The upper end is “maximal positive dependence” (one event inside the other); the lower end is “maximal negative dependence” (the events overlap only as much as forced). In risk management the same bounds give best/worst cases for the chance two positions default together when only the individual default probabilities are known.

5 Arrangements with Identical Items

Multinomial Coefficient

Arrange n items where type i appears n_i times ($\sum n_i = n$):

$$\text{Distinct arrangements} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

Why: Start with $n!$ permutations of all items as if distinct. Divide by $n_i!$ for each group to cancel identical rearrangements.

Special Case: Binomial Coefficient

$$\binom{n}{k} = \frac{n!}{k! (n-k)!}$$

Classic Example: MISSISSIPPI

11 letters: 1 M, 4 I, 4 S, 2 P.

$$\frac{11!}{1! \cdot 4! \cdot 4! \cdot 2!} = \frac{39,916,800}{1 \times 24 \times 24 \times 2} = 34,650.$$

Multinomial Theorem

$$(x_1 + \cdots + x_k)^n = \sum_{\substack{n_1 + \cdots + n_k = n \\ n_i \geq 0}} \frac{n!}{n_1! \cdots n_k!} x_1^{n_1} \cdots x_k^{n_k}.$$

6 Expected Flips for Consecutive Heads

Problem: How many fair-coin flips are needed (on average) to see 2 consecutive heads?

Recursive Formula for HH

Let E = expected flips. Condition on the first few outcomes:

Outcome	Prob.	Contribution
T	$\frac{1}{2}$	Wasted 1 flip, restart: $\frac{1}{2}(1 + E)$
H then T	$\frac{1}{4}$	Wasted 2 flips, restart: $\frac{1}{4}(2 + E)$
H then H	$\frac{1}{4}$	Done in 2 flips: $\frac{1}{4}(2)$

$$E = \frac{1}{2}(1 + E) + \frac{1}{4}(2 + E) + \frac{1}{4}(2) = \frac{3}{2} + \frac{3E}{4}$$

$$\Rightarrow \frac{E}{4} = \frac{3}{2}$$

$$E[\text{flips for HH}] = 6$$

Extending to HHH

$$E = \frac{1}{2}(1 + E) + \frac{1}{4}(2 + E) + \frac{1}{8}(3 + E) + \frac{1}{8}(3)$$

$$= \frac{14}{8} + \frac{7E}{8} \Rightarrow \frac{E}{8} = \frac{14}{8}$$

$$E[\text{flips for HHH}] = 14$$

General Pattern

Expected flips to get k consecutive heads on a fair coin:

$$E_k = 2^{k+1} - 2$$

Check: $E_1 = 2$, $E_2 = 6$, $E_3 = 14$, $E_4 = 30$, ...

For a **biased coin** with $P(H) = p$:

$$E_k = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k} = \sum_{j=1}^k \frac{1}{p^j}.$$

Verify at $p = \frac{1}{2}$: $E_2 = 2 + 4 = 6$. ✓

Why This Method Matters

- Works for *any* target pattern (HT, HTH, HTHH, ...).
- Handles biased coins by replacing $\frac{1}{2}$ with p or $1 - p$.
- The key insight: on any “broken” sequence, identify which state you reset to and write an equation for that state.

7 Pattern Stopping Times (Penney's Game)

Problem: You repeatedly flip a fair coin until either **HHT** or **HTT** appears. If you may bet on which pattern shows up first, which should you choose, and with what probability does it win?

Set Up a Markov Chain on “Progress” States

A **Markov chain** is a random process that hops between a finite set of *states*, where the probability of the next move depends *only on the current state* — not on the full history of how you arrived there. This is the “memoryless” property. The art is to define the states so that this actually holds: here we track only the useful *suffix* of recent flips (the tail that could still grow into a target) and discard everything before it, because only that suffix affects what happens next.

Both targets begin with H, so the relevant states are S — the **start state**, meaning no useful suffix yet — together with H , HH , and HT . Let $p_\bullet$ be the probability that **HHT** appears before **HTT** starting from that state. (This head-to-head race is the classic [Penney's game](#).)

State	One-step equation
S	$p_S = \frac{1}{2}p_H + \frac{1}{2}p_S \Rightarrow p_S = p_H$
H	$p_H = \frac{1}{2}p_{HH} + \frac{1}{2}p_{HT}$
HH	$p_{HH} = \frac{1}{2}(1) + \frac{1}{2}p_{HH} \Rightarrow p_{HH} = 1$ (a T completes HHT)
HT	$p_{HT} = \frac{1}{2}(0) + \frac{1}{2}p_H = \frac{1}{2}p_H$ (a T completes HTT, a loss)

Substitute $p_{HH} = 1$ and $p_{HT} = \frac{1}{2}p_H$ into the H equation:

$$p_H = \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}p_H\right) = \frac{1}{2} + \frac{1}{4}p_H \Rightarrow \frac{3}{4}p_H = \frac{1}{2} \Rightarrow p_H = \frac{2}{3}.$$

Since $p_S = p_H$:

$P(\text{HHT before HTT}) = \frac{2}{3}$, so bet on HHT (twice as likely to appear first as HTT).

The Paradox: Equal Waiting Times, Unequal Race

A natural guess is that the pattern with the shorter *expected wait* wins the race. That intuition is wrong. Both HHT and HTT have the *same* expected waiting time of 8 flips, yet HHT wins the head-to-head 2/3 of the time.

$E[\text{flips to HHT}]$. Let a, b, c be expected flips from states $\emptyset, H, HH$:

$$c = 1 + \frac{1}{2}c \Rightarrow c = 2, \quad b = 1 + \frac{1}{2}c + \frac{1}{2}a = 2 + \frac{1}{2}a, \quad a = 1 + \frac{1}{2}b + \frac{1}{2}a \Rightarrow a = 2 + b.$$

So $a = 2 + (2 + \frac{1}{2}a) = 4 + \frac{1}{2}a \Rightarrow a = 8$.

$E[\text{flips to HTT}]$. With states $\emptyset, H, HT$:

$$c = 1 + \frac{1}{2}b \text{ (a H returns to } H), \quad b = 1 + \frac{1}{2}b + \frac{1}{2}c \Rightarrow b = 2 + c, \quad a = 1 + \frac{1}{2}b + \frac{1}{2}a \Rightarrow a = 2 + b,$$

giving $b = 2 + (1 + \frac{1}{2}b) \Rightarrow b = 6$, then $a = 2 + 6 = 8$.

Both wait 8 flips on average, but HHT beats HTT 2/3 vs. 1/3 head-to-head. “Which comes first on average” and “which comes first in a direct race” are different questions — overlaps between the patterns, not the mean wait, decide the race.

Concept Summary and Applications

- **Method:** model the race as a finite Markov chain whose states are the longest current suffix that is a prefix of a target. Absorbing states are the two completed patterns. Solve the linear system for the win probability.
- **Penney’s game:** these races are **non-transitive** — for any length-3 pattern the first player picks, the second player can pick one that beats it. There is no “best” pattern.
- **Expected waiting time** of a pattern is governed by its self-overlaps (Conway’s leading-number / correlation method): for a fair coin $E = \sum 2^k$ over the overlap lengths k where a proper prefix equals a suffix, plus the full length. E.g. HHT and HTT have no self-overlap, so $E = 2^3 = 8$; HTH overlaps in 1, so $E = 2^3 + 2^1 = 10$; HHH overlaps in 1, 2, so $E = 2 + 4 + 8 = 14$.
- **This generalises the “Expected Flips for Consecutive Heads” section.** A run of k heads is the special case where *every* proper prefix of $H \cdots H$ is also a suffix, so *all* overlaps $k, k-1, \dots, 1$ are present and $E = 2^k + 2^{k-1} + \dots + 2^1 = 2^{k+1} - 2$ (recovering $HH \rightarrow 6$, $HHH \rightarrow 14$). A general pattern simply keeps only the overlaps it actually has — the earlier recursion is this same idea specialised to a single, maximally self-overlapping target.
- **Applications:** genetics/DNA motif waiting times, quality-control run detection, and any “which signal fires first” comparison.

Additional Problem 1 — HHT vs. THH

Problem: Race HHT against THH. Which is favored?

Same state method (states $\emptyset, H, T, HH, TH$; $p = P(\text{HHT first})$):

$$p_{HH} = 1, \quad p_{TH} = \frac{1}{2}(0) + \frac{1}{2}p_T, \quad p_T = \frac{1}{2}p_{TH} + \frac{1}{2}p_T \Rightarrow p_T = p_{TH}.$$

Then $p_T = \frac{1}{2}p_T \Rightarrow p_T = 0$, so $p_H = \frac{1}{2}p_{HH} + \frac{1}{2}p_T = \frac{1}{2}$ and $p_\emptyset = \frac{1}{2}p_H + \frac{1}{2}p_T = \frac{1}{4}$.

$$P(\text{HHT before THH}) = \frac{1}{4} \text{ — i.e. THH beats HHT 3 to 1.}$$

Why so lopsided: once any HH appears it was almost surely preceded by a T (from an earlier tail), completing THH before the trailing T of HHT can land. This is the textbook Penney’s-game upset.

Additional Problem 2 — HHT vs. HTH

Problem: Race HHT against HTH (fair coin, then a biased coin with $P(H) = p$).

States $\emptyset, H, HH, HT$; with $P(H) = p$, $q_{HH} = 1$, $q_{HT} = (1-p)q_{\emptyset}$, and $q_{\emptyset} = q_H = p \cdot 1 + (1-p)q_{HT}$. Substituting,

$$q_{\emptyset} = p + (1-p)^2 q_{\emptyset} \implies q_{\emptyset}(1 - (1-p)^2) = p, \quad 1 - (1-p)^2 = p(2-p),$$

so $q_{\emptyset} = \frac{1}{2-p}$.

$$P(\text{HHT before HTH}) = \frac{1}{2-p}; \quad \text{for a fair coin } \left(p = \frac{1}{2}\right) \text{ this is } \frac{2}{3}.$$

8 Alternating Coin Games (Geometric Series)

Problem: We play a game and *you go first*. You flip a coin: heads, I pay you \$30. Tails, you hand me the coin and I flip: heads, you pay me \$30; tails, the coin comes back to you. We continue until someone flips heads. What is the most you would pay to go first?

Reduce to “Who Gets the First Heads?”

Nothing is paid until a head appears, so the whole game collapses to a single question: on whose turn does the first head land? You win +\$30 if it is yours, lose \$30 if it is mine.

Let $p = P(\text{you get the first heads})$. On your turn you either flip heads now $\left(\frac{1}{2}\right)$, or both of us flip tails $\left(\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}\right)$ and the game resets to the start:

$$p = \frac{1}{2} + \frac{1}{4}p \implies \frac{3}{4}p = \frac{1}{2} \implies p = \frac{2}{3}.$$

Equivalently, as a geometric series over your turns (flips 1, 3, 5, ...):

$$p = \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} + \left(\frac{1}{4}\right)^2 \cdot \frac{1}{2} + \dots = \frac{1/2}{1 - 1/4} = \frac{2}{3}.$$

Expected Value

$$E[X] = 30 \cdot \frac{2}{3} + (-30) \cdot \frac{1}{3} = 20 - 10 = \$10.$$

Going first is worth $E[X] = \$10$, so pay up to \$10. The first mover gets the first heads with probability $2/3$.

Concept Summary and Applications

- **Memoryless reset:** when a game returns to an identical state after a fixed-probability “both miss” event, write a self-referential equation $V = (\text{immediate term}) + (\text{reset prob}) \cdot V$ and solve for V . This is the one-line substitute for summing an infinite series.
- **Geometric-series view:** you win the instant *your* flip is the first head, and your turns are rounds 1, 2, 3, ... To win on your k -th turn, two things must happen: (i) the game survives the previous $k - 1$ rounds — both players flip tails each time, probability r^{k-1} with $r = P(\text{both miss}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; then (ii) you flip heads now, probability $\frac{1}{2}$. Summing over all your turns,

$$p = \sum_{k \geq 1} \underbrace{r^{k-1}}_{\text{reach turn } k} \cdot \underbrace{\frac{1}{2}}_{\text{win now}} = \frac{\frac{1}{2}}{1 - r} = \frac{\frac{1}{2}}{3/4} = \frac{2}{3}.$$

The self-referential equation $p = \frac{1}{2} + r p$ above is just this geometric sum in closed form.

- **First-mover advantage** is real whenever a round can end on the first player's action: going first is worth a strictly positive premium (here \$10 on a $\pm\$30$ game).
- **Applications:** auctions and races decided by "first success," turn-based duels, and any renewal game with a geometric number of rounds.

9 Elegant Symmetry: $n+1$ vs. n Coins

Problem: Player A tosses $n + 1$ fair coins and player B tosses n fair coins. B wins if he gets *at least as many* heads as A . Find $P(B \text{ wins})$. (Equivalently, A gets *strictly more* heads with the complementary probability.)

Solution: Condition on A 's Extra Coin

Peel off A 's one extra coin. Write

$$A' = \# \text{heads among } A \text{'s first } n \text{ coins,} \quad X = A \text{'s extra coin (1 if head, 0 if tail),}$$

so A 's total is $A = A' + X$. The key facts: A' and B are **identically distributed** (each is the head count of n fair coins), and X is an independent fair bit.

B wins when $B \geq A = A' + X$. Condition on the extra coin:

- $X = 0$ (prob $\frac{1}{2}$): B wins iff $B \geq A'$;
- $X = 1$ (prob $\frac{1}{2}$): B wins iff $B \geq A' + 1$, i.e. $B > A'$.

Therefore

$$P(B \text{ wins}) = \frac{1}{2} P(B \geq A') + \frac{1}{2} P(B > A').$$

Now use that A' and B are i.i.d., so relabelling $A' \leftrightarrow B$ gives $P(A' > B) = P(B > A')$. Hence

$$P(B \geq A') = 1 - P(A' > B) = 1 - P(B > A'),$$

i.e. the two terms $P(B \geq A')$ and $P(B > A')$ are complementary and sum to 1. Thus

$$P(B \text{ wins}) = \frac{1}{2} [P(B \geq A') + P(B > A')] = \frac{1}{2} \cdot 1 = \frac{1}{2}.$$

$P(B \text{ has at least as many heads}) = \frac{1}{2}$, and hence $P(A \text{ strictly more}) = \frac{1}{2}$ — *independent of n* . The lone extra coin is exactly what removes the tie and splits the two outcomes evenly.

The Heads–Tails Symmetry Shortcut

The same $\frac{1}{2}$ drops out in one line, with no conditioning. Let H_A, H_B be the head counts and T_A, T_B the tail counts. Because A has exactly **one more coin** than B , exactly one of the following always holds:

$$H_A > H_B \quad \text{or} \quad T_A > T_B,$$

mutually exclusive and exhaustive because

$$T_A > T_B \iff (n+1) - H_A > n - H_B \iff H_A \leq H_B,$$

which is precisely the negation of $H_A > H_B$. Now **flip every coin**, swapping heads and tails: this is a probability-preserving symmetry, so

$$P(H_A > H_B) = P(T_A > T_B).$$

Two equal probabilities that sum to 1 are each $\frac{1}{2}$, so $P(A \text{ strictly more heads}) = \frac{1}{2}$. No binomial sums and no casework — the same answer as the conditioning argument, but the single extra coin and one flip symmetry do all the work.

Illustrative Verification (5 vs. 4 coins)

Take $n = 4$, so $X = H_A \sim \text{Bin}(5, \frac{1}{2})$ and $Y = H_B \sim \text{Bin}(4, \frac{1}{2})$, independent. We check $P(Y \geq X) = \frac{1}{2}$ by brute force. The tail probabilities of Y (out of 16) are

$$P(Y \geq 0, 1, 2, 3, 4) = \frac{16}{16}, \frac{15}{16}, \frac{11}{16}, \frac{5}{16}, \frac{1}{16}, \quad P(Y \geq 5) = 0,$$

and $P(X = k) = \frac{1, 5, 10, 10, 5, 1}{32}$ for $k = 0, \dots, 5$. Then

$$P(Y \geq X) = \sum_{k=0}^5 P(X = k) P(Y \geq k) = \frac{16 + 75 + 110 + 50 + 5 + 0}{512} = \frac{256}{512} = \frac{1}{2}. \quad \checkmark$$

The direct computation confirms the symmetry answer — but the one-line symmetry argument is what you give in an interview.

Concept Summary and Applications

- **Pattern:** whenever two symmetric processes differ by exactly one extra “trial,” look for a parity/complement dichotomy that the extra trial resolves, then invoke a symmetry (here head $\leftrightarrow$ tail) to split the remaining probability evenly.
- **Key identity:** with one extra coin, “ A leads in heads” and “ A leads in tails” partition all outcomes — there are no ties to worry about, which is what makes the split exactly $\frac{1}{2}$.
- **Applications:** best-of races with a first-mover extra draw, “leader after one more sample” problems, and any comparison where an odd total rules out ties.

10 Parity of Heads

Problem: You flip n fair coins. What is the probability that the number of heads is **even** (i.e. $0, 2, 4, \dots$; note 0 counts as even)?

Background

Each flip is heads or tails, so n flips produce one of 2^n equally likely outcome sequences (e.g. HTTH). The number of sequences with exactly k heads is the **binomial coefficient** $\binom{n}{k}$ (the number of ways to choose which k of the n positions are heads). “Even number of heads” just means $k \in \{0, 2, 4, \dots\}$. We want the total probability of all such outcomes.

Fair Coin — Method 1: Count with the Binomial Theorem

Since every one of the 2^n sequences is equally likely,

$$P(\text{even}) = \frac{\#\{\text{sequences with an even number of heads}\}}{2^n} = \frac{1}{2^n} \sum_{k \text{ even}} \binom{n}{k}.$$

To evaluate $\sum_{k \text{ even}} \binom{n}{k}$, apply the **binomial theorem** $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$ at two values of x :

$$x = 1 : \quad (1+1)^n = \sum_{k=0}^n \binom{n}{k} = 2^n \quad (\text{all sequences}),$$

$$x = -1 : \quad (1-1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k = \sum_{k \text{ even}} \binom{n}{k} - \sum_{k \text{ odd}} \binom{n}{k} = 0.$$

The second line says the even- k and odd- k binomial coefficients sum to the *same* amount. Adding the two equations,

$$2 \sum_{k \text{ even}} \binom{n}{k} = 2^n \implies \sum_{k \text{ even}} \binom{n}{k} = 2^{n-1},$$

so

$$P(\text{even}) = \frac{2^{n-1}}{2^n} = \frac{1}{2}.$$

Check, $n = 4$: even-head counts 0, 2, 4 give $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 1 + 6 + 1 = 8 = 2^3$, while odd counts 1, 3 give $\binom{4}{1} + \binom{4}{3} = 4 + 4 = 8$; indeed $8/16 = \frac{1}{2}$.

Fair Coin — Method 2: The One-Flip Bijection

Here is the same answer in one line. Pair each outcome with the outcome you get by *toggling the first coin* ($H \leftrightarrow T$). That single toggle changes the head count by ± 1 , so it flips the parity — turning every even-head outcome into an odd-head one, and vice versa. This is a perfect one-to-one pairing of even outcomes with odd outcomes, so exactly half of all outcomes are even.

$$P(\text{even number of heads}) = \frac{1}{2}, \text{ for every } n \geq 1.$$

Additional 1 — All Coins Biased with $P(H) = p$

Now every coin lands heads with the *same* probability p (tails $q = 1 - p$), independently. The counting shortcut above breaks because sequences are no longer equally likely: a sequence with k heads has probability $p^k q^{n-k}$, so

$$P(k \text{ heads}) = \binom{n}{k} p^k q^{n-k}.$$

We again isolate the even terms, but now with a **signed (alternating) sum**. Consider

$$S = \sum_{k=0}^n (-1)^k P(k \text{ heads}) = \sum_{k=0}^n \binom{n}{k} (-p)^k q^{n-k}.$$

By the binomial theorem $\sum_k \binom{n}{k} a^k b^{n-k} = (a + b)^n$ with $a = -p$, $b = q$,

$$S = (q - p)^n = (1 - 2p)^n.$$

On the other hand, grouping the sum S by parity, the even terms carry $(-1)^k = +1$ and the odd terms carry -1 :

$$S = \underbrace{\sum_{k \text{ even}} P(k)}_{P(\text{even})} - \underbrace{\sum_{k \text{ odd}} P(k)}_{P(\text{odd})} = P(\text{even}) - P(\text{odd}).$$

Together with the total-probability equation $P(\text{even}) + P(\text{odd}) = 1$, we have a simple 2×2 system:

$$P(\text{even}) - P(\text{odd}) = (1 - 2p)^n, \quad P(\text{even}) + P(\text{odd}) = 1.$$

Adding and dividing by 2:

$$P(\text{even}) = \frac{1 + (1 - 2p)^n}{2}, \quad P(\text{odd}) = \frac{1 - (1 - 2p)^n}{2}.$$

Sanity checks. A fair coin has $p = \frac{1}{2}$, so $1 - 2p = 0$ and $P(\text{even}) = \frac{1+0}{2} = \frac{1}{2}$, recovering the earlier result. A single coin ($n = 1$) gives $\frac{1+(1-2p)}{2} = 1 - p = P(0 \text{ heads})$, as it must. (Equivalently, this is the generating function $f(x) = (q + px)^n$ evaluated at the two parity-filtering points: $P(\text{even}) = \frac{1}{2}[f(1) + f(-1)]$, since $f(1) = 1$ and $f(-1) = (q - p)^n$. With *different* biases p_i the same argument gives $P(\text{even}) = \frac{1}{2}[1 + \prod_i (1 - 2p_i)]$.)

Additional 2 — One Biased Coin Among Fair Ones

Suppose one coin is biased with $P(H) = p$ and the other $n - 1$ coins are fair ($n \geq 2$). Split the total head count into the biased coin plus the fair group, and condition on the biased coin's outcome:

$$P(\text{even total}) = \underbrace{P(\text{biased} = T)}_{1-p} \cdot P(\text{fair group even}) + \underbrace{P(\text{biased} = H)}_p \cdot P(\text{fair group odd}).$$

(If the biased coin is a tail it adds 0 heads, so the total is even exactly when the fair group is even; if it is a head it adds 1, so the total is even exactly when the fair group is *odd*.) But the fair group has $n - 1 \geq 1$ fair coins, and by the fair-coin result its head count is even or odd each with probability $\frac{1}{2}$. Hence

$$P(\text{even total}) = (1 - p) \cdot \frac{1}{2} + p \cdot \frac{1}{2} = \frac{1}{2}.$$

The bias p cancels completely: whatever the biased coin does, the fair group has already randomised the parity.

Additional 3 — Only One Fair Coin

Now flip the situation: exactly *one* coin is fair and the other $n - 1$ are biased with *arbitrary* (possibly different, possibly unknown) probabilities. Let the biased group have head-count parity “even” with some probability e and “odd” with probability $1 - e$ — we will not need e . Condition on the single fair coin:

$$P(\text{even total}) = \underbrace{P(\text{fair} = T)}_{1/2} P(\text{biased group even}) + \underbrace{P(\text{fair} = H)}_{1/2} P(\text{biased group odd}) = \frac{1}{2}e + \frac{1}{2}(1 - e) = \frac{1}{2}.$$

The lone fair coin is a perfect parity randomiser: mixing any fixed parity distribution with an independent fair ± 1 flip yields $\frac{1}{2}$, no matter what the biased coins do.

One fair coin is enough. As long as at least one coin is fair, the total parity is exactly 50/50 — that single fair coin washes out everything else. Bias only matters when every coin is biased, giving $P(\text{even}) = \frac{1 + \prod_i (1 - 2p_i)}{2}$; its deviation from $\frac{1}{2}$ is the product $\prod_i (1 - 2p_i)$, which collapses to 0 the instant any one $p_i = \frac{1}{2}$.

Concept Summary and Applications

- **Two tools for the fair case:** count directly with $\sum_{k \text{ even}} \binom{n}{k} = 2^{n-1}$ (from the binomial theorem at $x = \pm 1$), or use the one-flip *bijection*. Both give $\frac{1}{2}$.
- **Parity filter for the biased case:** the signed sum $\sum_k (-1)^k P(k)$ equals $P(\text{even}) - P(\text{odd})$ and, by the binomial theorem, equals $(1 - 2p)^n$; combine with $P(\text{even}) + P(\text{odd}) = 1$ to solve. In generating-function language this is $\frac{1}{2}[f(1) \pm f(-1)]$ with $f(x) = (q + px)^n$ — the “roots-of-unity filter,” which extends to any modulus.
- **Key intuition:** a single fair coin forces parity to $\frac{1}{2}$; bias shows up only through the product $\prod (1 - 2p_i)$.
- **Applications:** error-detecting parity bits, the XOR of random bits, and any “is the count even or odd?” question over independent trials.

11 Martingales, Random Walks, and Gambler's Ruin

Problem: A fair ± 1 random walk starts at 0. You *win* if it reaches +20 and *lose* if it reaches -10, whichever happens first. What is the probability that you lose?

Background: What Is a Martingale?

A **martingale** is the mathematical model of a *fair game*: a sequence of random values $X_0, X_1, X_2, \dots$ — picture your running bankroll — with the property that, no matter what has happened so far, your expected value after the next step equals your value right now:

$$\mathbb{E}[X_{n+1} \mid X_0, X_1, \dots, X_n] = X_n.$$

In words: there is **no drift**. On average you neither gain nor lose, whatever the history. A running total of fair ± 1 coin flips is the archetype; a *biased* game drifts and is not a martingale.

The reason martingales are so useful is the **Optional Stopping Theorem (OST)**: if you stop a fair game at a random time T chosen by a reasonable rule (e.g. “stop when I first hit a boundary”), then the expected value *at that stopping time* still equals the start,

$$\mathbb{E}[X_T] = \mathbb{E}[X_0].$$

This is powerful because it lets us compute where a walk ends up *without* solving any recursion — we just balance the possible endings against the mean.

Solution via Optional Stopping (Step by Step)

Let X_n be the position after n steps, starting at $X_0 = 0$, each step $+1$ or -1 with probability $\frac{1}{2}$. We want $p = P(\text{hit } -10 \text{ before } +20)$.

Step 1 — X_n is a martingale. Each step adds ± 1 with equal probability, independent of the past, so

$$\mathbb{E}[X_{n+1} \mid \text{past}] = X_n + \underbrace{\left(\frac{1}{2}(+1) + \frac{1}{2}(-1)\right)}_{=0} = X_n.$$

The zero-mean step is exactly where the *fair* coin is used.

Step 2 — the mean is preserved. Taking expectations, $\mathbb{E}[X_{n+1}] = \mathbb{E}[X_n]$, so $\mathbb{E}[X_n] = \mathbb{E}[X_0] = 0$ for all n .

Step 3 — apply OST at the stopping time. Let T be the first time the walk hits -10 or $+20$. The walk is trapped in the bounded band $[-10, 20]$ and T is finite with probability 1, so OST applies and the mean survives the random stopping:

$$\mathbb{E}[X_T] = \mathbb{E}[X_0] = 0.$$

Step 4 — evaluate the two endings. At time T the walk sits on exactly one barrier: $X_T = -10$ (you lose, probability p) or $X_T = +20$ (you win, probability $1 - p$), so

$$\mathbb{E}[X_T] = (-10)p + (20)(1 - p).$$

Step 5 — solve. Set this equal to 0:

$$(-10)p + 20(1 - p) = 0 \implies 20 - 30p = 0 \implies p = \frac{20}{30} = \frac{2}{3}.$$

$$P(\text{lose, i.e. hit } -10 \text{ first}) = \frac{2}{3}, \quad P(\text{win, hit } +20 \text{ first}) = \frac{1}{3}.$$

(Equivalently, shift to $Y_n = X_n + 10$: the barriers become 0 and 30 and the start is 10, the textbook symmetric *gambler's ruin* on $\{0, 1, \dots, 30\}$.)

Gambler's Ruin — General Formula

For a symmetric walk with an *absorbing* barrier a distance a below and b above the start, each barrier's hit probability is the distance to the **opposite** barrier over the total width:

$$P(\text{hit lower first}) = \frac{b}{a+b}, \quad P(\text{hit upper first}) = \frac{a}{a+b}.$$

Here $a = 10$, $b = 20$, so $P(\text{hit } -10) = \frac{20}{30} = \frac{2}{3}$. Intuitively $E[X_T] = 0$ forces the outcomes to balance around the start: the far barrier (+20) has more leverage, so the near barrier (−10) must carry proportionally more probability.

Biased Walk: the $(q/p)^{X_n}$ Martingale

If steps are +1 with probability $p \neq \frac{1}{2}$ and −1 with probability $q = 1 - p$, the walk **drifts**: the mean step is $p(+1) + q(-1) = p - q \neq 0$, so X_n itself is *not* a martingale and the balancing trick fails on it directly. The remedy is to find a *function* of the walk that *is* a martingale. The right choice is

$$M_n = (q/p)^{X_n}.$$

Verify the martingale property from a generic position $X_n = x$ (a step takes $x \rightarrow x + 1$ w.p. p or $x \rightarrow x - 1$ w.p. q):

$$\mathbb{E}[M_{n+1} \mid X_n = x] = p(q/p)^{x+1} + q(q/p)^{x-1} = (q/p)^x \left[p \cdot \frac{q}{p} + q \cdot \frac{p}{q} \right] = (q/p)^x [q + p] = (q/p)^x = M_n.$$

The bracket collapses to $p + q = 1$ — *that* is why the base q/p is the magic value: it makes the up- and down-contributions cancel, so M_n has no drift even though X_n does.

Numerical illustration. Take $p = \frac{2}{3}$, $q = \frac{1}{3}$, so $q/p = \frac{1}{2}$ and $M_n = (\frac{1}{2})^{X_n}$. From any level x ,

$$\mathbb{E}[M_{n+1} \mid X_n = x] = \frac{2}{3}(\frac{1}{2})^{x+1} + \frac{1}{3}(\frac{1}{2})^{x-1} = (\frac{1}{2})^x \left[\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 2 \right] = (\frac{1}{2})^x \left[\frac{1}{3} + \frac{2}{3} \right] = (\frac{1}{2})^x = M_n.$$

Even though the walk drifts *upward* (heads twice as likely as tails), the quantity $(\frac{1}{2})^{X_n}$ keeps a constant expectation — a martingale, confirmed numerically.

Gambler's ruin from it. Applying OST to M_n between barriers 0 and N gives $\mathbb{E}[M_T] = M_0 = (q/p)^k$ when starting at k ; splitting into the two endings ($M = 1$ at level 0, $M = (q/p)^N$ at level N) and solving,

$$P(\text{ruin from } k) = P(\text{hit 0 before } N) = \frac{(q/p)^k - (q/p)^N}{1 - (q/p)^N}.$$

For instance with $p = \frac{2}{3}$ ($q/p = \frac{1}{2}$), start $k = 1$, barriers 0 and $N = 3$:

$$P(\text{ruin}) = \frac{(\frac{1}{2})^1 - (\frac{1}{2})^3}{1 - (\frac{1}{2})^3} = \frac{\frac{1}{2} - \frac{1}{8}}{1 - \frac{1}{8}} = \frac{3/8}{7/8} = \frac{3}{7} \approx 0.43,$$

comfortably below $\frac{1}{2}$ — the upward drift makes reaching +3 before 0 the more likely outcome, exactly as it should.

Additional Problem — 52-Card Walk *With* Replacement

Problem: Draw from a shuffled 52-card deck (26 black, 26 red): black = +1 point, red = -1 point. After each draw you **replace and reshuffle**, so you may draw indefinitely. What strategy maximises your total?

The words “with replacement” are the whole problem. Because every draw is from the full deck, the draws are i.i.d. ± 1 with probability $\frac{1}{2}$ each, and the running score $S_n = X_1 + \dots + X_n$ is a **simple symmetric random walk** — a martingale. Two facts now pull in opposite directions.

Fact 1 (fairness). For any stopping rule τ with finite expected duration $\mathbb{E}[\tau] < \infty$, optional stopping gives $\mathbb{E}[S_\tau] = 0$: no time-bounded strategy earns a positive *expected* score.

Fact 2 (recurrence). A symmetric walk on the integers visits *every* integer with probability 1. So *any* target level is reached almost surely — not only +1, but +2, +100, +100⁹, and even the astronomically distant negative level -100¹⁰: each of these is hit eventually with probability 1 (the walk wanders arbitrarily far in *both* directions).

Optimal strategy. Pick a positive target and simply wait for it. The cleanest rule is

quit the first moment you are ahead ($S_n = +1$),

which finishes with a guaranteed +1. Since +2, +100, +100⁹, ... are all reached a.s. as well, **there is no finite maximum**: the supremum of what you can guarantee is $+\infty$, and no single strategy attains it.

The catch. The free lunch is paid for in *time*. The expected number of draws to reach any positive level is infinite, $\mathbb{E}[\tau_{+1}] = \infty$. That is precisely why guaranteeing $+T$ almost surely does *not* contradict Fact 1: the winning rule’s stopping time has infinite mean, so the theorem “ $\mathbb{E}[S_\tau] = 0$ ” (which *requires* $\mathbb{E}[\tau] < \infty$) does not apply to it.

With replacement: stop the first time you are ahead — you reach +1 (or any target you name) with probability 1, so you can guarantee an arbitrarily large gain, but only at the cost of an unbounded expected number of draws. Every strategy that must finish in finite *expected* time has value \$0.

With vs. Without Replacement — Why It Matters

- **With replacement** (this problem): the total is a *free* random walk that can wander arbitrarily far in either direction. A martingale with no endpoint constraint has $EV = 0$ — stopping cannot manufacture value.
- **Without replacement** (finite deck, next section): the walk is *pinned* to end at 0 (26 black — 26 red), a **martingale bridge**. That pinning — knowing the leftover composition — is exactly what creates a real, finite “option value” ($\approx \$2.62$ for a full deck). Removing replacement turns a worthless game into a solvable one.

Concept Summary and Applications

- **Recipe:** spot a martingale (fair walk X_n ; or $(q/p)^{X_n} / X_n^2 - n$ for biased or variance problems), then apply optional stopping $E[X_T] = X_0$ and read off the barrier probabilities.
- **Applications:** risk of ruin and bankroll survival, first-passage and hitting probabilities, sequential-testing boundaries, and pricing by no-arbitrage (a fairly priced game is a martingale).

12 Finite-Deck Optimal Stopping (Backward Induction)

Problem: A bag holds 3 black and 2 white balls. You draw one at a time *without replacement* and may **stop whenever you like**. Your payoff is $B - W$: (black balls drawn) – (white balls drawn). When should you stop, and what is the game worth?

Background: Why This Needs Backward Induction

Each black ball drawn is worth +1 and each white ball –1, and you may quit at any point keeping what you have. The order is random, so at every step you face a choice — stop now, or draw again hoping for more black than white from what remains. The catch is that the *right* choice now depends on what you would do later, which depends on still-later choices. The clean way to untangle this is to solve **backward**: figure out the value of the near-empty bag first, then use it to value fuller bags, and so on back to the start. That is **dynamic programming** (a.k.a. backward induction).

State and the Bellman Equation

Describe the situation by what is *left* in the bag: b black and w white. Let $V(b, w)$ be the best expected *additional* payoff achievable from that point. You compare two options:

- **Stop:** take 0 more.
- **Draw:** the next ball is black w.p. $\frac{b}{b+w}$ (+1, then state $(b-1, w)$) or white w.p. $\frac{w}{b+w}$ (–1, then state $(b, w-1)$).

Taking the better of the two gives the **Bellman equation**:

$$V(b, w) = \max\left\{0, \frac{b}{b+w}(1 + V(b-1, w)) + \frac{w}{b+w}(-1 + V(b, w-1))\right\},$$

$$V(b, 0) = b \text{ (all black left — take them all),} \quad V(0, w) = 0 \text{ (all white left — stop).}$$

Solve by Working Upward

Start from the boundary and build up (each “continue” value is the bracket above; the state’s V is that value if positive, else 0):

$$V(1, 0) = 1, \quad V(2, 0) = 2, \quad V(3, 0) = 3, \quad V(0, 1) = V(0, 2) = 0.$$

$$V(1, 1) : \frac{1}{2}(1 + \underbrace{V(0, 1)}_0) + \frac{1}{2}(-1 + \underbrace{V(1, 0)}_1) = \frac{1}{2} + 0 = \frac{1}{2} > 0 \Rightarrow V(1, 1) = \frac{1}{2}.$$

$$V(2, 1) : \frac{2}{3}(1 + \frac{1}{2}) + \frac{1}{3}(-1 + 2) = 1 + \frac{1}{3} = \frac{4}{3} \Rightarrow V(2, 1) = \frac{4}{3}.$$

$$V(1, 2) : \frac{1}{3}(1 + 0) + \frac{2}{3}(-1 + \frac{1}{2}) = \frac{1}{3} - \frac{1}{3} = 0 \Rightarrow V(1, 2) = 0 \text{ (indifferent — stop).}$$

$$V(2, 2) : \frac{1}{2}(1 + \underbrace{V(1, 2)}_0) + \frac{1}{2}(-1 + \underbrace{V(2, 1)}_{4/3}) = \frac{1}{2} + \frac{1}{6} = \frac{2}{3} \Rightarrow V(2, 2) = \frac{2}{3}.$$

$$V(3, 1) : \frac{3}{4}(1 + \frac{4}{3}) + \frac{1}{4}(-1 + 3) = \frac{7}{4} + \frac{1}{2} = \frac{9}{4} \Rightarrow V(3, 1) = \frac{9}{4}.$$

$$V(3, 2) : \frac{3}{5}(1 + \underbrace{V(2, 2)}_{2/3}) + \frac{2}{5}(-1 + \underbrace{V(3, 1)}_{9/4}) = \frac{3}{5} \cdot \frac{5}{3} + \frac{2}{5} \cdot \frac{5}{4} = 1 + \frac{1}{2} = \frac{3}{2}.$$

Optimal Stopping Rule and Answer

Continue exactly when the continuation value is positive. Scanning the states, the only reachable one where it is not is (1, 2) (value 0), plus any $b = 0$. So the rule from the starting bag (3, 2) is:

Keep drawing; stop only if the remaining bag becomes (1 black, 2 white) or runs out of black. The game is worth $V(3, 2) = \frac{3}{2}$.

Generalization

Nothing was special about (3, 2): for *any* starting counts (b, w) the same recursion

$$V(b, w) = \max\left\{0, \frac{b}{b+w}(1 + V(b-1, w)) + \frac{w}{b+w}(-1 + V(b, w-1))\right\}, \quad V(b, 0) = b, \quad V(0, w) = 0,$$

computed backward from the empty bag, determines both the value and the optimal rule (continue iff the bracket is positive). The hard part is purely the bookkeeping of filling in the $V(b, w)$ table.

Additional Problem 1 — Full Deck (26 Black, 26 Red)

Problem: A standard deck (26 black, 26 red) is shuffled face down; black pays +\$1, red −\$1; stop any time and keep your running profit. Value?

This is the *same* dynamic program at full scale — too big for hand computation, but everything above still applies.

Reframe (a bridge pinned to zero). Your running profit is (blacks turned) − (reds turned). Turning the *whole* deck always ends at $26 - 26 = 0$, so the profit is a symmetric random walk **pinned to 0 at both ends**. A “never stop” strategy is therefore worth exactly \$0 — **all the value is in the right to quit early** on a temporary lead. (Handy identity: current profit = reds remaining − blacks remaining, so “ahead” means “more red than black still buried.”)

The trap. The myopic rule “the next card is worth $\frac{b-r}{b+r}$, so stop as soon as $b \leq r$ ” is **wrong**: at the start $b = r = 26$ it says to stop for a value of \$0. That ignores **option value** — even when black and red remaining are *equal*, so the next flip has zero immediate expectation, the *right to stop after a good flip* makes continuing strictly worthwhile. Indeed $V(26, 26) > 0$, and the optimum keeps drawing at every balanced state (26−26, 10−10, ...); the true boundary is whatever the DP gives, and it is more patient than “stop the instant red leads.”

Same backward-induction DP. With b black and r red cards left,

$$V(b, r) = \max\left\{0, \frac{b}{b+r}(1 + V(b-1, r)) + \frac{r}{b+r}(-1 + V(b, r-1))\right\}, \quad V(0, r) = 0, \quad V(b, 0) = b,$$

solved up to the full deck, gives

$$V(26, 26) \approx \$2.62 \quad (2.6245\dots).$$

The value grows slowly: $V(n, n) = \frac{1}{2}, \frac{2}{3}, 0.85, 1.12, 1.84, 2.62$ for $n = 1, 2, 3, 5, 13, 26$ (and $V(1, 1) = \frac{1}{2}$ is the one-pair game you can do in your head).

Estimating \$2.62 by hand.

- **Order of magnitude:** a sum of $\sim k \pm 1$ steps has typical size $\sqrt{k}$, largest mid-deck: $SD \approx \sqrt{26^2/51} \approx \sqrt{13} \approx 3.6$ — so the answer is a *few dollars*.
- **Scaling law, calibrated at $n = 1$:** you cannot stop exactly at the peak, so you capture a fraction of that swing, giving $V(n, n) \approx \frac{1}{2}\sqrt{n}$ (exact at $n = 1$). A full deck: $\frac{1}{2}\sqrt{26} \approx \2.5 .
- **Sharper:** the asymptotic $V(n, n) \approx \sqrt{\pi n/12} = \sqrt{6.8} \approx \2.6 , essentially exact.

Play by backward induction (continue while $V(b, r) > 0$; roughly, stop once red sufficiently outnumbers black). Expected winnings $\approx \$2.62$.

Additional Problem 2 — Stop and Match the Next Card's Colour (Q405)

Problem: A shuffled deck has 26 red and 26 black cards. You repeatedly look at the top card and either *discard* it or *end the game*. When you end the game, you win if the *next* card (the one directly below the visible top card) has the **same colour** as the top card; otherwise you lose. What is the optimal strategy, and what is the win probability?

(You must end the game by the time only 2 cards remain — discarding then would leave no “next card.”)

Same idea, different payoff. As in the main problem, the cards are drawn *without replacement*, so the state is the remaining composition and we solve the game backward. The difference is the payoff: not a running profit, but a *win probability* for a colour match.

Step 1 — State and the two actions. Let $V(r, b)$ be the best achievable win probability when r red and b black cards remain (top card not yet revealed), and let $n = r + b$. Suppose the top card turns out **red** (probability r/n). Below it there are $r - 1$ red and b black cards.

- **Stop:** win iff the next card is red: $S_R = \frac{r-1}{n-1}$.
- **Discard:** the game continues from $(r-1, b)$: $C_R = V(r-1, b)$.

A black top card is the same with colours swapped: $S_B = \frac{b-1}{n-1}$, $C_B = V(r, b-1)$.

Step 2 — Bellman recursion. For $n \geq 3$,

$$V(r, b) = \frac{r}{n} \max \left\{ \frac{r-1}{n-1}, V(r-1, b) \right\} + \frac{b}{n} \max \left\{ \frac{b-1}{n-1}, V(r, b-1) \right\},$$

with the forced-stop boundary at $n = 2$: $V(2, 0) = V(0, 2) = 1$, $V(1, 1) = 0$.

Step 3 — Claim.

$$V(r, b) = 1 - \frac{rb}{n(n-1)} \quad (n = r + b \geq 3),$$

and for $n \geq 4$ discarding is always at least as good as stopping.

Step 4 — Base case $n = 3$ (by hand).

- $(3, 0)$: every card matches, $V = 1 = 1 - 0$. ✓
- $(2, 1)$, *red on top*: stop wins $\frac{1}{2}$; discarding leaves one red and one black, which cannot match $(0) \Rightarrow$ **stop**.
Black on top: stop wins 0; discarding leaves two reds $(1) \Rightarrow$ **discard**.
 $V(2, 1) = \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 = \frac{2}{3} = 1 - \frac{2}{6}$. ✓

This is the one place where stopping can be *strictly* better.

Step 5 — Inductive step ($n \geq 4$): continue vs. stop. Assume the formula for decks of size $n - 1$ (≥ 3). With red on top,

$$C_R = 1 - \frac{(r-1)b}{(n-1)(n-2)}, \quad S_R = \frac{r-1}{n-1}.$$

Multiply the difference by the common denominator $(n-1)(n-2)$:

$$\begin{aligned} (n-1)(n-2)(C_R - S_R) &= (n-1)(n-2) - (r-1)b - (r-1)(n-2) \\ &= (n-2)[(n-1) - (r-1)] - (r-1)b \\ &= (n-2)b - (r-1)b \quad (n-r=b) \\ &= b[(n-2) - (r-1)] = b(b-1) \quad (n-1-r=b-1). \end{aligned}$$

$$C_R - S_R = \frac{b(b-1)}{(n-1)(n-2)} \geq 0, \quad b = \text{number of opposite-colour cards below the top card.}$$

opposite cards b	$C_R - S_R$	conclusion
0	0	tie — every remaining card matches
1	0	tie — both win with probability $\frac{n-2}{n-1}$
≥ 2	> 0	waiting is strictly better

Why $b = 0$ or $b = 1$ is a tie. Let $m = n - 1$ be the number of cards below the red top card.

- $b = 0$: all m cards below are red. Stopping wins surely; continuing also wins surely (whatever ends on top, the next card is the same colour).
- $b = 1$: the single black card is equally likely to sit in any of the m positions below. *Stop now*: lose only if it is the very next card, probability $\frac{1}{m}$. *Keep waiting*: you lose only if the last three cards are R B R (with 3 left you see a red top card, stop, and the next card is black) — i.e. the black card is second-to-last, again probability $\frac{1}{m}$. Anywhere else you win: it surfaces earlier (discard it; all remaining cards are red) or the ending is R R B (stop, next is R) or B R R (discard B). Same win probability $\frac{n-2}{n-1}$.

- $b \geq 2$: stopping loses whenever *any* of the b black cards is next; waiting keeps extracting information about the composition.

Examples: 5R, 1B, red on top: stop $\frac{4}{5}$, wait $V(4, 1) = 1 - \frac{4}{20} = \frac{4}{5}$ (tie). 4R, 2B, red on top: stop $\frac{3}{5}$, wait $V(3, 2) = 1 - \frac{6}{20} = \frac{7}{10}$ (wait better).

Step 6 — The formula satisfies the recursion. Since each max equals its continuation value when $n \geq 4$,

$$\begin{aligned} V(r, b) &= \frac{r}{n} \left[1 - \frac{(r-1)b}{(n-1)(n-2)} \right] + \frac{b}{n} \left[1 - \frac{r(b-1)}{(n-1)(n-2)} \right] \\ &= 1 - \frac{rb[(r-1) + (b-1)]}{n(n-1)(n-2)} = 1 - \frac{rb(n-2)}{n(n-1)(n-2)} = 1 - \frac{rb}{n(n-1)}. \end{aligned}$$

By induction the formula holds for every $n \geq 3$, and the strategy attains it, so it is optimal.

Step 7 — When do you lose? Under the waiting strategy the only decision is at the last three cards: all one colour $\rightarrow$ win; $XXY \rightarrow$ stop on X , win; $YXX \rightarrow$ discard Y , win; $XYX \rightarrow$ stop on X , **lose**. So you lose only if the deck ends in an alternating triple (RBR or BRB):

$$P(\text{lose}) = \frac{rb(r-1) + br(b-1)}{n(n-1)(n-2)} = \frac{rb}{n(n-1)}.$$

Discard until 3 cards remain (stopping earlier only ties, and only when at most one opposite-colour card is left; e.g. “stop once ≤ 1 opposite card remains” is equally optimal). With 3 left, stop iff the top card’s colour appears twice among the three. Win probability

$$V(26, 26) = 1 - \frac{26 \cdot 26}{52 \cdot 51} = 1 - \frac{13}{51} = \frac{38}{51} \approx 74.5\%.$$

Common wrong answers: stop on the first card $\rightarrow \frac{25}{51}$; never stop (last two cards decide) $\rightarrow \frac{25}{51}$; “stop as soon as the top card’s colour is the majority of the remaining deck” $\rightarrow$ only $\frac{26}{51}$, because it stops with $b \geq 2$, which Step 5 shows is strictly worse. **Lesson (same as the main problem):** in finite-deck stopping problems, the option to wait has value; check the continue-minus-stop gap rather than acting on the immediate odds.

Additional Problem 3 — Blue and Red Balls: A Simpler Rule and the Value of Stopping (Q419)

Problem: A bag holds 3 blue and 2 red balls. You draw without replacement, winning +\$1 for each blue and −\$1 for each red, and may stop at any time. What is the expected value and the optimal strategy?

Same game as the Representative Problem. Relabel blue $\rightarrow$ black and red $\rightarrow$ white: this is exactly the 3 black / 2 white game above (and also Q285 and Q367). So the Bellman recursion and the table of values carry over unchanged, and the game is worth $V(3, 2) = \frac{3}{2}$. What this problem adds is a cleaner way to *state* the strategy and to *see* where the value comes from.

Step 1 — every reachable state. From the values computed above:

State (blue, red) left	V	Action
$(3, 2), (2, 2), (3, 1), (2, 1), (1, 1)$	$\frac{3}{2}, \frac{2}{3}, \frac{9}{4}, \frac{4}{3}, \frac{1}{2}$	continue (value of drawing > 0)
$(b, 0)$ — only blue left	b	continue (every draw is $+1$)
$(1, 2)$	0	indifferent: stop or continue
$(0, 1), (0, 2)$ — no blue left	0	stop (every draw is -1)

Step 2 — the rule in plain words. Since stopping and continuing tie at $(1, 2)$, one optimal rule is simply:

Keep drawing until all 3 blue balls are out, then stop.

The state $(1, 2)$ is reached only when your first two draws are both blue (you are up $+\$2$); stopping there with $\$2$ is equally good, but not necessary.

Step 3 — where the extra value comes from. If you had to draw all five balls you would always finish with $3 - 2 = +\$1$. Under the rule above you still collect all 3 blues, but you *skip every red ball that comes after the last blue* (the “trailing reds”):

$$\text{payoff} = 3 - (\text{\#trailing reds}) = 1 + \text{\#trailing reds}.$$

The last ball is red with probability $\frac{2}{5}$; the last two are both red with probability $\frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$. So

$$\mathbb{E}[\text{\#trailing reds}] = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}, \quad \mathbb{E}[\text{payoff}] = 1 + \frac{1}{2} = \frac{3}{2},$$

which matches $V(3, 2)$ from the Bellman recursion — an independent check.

Value $\frac{3}{2} = \$1.50$ (versus $\$1$ if you must draw every ball). Optimal strategy: draw until all three blue balls are out, then stop — never draw when only red balls remain; at 1 blue and 2 red left, stopping and continuing are equally good. The option to stop is worth exactly the expected number of red balls you avoid, $\$0.50$.

Takeaway. “Stop after the last good ball” is optimal *here* because the bag is small and tilted toward blue. For bigger or balanced decks (e.g. Additional Problem 1) the optimal rule stops earlier, on a temporary lead, and you need the full backward-induction table.

Contrast with the *With-Replacement* Walk

If the cards were **replaced** after each draw (previous section), the total would be a *free* recurrent random walk with $\text{EV} = 0$: no stopping rule helps. What makes *these* games have positive value is exactly that the bag or deck is *finite and not replaced*: the walk is pinned to a fixed endpoint (0 for the balanced deck), and knowing the leftover composition (b vs. w or r) gives you a real, computable edge. **Removing replacement turns a worthless game into a solvable one** — and makes the backward-induction DP well-posed, because the horizon is finite (the bag empties).

Concept Summary and Applications

- **Method:** finite horizon + sampling *without* replacement $\Rightarrow$ let the state be the remaining composition and solve $V = \max(\text{stop}, \text{expected continue})$ backward from the terminal (empty) states. Small cases (like 3B/2W) are hand-computable; large ones (a full deck) need a computer but obey the identical rule.
- **Watch for option value:** a myopic “stop when the next draw is non-positive” rule under-values the game, because the right to stop later is itself worth something.
- **Applications:** card-counting / advantage play, secretary-style hiring from a known pool, inventory drawdown, and any “draw without replacement, stop when ahead” problem.

13 Optimal Stopping with a Reroll Cost

Problem: You roll a 100-sided die. You may accept the face value in dollars, or pay \$1 to re-roll — as many times as you like. What is the optimal strategy, and what is the fair value of the game?

Before any formulas, get a feel for the game. The faces run 1 to 100, so a blind single roll averages 50.5. But you are not forced to keep a bad roll: for just \$1 you can throw it back and try again. Because a reroll is so cheap compared with the spread of possible values (up to \$100), you should be *picky* — keep only high rolls and pay the small fee to escape low ones. The whole problem is figuring out exactly *how* picky: what is the cutoff above which you keep, and below which you pay to reroll?

Step 1 — The Value Equation (a fixed point)

Let V be the value of the game: your expected earnings when you play optimally starting from a fresh roll. The trick is that V refers to itself. Suppose you roll and see a face x . You have two choices:

- **keep it** — you walk away with x dollars; or
- **reroll** — you pay \$1 and are back to a fresh roll, which by definition is worth V . So rerolling is worth $V - 1$ (the continuation value minus the \$1 fee).

A rational player takes whichever is larger, so the outcome of seeing face x is worth $\max(x, V - 1)$. Since each face $1, \dots, 100$ is equally likely, average over all of them:

$$V = \frac{1}{100} \sum_{x=1}^{100} \max(x, V - 1).$$

This is a **fixed-point equation**: V appears on both sides. That is not a problem — it is exactly what pins down the value.

Step 2 — Why the Rule Is a Threshold

Look at $\max(x, V - 1)$. Write $C = V - 1$ for the continuation value. If x is large it beats C and you keep it; if x is small, C wins and you reroll. So the optimal policy is automatically a **cutoff**: there is an integer k with

$$\text{reroll on } 1, 2, \dots, k, \quad \text{keep on } k + 1, \dots, 100,$$

and the switch happens where x crosses C , i.e.

$$k < C \leq k + 1, \quad C = V - 1.$$

Now rewrite the value equation using this structure. The k rerolled faces each contribute the continuation value $C = V - 1$ (*not* V — rerolling costs a dollar), and the kept faces contribute

their own value:

$$V = \frac{1}{100} \left(\underbrace{k(V-1)}_{k \text{ rerolled faces}} + \underbrace{\sum_{x=k+1}^{100} x}_{\text{kept faces}} \right).$$

Step 3 — The Value for a Given Cutoff

Clear the fraction and the sum. Multiply through by 100:

$$100V = k(V-1) + \sum_{x=k+1}^{100} x \implies (100-k)V = \sum_{x=k+1}^{100} x - k.$$

The remaining sum is the total $1 + \dots + 100 = 5050$ minus the first k terms:

$$\sum_{x=k+1}^{100} x = 5050 - \frac{k(k+1)}{2}.$$

Substituting and combining the two k -terms $(-\frac{k(k+1)}{2} - k = -\frac{k(k+3)}{2})$ gives the key equation

$$(100-k)V = 5050 - \frac{k(k+3)}{2}. \quad (\star)$$

This is exact once we know the correct integer k .

Step 4 — Solve for the Cutoff with a Quadratic

We could test values of k one by one, but there is a cleaner route. Recall the consistency condition $k < V-1 \leq k+1$. Subtract 1 throughout:

$$k-1 < V-2 \leq k,$$

so the integer k and the continuous quantity $V-2$ differ by less than 1. At the threshold they essentially coincide, so **relax k to its continuous value** and set $k = V-2$. Substitute into $(\star)$, using $100-k = 102-V$ and $\frac{k(k+3)}{2} = \frac{(V-2)(V+1)}{2}$:

$$V(102-V) = 5050 - \frac{(V-2)(V+1)}{2}.$$

Multiply by 2 and expand $(V-2)(V+1) = V^2 - V - 2$:

$$204V - 2V^2 = 10100 - (V^2 - V - 2) = 10102 - V^2 + V,$$

which collapses to a single quadratic:

$$V^2 - 203V + 10102 = 0.$$

By the quadratic formula,

$$V = \frac{203 \pm \sqrt{203^2 - 4 \cdot 10102}}{2} = \frac{203 \pm \sqrt{801}}{2} = \frac{203 \pm 28.30}{2}.$$

The upper root ≈ 115.7 is impossible (a die value can be at most 100), so the relevant root is $V = \frac{203-28.30}{2} \approx 87.35$.

Step 5 — Pin the Integer Cutoff and Get the Exact Value

The relaxed answer $V \approx 87.35$ tells us where the cutoff sits: keep $x \geq C = V - 1 \approx 86.35$, i.e. keep $x \geq 87$ and reroll $x \leq 86$, so the true integer count of rerolled faces is $k = 86$. Put $k = 86$ back into the *exact* equation ($\star$) to remove the continuous approximation:

$$(100 - 86)V = 5050 - \frac{86 \cdot 89}{2} \implies 14V = 5050 - 3827 = 1223 \implies V = \frac{1223}{14} \approx 87.357.$$

The quadratic's 87.35 and the exact $1223/14 = 87.357$ agree to two decimals; the tiny gap is only the continuous relaxation of the integer k .

Keep any roll of 87 or higher; otherwise pay \$1 and reroll. Fair value $V = \frac{1223}{14} \approx \87.36 .

Intuition. You stop only on the top 14 faces (about the top 14%). A reroll is cheap — \$1 against a spread up to \$100 — so it pays to hold out for a value far above the naive mean of 50.5; the \$1 fee per reroll is exactly what stops you from waiting forever for a 100.

Concept Summary and Applications

- **Method:** the value solves the fixed point $V = \mathbb{E}[\max(x, V - c)]$, where c is the reroll cost. The optimal policy is always a threshold; the cutoff is where the continuation value $C = V - c$ falls between two consecutive faces.
- **The quadratic shortcut:** write the value equation for a general cutoff, then *relax the integer cutoff to its continuous value* (set it equal to the continuation value). This turns a search into one quadratic; solve it, discard the root that exceeds the maximum face, and round to pin the exact integer cutoff.
- **Intuition:** cheap rerolls $\Rightarrow$ a high, picky threshold (hold out for a top value); expensive rerolls $\Rightarrow$ settle sooner.
- **Applications:** job-offer / house-hunting search with a search cost, “keep or redraw” games, and reservation-price setting.

Additional Problem 1 — 64-Sided Die, \$1 Reroll

Problem: Same game, but with a 64-sided die and a \$1 reroll cost.

The setup is identical. With cutoff m (keep $x \geq m$, reroll $x \leq m-1$), the $m-1$ rerolled faces each contribute $V-1$, so

$$64V = \sum_{x=m}^{64} x + (m-1)(V-1) \implies (65-m)V = \sum_{x=m}^{64} x - (m-1).$$

Writing the sum as $\sum_{x=m}^{64} x = 2080 - \frac{m(m-1)}{2}$ (the total $1 + \dots + 64 = 2080$ minus the first $m-1$ terms) and combining k -terms gives the key equation

$$(65-m)V = 2080 - \frac{(m-1)(m+2)}{2}. \quad (\star)$$

The cutoff is self-consistent when $m-1 < V-1 \leq m$, so relax m to its continuous value and set $m = V-1$. Then $65-m = 66-V$ and $(m-1)(m+2) = (V-2)(V+1)$, and $(\star)$ becomes $V(66-V) = 2080 - \frac{(V-2)(V+1)}{2}$. Multiplying by 2 and simplifying yields

$$V^2 - 131V + 4162 = 0.$$

The discriminant is $131^2 - 4 \cdot 4162 = 513$, so $V = \frac{131 \pm \sqrt{513}}{2} = \frac{131 \pm 22.65}{2}$. The upper root ≈ 76.8 exceeds 64 and is spurious; the relevant root is $V \approx 54.18$. This places the cutoff at $m = 54$ (keep $x \geq 54$, since $V-1 \approx 53.18$). Putting $m = 54$ back into $(\star)$ gives the exact value:

$$(65-54)V = 2080 - \frac{53 \cdot 56}{2} = 596 \implies 11V = 596 \implies V = \frac{596}{11} \approx 54.18.$$

Keep any roll of 54 or higher; value $V = \frac{596}{11} \approx \54.18 .

Additional Problem 2 — n -Sided Die, Reroll Costs $1/n$

Problem: An n -sided die, but now each reroll costs only $\frac{1}{n}$ dollars. What is the value of the game?

Everything is the same, with the reroll cost 1 replaced by $c = \frac{1}{n}$. After seeing k you take $\max(k, V - \frac{1}{n})$, so

$$V = \frac{1}{n} \sum_{k=1}^n \max\left(k, V - \frac{1}{n}\right).$$

With cutoff m (keep $x \geq m$), the $m-1$ rerolled faces each contribute $V - \frac{1}{n}$, and the self-consistency condition is $m-1 < V - \frac{1}{n} \leq m$. Writing out the value equation and the sum $\sum_{k=m}^n k = \frac{n(n+1)}{2} - \frac{(m-1)m}{2}$ gives, after moving the $(m-1)V$ term left,

$$(n-m+1)V = \frac{n(n+1)}{2} - \frac{(m-1)m}{2} - \frac{m-1}{n}. \quad (\star)$$

(The last term $-\frac{m-1}{n}$ is the total reroll cost: $m-1$ rerolled faces, each paying $\frac{1}{n}$.)

The quadratic shortcut. Relax the cutoff to its continuous value and set $m = V - \frac{1}{n} =: u$. It is cleanest to substitute *before* dividing: from the undivided value equation $nV = (m - 1)(V - \frac{1}{n}) + \frac{n(n+1)}{2} - \frac{(m-1)m}{2}$, both mixed terms become $(u - 1)u$ (since $m = u$ and $V - \frac{1}{n} = u$), giving

$$nV = (u - 1)u + \frac{n(n + 1)}{2} - \frac{(u - 1)u}{2} = \frac{(u - 1)u}{2} + \frac{n(n + 1)}{2}.$$

Multiply by 2 and use $2nV = 2n(u + \frac{1}{n}) = 2nu + 2$:

$$2nu + 2 = (u - 1)u + n(n + 1) = u^2 - u + n^2 + n,$$

which collapses to a remarkably clean quadratic in u :

$$u^2 - (2n + 1)u + (n + 2)(n - 1) = 0.$$

Its discriminant is a perfect square, $(2n + 1)^2 - 4(n + 2)(n - 1) = 9$, so

$$u = \frac{(2n + 1) \pm 3}{2} \in \{n - 1, n + 2\}.$$

The root $u = n + 2$ exceeds n and is spurious; the valid one is $u = n - 1$. Hence

$$V = u + \frac{1}{n} = n - 1 + \frac{1}{n}.$$

Here the relaxation is not even an approximation: $u = V - \frac{1}{n} = n - 1$ is already an integer, so the continuous cutoff coincides exactly with the true integer cutoff $m = n - 1$ (keep the top two faces $n - 1, n$; equivalently keep only n , since at $n - 1$ you are exactly indifferent).

With reroll cost $1/n$: value $V = n - 1 + \frac{1}{n} \approx n - 1$ (hold out for the top). For example $n = 100$ gives $V = 99.01$; $n = 6$ gives $V = 5\frac{1}{6} \approx 5.17$.

14 The Pigeonhole Principle

Problem: What is the minimum number of people in a room to guarantee that at least 5 of them share the same birth month?

Fill the Holes as Evenly as Possible

The 12 months are the “holes.” To *avoid* 5 in any single month, the most you can place is 4 per month:

$$12 \times 4 = 48 \text{ people with no month reaching } 5.$$

The very next person forces some month to 5. Hence

$12 \times 4 + 1 = 49$ people are needed to guarantee 5 share a birth month.

The Principle

If n items are placed into k boxes, some box holds at least $\lceil n/k \rceil$ items.

To **force** at least m items in some box, you need $k(m - 1) + 1$ items.

Here $k = 12$, $m = 5$, giving $12(5 - 1) + 1 = 49$. (The same formula gives 73 for “7 share a month.”)

Additional Problem — The Handshake Pigeonhole

Problem: At a party of $n \geq 2$ people, show that at least two people have shaken the same number of hands.

Each person shakes somewhere between 0 and $n - 1$ hands — that looks like n possible values for n people, not enough for pigeonhole on its own. The trick: 0 and $n - 1$ **cannot both occur**. If someone shook all $n - 1$ others, then nobody shook 0 hands; if someone shook 0, nobody shook everyone. So the handshake counts actually lie in a set of only $n - 1$ possible values, shared among n people.

n people into $n - 1$ possible handshake counts $\Rightarrow$ by pigeonhole, at least two people have the same handshake number.

Concept Summary and Applications

- **Existence, not construction:** pigeonhole proves *that* a coincidence must exist without finding it. The art is choosing the right “holes,” and sometimes shrinking their count (as with $0/n-1$ above).

- **Applications:** guaranteed collisions (hashing, birthdays), approximation results (two of $n+1$ reals within $1/n$), the existence of monochromatic structures (Ramsey-type arguments), and “some substring/subsum must repeat” proofs.

15 Coin Simulation and Randomness Extraction

Problem: You have a *biased* coin (heads with unknown probability p). How can you simulate a perfectly **fair** coin?

Von Neumann's Trick — Pair the Flips

Flip the biased coin **twice** and look at the ordered pair:

- $HT \Rightarrow$ output **H**;
- $TH \Rightarrow$ output **T**;
- HH or $TT \Rightarrow$ discard both and repeat.

Why it is exactly fair: the two kept patterns have *equal* probability,

$$P(HT) = p(1-p) = P(TH),$$

so conditioning on “ HT or TH ” gives $P(\text{output } H) = P(\text{output } T) = \frac{1}{2}$ — regardless of the unknown bias p .

Output H on HT , T on TH , discard HH/TT and repeat. Each output bit is fair for *any* bias $p \in (0, 1)$.

Concept Summary and Applications

Two complementary techniques cover almost every “simulate one source from another” problem:

- **Extraction by symmetry (von Neumann):** turn a biased or unknown source into fair bits using $P(HT) = P(TH)$ — no knowledge of p required. The same “compare two independent draws” idea makes a fair coin from *any* continuous source.
- **Rejection sampling:** build a target uniform distribution from an easy, *larger* equiprobable source by discarding out-of-range outcomes and retrying. Every kept outcome stays equally likely, so the result is exactly uniform.
- **Applications:** de-biasing hardware RNGs, fair decisions from unfair sources, simulating dice/-cards from coins (and vice versa), and Monte-Carlo seeding.

Additional Problem 1 — Fair d6 from a Fair Coin (Rejection Sampling)

Problem: Simulate a fair six-sided die using only fair coin flips.

Flip 3 coins to get one of $2^3 = 8$ equally likely outcomes; read them as a binary number 0–7. Keep 1–6 as the die face; **reject** 0 (000) and 7 (111) and reflip. Each kept value has probability $\frac{1}{8}$, and conditioning on acceptance gives $\frac{1/8}{6/8} = \frac{1}{6}$ per face — a fair die.

$$P(\text{accept per round}) = \frac{6}{8} = \frac{3}{4}, \quad E[\text{rounds}] = \frac{8}{6} = \frac{4}{3}, \quad E[\text{flips}] = 3 \cdot \frac{4}{3} = 4.$$

Three coin flips $\rightarrow$ a number 1–6 (reject 0 and 7, reflip). Uses 4 flips on average.

Because 6 is not a power of 2, no fixed-length scheme can be exactly fair — rejection (unbounded flips) is unavoidable. The simple 3-flip method uses 4 flips because it throws away the two rejected outcomes' entropy; recycling them (the Knuth–Yao construction) averages $\frac{11}{3} \approx 3.67$, and the information floor is $\log_2 6 \approx 2.585$ flips.

Additional Problem 2 — Fair Coin from a Continuous Source

Problem: Given i.i.d. draws $U_1, U_2, \dots$ from some continuous distribution, simulate a fair coin.

The distribution does not need to be uniform. This is the key point, so let us be explicit about it. The draws could come from *any* continuous distribution at all — Uniform[0, 1], a bell-shaped Normal, a lop-sided Exponential, a heavily skewed heavy-tailed law — and the method below is still *exactly* fair. You do not need to know which distribution it is, and you do not need it to be symmetric or centred anywhere in particular.

Draw two values and compare: output **H** if $U_1 < U_2$, **T** if $U_1 > U_2$ (the tie $U_1 = U_2$ has probability 0 for a continuous law; redraw in the measure-zero event it occurs).

Why it is fair for any continuous law. The only fact we use is that U_1 and U_2 are *independent and identically distributed*. Because they share the same distribution, relabelling “draw 1” and “draw 2” cannot change any probability — the pair (U_1, U_2) and the swapped pair (U_2, U_1) are statistically indistinguishable. This symmetry (called *exchangeability*) forces

$$P(U_1 < U_2) = P(U_2 < U_1),$$

and since a continuous law makes ties impossible, these two probabilities are the complementary halves of certainty:

$$P(U_1 < U_2) + P(U_1 > U_2) = 1 \implies P(U_1 < U_2) = P(U_1 > U_2) = \frac{1}{2}.$$

Notice the argument never mentions the *shape* of the distribution — only that both draws follow the same one. That is exactly why uniformity is irrelevant. It is the same von Neumann pairwise comparison as before, now on real numbers instead of coin pairs.

Compare two independent draws from *any* continuous distribution (need not be uniform, or even symmetric): output **H** if the first is smaller, **T** if larger. Fair for every continuous source, with no knowledge of it required.

16 Poisson Process and Arrival Rates

Problem: The probability of seeing at least one car in an hour is $\frac{544}{625}$. What is the probability of seeing *no* cars in a 15-minute interval?

Background: the Poisson Process

A **Poisson process** models events (arrivals) occurring in continuous time at a constant average rate λ per unit time, with three defining properties:

- **Independent increments:** counts in disjoint time intervals are independent.
- **Stationary increments:** the count in an interval depends only on its *length*, not its location.
- **Orderliness:** arrivals occur one at a time (no simultaneous events).

Then the number of arrivals in an interval of length t is $\text{Poisson}(\lambda t)$, the waiting time between arrivals is $\text{Exponential}(\lambda)$, and in particular

$$P(\text{no arrivals in length } t) = e^{-\lambda t}.$$

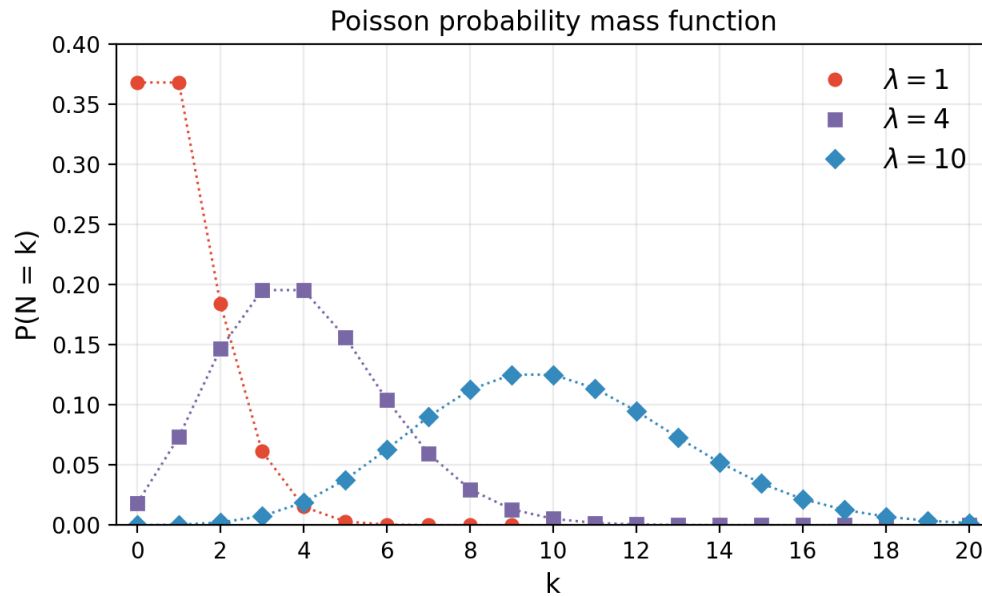
Background: the Poisson Distribution

The Poisson distribution describes a **count** — how many times something happens — when the events are rare per opportunity but there are many opportunities. A single number $\mu > 0$, the expected count, controls it completely:

$$N \sim \text{Poisson}(\mu): \quad P(N = k) = e^{-\mu} \frac{\mu^k}{k!}, \quad k = 0, 1, 2, \dots$$

Mean = Variance = μ .

Reading the formula: $\mu^k/k!$ grows then shrinks as k increases, and the $e^{-\mu}$ prefactor is exactly the constant that makes all the probabilities sum to 1 (it is the Taylor series $\sum_k \mu^k/k! = e^\mu$). The distribution peaks near $k = \mu$ and, as μ grows, spreads out and becomes bell-shaped:



For $\mu = 1$ the mass sits on the smallest counts (most likely 0 or 1); for $\mu = 4$ it centres near 4; for $\mu = 10$ it is a wide, near-symmetric hump around 10. In every case the mean and the variance both equal μ , so the spread (standard deviation $\sqrt{\mu}$) grows only like the square root of the mean.

The Law of Rare Events

Where does this formula come from? Imagine chopping a time interval into n tiny slots, each of which independently either contains one arrival (probability p) or none. The total count is then $\text{Binomial}(n, p)$, with mean $\mu = np$. Now let the slots become finer and finer: $n \rightarrow \infty$ while each slot gets rarer, $p \rightarrow 0$, holding the average $np = \mu$ fixed. In that limit the binomial probabilities converge to the Poisson formula:

$$\binom{n}{k} p^k (1-p)^{n-k} \xrightarrow[n \rightarrow \infty, p \rightarrow 0]{np \rightarrow \mu} e^{-\mu} \frac{\mu^k}{k!}.$$

This is the **law of rare events**: a Poisson count is what you get from a huge number of independent trials, each individually very unlikely to fire, when the *expected* number of firings is moderate. The three binomial ingredients map cleanly onto the limit: $\binom{n}{k} p^k \rightarrow \mu^k/k!$ counts the ways to place k successes, and $(1-p)^{n-k} \rightarrow e^{-\mu}$ is the chance the other slots stay empty (using $(1 - \frac{\mu}{n})^n \rightarrow e^{-\mu}$).

Numeric example. Take $\mu = 2$ and watch $\text{Binomial}(n, \frac{2}{n})$ approach $\text{Poisson}(2)$ for the probability of exactly $k = 1$ event:

	$n = 5, p = 0.4$	$n = 20, p = 0.1$	$n = 100, p = 0.02$	$\text{Poisson}(2)$
$P(N = 1)$	0.2592	0.2702	0.2707	$2e^{-2} = 0.2707$

Even by $n = 100$ the binomial value is indistinguishable from the Poisson limit to four decimals. A classic real-world instance: the yearly number of Prussian cavalymen killed by horse kicks per

army corps (a rare event with many soldier-years of exposure, averaging about 0.7 deaths) follows a Poisson distribution almost exactly.

Rule of thumb. The approximation $\text{Binomial}(n, p) \approx \text{Poisson}(np)$ is already excellent when $n \geq 20$ and $p \leq 0.05$, and it turns an awkward binomial (large n , tiny p , huge factorials) into a one-line Poisson evaluation. In our arrivals problem, $\mu = \lambda t$ is the expected number of arrivals in an interval of length t .

Solving the Car Problem

Let $p = P(\text{no cars in 15 min})$. An hour is four disjoint 15-minute intervals, and by **independent increments** the “no car” events multiply:

$$P(\text{no cars in an hour}) = p^4.$$

We are given $P(\geq 1 \text{ car in an hour}) = \frac{544}{625}$, so

$$p^4 = P(\text{no cars in an hour}) = 1 - \frac{544}{625} = \frac{81}{625} \implies p = \left(\frac{81}{625}\right)^{1/4} = \frac{3}{5}.$$

$$P(\text{no cars in 15 min}) = \frac{3}{5}.$$

Equivalently in Poisson terms: $e^{-\lambda} = \frac{81}{625}$, so $e^{-\lambda/4} = (e^{-\lambda})^{1/4} = \frac{3}{5}$ — the “no arrival” probability over a quarter interval is the fourth root of the hourly one.

Concept Summary and Applications

- **Key handle:** $P(0 \text{ in } t) = e^{-\lambda t}$, and disjoint intervals are independent — so “no event” probabilities multiply and rates add. Splitting/merging Poisson streams keeps them Poisson.
- **Rate scaling:** halving the interval takes the “no arrival” probability to its square root; the count’s mean scales linearly with time.
- **Applications:** order-flow and trade-arrival modelling, queueing and call centres, radioactive decay, and defect/failure counts.

17 Simpson's Paradox

Problem: Is it possible for one option to be better than another in *every* subgroup, yet worse when the subgroups are pooled? Give an example.

Yes — Two Kidney-Stone Treatments

Yes. A comparison that holds *within every subgroup* can reverse once the groups are combined, because a lurking variable is spread unevenly. Recovery rates:

	Small stones	Large stones	Overall
Treatment A	$\frac{81}{87} = \mathbf{93\%}$	$\frac{192}{263} = \mathbf{73\%}$	$\frac{273}{350} = 78\%$
Treatment B	$\frac{234}{270} = 87\%$	$\frac{55}{80} = 69\%$	$\frac{289}{350} = \mathbf{83\%}$

Treatment A wins in *both* subgroups (93% vs. 87% and 73% vs. 69%), yet pooled, Treatment B looks better (83% vs. 78%).

Lurking Variables and Confounders

The reversal is driven by a third variable hiding behind the table, so it is worth defining the terms precisely before explaining the mechanism.

A **lurking variable** is a variable that influences the outcome but is *not shown* in the analysis — it “lurks” outside the two columns you are comparing. Here the lurking variable is **stone size**. The headline comparison (Treatment A vs. B, overall) never mentions stone size, yet stone size strongly affects whether a patient recovers: small stones are easy to cure ($\sim 90\%$), large stones are hard ($\sim 70\%$). A variable can lurk simply because the analyst forgot it, never measured it, or chose to average over it.

A **confounder** (or confounding variable) is the important special case of a lurking variable that is associated with *both* the “cause” being studied *and* the outcome. It thereby tangles up (“confounds”) the apparent relationship between them. For stone size to be a confounder here, it must be linked to both:

- **linked to the treatment:** the doctors did not assign treatments at random. Surgery (Treatment A) was chosen mostly for the *severe*, large-stone cases (263 of A’s 350 patients), while B was reserved mostly for the *easy*, small-stone cases (270 of 350).
- **linked to the outcome:** large stones are harder to cure regardless of treatment.

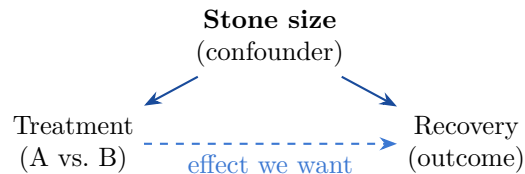
Because stone size is tied to *both* the treatment received and the chance of recovery, it confounds the raw A-vs-B comparison: Treatment A looks worse overall not because it is worse, but because it was handed the harder patients.

Lurking variable: any unshown variable that affects the outcome.

Confounder: a lurking variable linked to *both* the treatment/group *and* the outcome, so it distorts their measured association.

Every confounder is a lurking variable; not every lurking variable is a confounder (it must touch *both* sides).

A quick picture of the causal wiring, with stone size sitting upstream of both:



The dashed arrow is the treatment effect we actually care about; the two solid arrows are the confounding path that contaminates the naive overall comparison. The fix is to “control for” the confounder — compare A vs. B *within* each stone-size group, which is exactly what the subgroup rows do.

Lurking vs. confounding, in one line: a lurking variable is merely absent from the analysis; it becomes a confounder only when it is correlated with both the grouping and the outcome. Randomised assignment (a coin flip deciding A or B) breaks the confounder’s link to the treatment, which is precisely why randomised trials defeat Simpson’s paradox.

Why It Happens — Weighted Averages

The overall rate is a **weighted average** of the subgroup rates, weighted by group sizes. The lurking variable is **stone size**, confounded with treatment choice: large stones are harder to cure, and Treatment A was used mostly on the *hard* cases (263 of its 350 patients), while B was used mostly on the *easy* ones (270 of 350):

$$A \approx \frac{87(93\%) + 263(73\%)}{350} \text{ (mostly the 73\%)}, \quad B \approx \frac{270(87\%) + 80(69\%)}{350} \text{ (mostly the 87\%).}$$

Better per-group rates lose to worse group weights.

Which number to trust? When the groups are a pre-existing confounder (as here), the subgroup comparison is honest: A is genuinely better, since severity — not treatment — drives the aggregate gap. If instead the grouping variable were a consequence of the choice, the pooled figure could be correct. It depends on the causal structure, not the numbers alone.

Concept Summary and Applications

- **Mechanism:** an aggregate is a size-weighted average of parts; reversal occurs when the weights are correlated with the rates in the opposite direction to the within-group ordering.
- **Red flag:** always ask whether a confounder is distributed unevenly across the things being pooled before trusting a headline average.

- **Applications:** A/B tests with unequal traffic mixes, drug-trial analysis, admissions/hiring fairness (the 1973 U.C. Berkeley case), and any KPI aggregated over heterogeneous segments.

Additional Problem — Batting Averages

Problem: Player A beats Player B in batting average in *both* the first and second halves of a season. Can B finish with the higher average overall?

Yes — the season average is $\frac{H_1+H_2}{N_1+N_2} = \frac{N_1a_1+N_2a_2}{N_1+N_2}$ (hits over at-bats), a weight-by-at-bats average. A real example (Justice vs. Jeter):

	Year 1	Year 2	Combined
Player A	$\frac{104}{411} = .253$	$\frac{45}{140} = .321$	$\frac{149}{551} = .270$
Player B	$\frac{12}{48} = .250$	$\frac{183}{582} = .314$	$\frac{195}{630} = \mathbf{.310}$

A hits better both years, but B — who took most of his at-bats in the higher-average year — wins the combined average.

Yes: overall averages are at-bat-weighted, so unequal weights let the season-long order reverse the within-season order. (This is Simpson's paradox.)

18 Circle Regions from Chords (Euler's Formula)

Background: Euler's Formula in 3D (Convex Polyhedra)

The most famous form of Euler's formula is about solid shapes. Take any convex polyhedron — a cube, a pyramid, a soccer-ball-shaped solid — and count its corners (**vertices** V), its **edges** E , and its flat **faces** F . Whichever convex polyhedron you pick, these three counts always satisfy

$$V - E + F = 2.$$

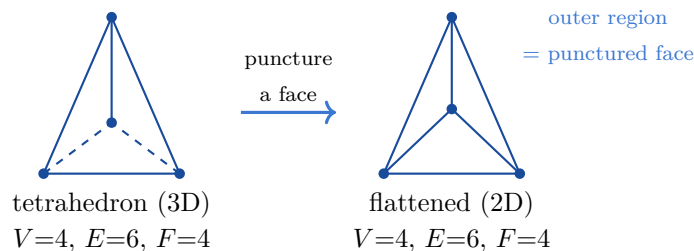
Check three familiar solids:

$$\text{cube } 8 - 12 + 6 = 2, \quad \text{tetrahedron } 4 - 6 + 4 = 2, \quad \text{octahedron } 6 - 12 + 8 = 2.$$

It is genuinely striking that the *same* fixed number 2 falls out of solids with wildly different vertex, edge, and face counts.

From Polyhedra to the Plane (2D Planar Graphs)

The identical law governs flat drawings, and a single picture bridges the two. Take a convex polyhedron, puncture one face, and stretch that hole open until the whole surface flattens into the plane — like cutting one face off a balloon and pulling its rim wide. No vertex, edge, or face is created or destroyed, so V , E , F are unchanged. The one face you punctured is now stretched around the *outside* of the drawing: it becomes the flat picture's single unbounded **outer face**.

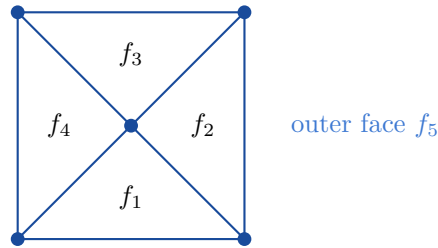


So a **planar graph** — dots (vertices) joined by non-crossing edges into one connected piece — obeys the very same identity, as long as we count *all* faces including the outer region:

$$V - E + F = 2 \quad (\text{connected planar graph, } F \text{ includes the outer face}).$$

Simplest check — a triangle. Three corners, three sides, and two faces (the inside and the outside): $V = 3$, $E = 3$, $F = 2$, so $3 - 3 + 2 = 2$. ✓

A slightly richer example — a square with both diagonals. Add the two diagonals of a square. In a planar graph edges may not cross, so the point where the diagonals meet becomes a *new vertex* in the middle. Now count:



Vertices: 4 corners + 1 centre = 5. Edges: 4 sides + 4 half-diagonals (each diagonal is split in two by the centre) = 8. Faces: 4 inner triangles + 1 outer = 5. Indeed $V - E + F = 5 - 8 + 5 = 2$. ✓ (This is exactly the $n = 4$ chord picture we return to below.)

The common value 2 is the **Euler characteristic of the sphere**, a topological invariant — a polyhedron's surface and a connected planar graph are both the sphere in disguise, which is why the count never depends on the exact drawing.

The Problem

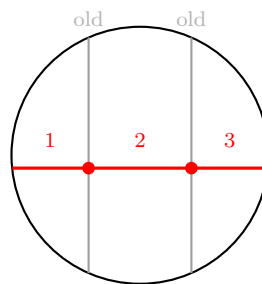
Problem: Place n points on a circle and draw all $\binom{n}{2}$ chords, in general position (no three chords meeting at one interior point, which maximises regions). Into how many regions is the disk divided?

Incremental Count — Why Each Chord and Each Crossing Adds a Region

Start with 1 region (the empty disk) and add the chords one at a time. Track how many *new* regions each chord creates as you sweep along it.

A chord with no crossings adds exactly one region. It enters the disk, runs through a single existing region, and slices that region into two — a net gain of one.

Each crossing the chord makes adds one more region. Every time the new chord crosses a chord already drawn, it passes from one region into the next, and that passage cuts an extra region in two. So a chord that crosses c existing chords is itself chopped into $c + 1$ little segments, and each segment splits a region: it adds $c + 1$ regions in all = 1 (for the chord) + c (one per crossing).



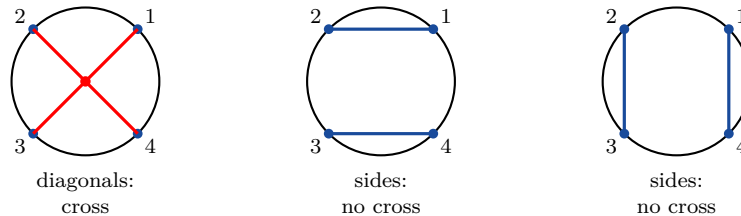
Here the new (red) chord crosses 2 old chords, so it is cut into 3 segments and creates 3 new regions = $1 + 2$. Summing over all chords, the total count is

$$\underbrace{1}_{\text{empty disk}} + \underbrace{(\text{number of chords})}_{\text{one region per chord}} + \underbrace{(\text{number of crossings})}_{\text{one region per crossing}}.$$

There are $\binom{n}{2}$ chords (one per pair of points), so all that remains is to count the interior crossings.

Why Exactly $\binom{n}{4}$ Crossings

The clean way to count crossings is to match each one with a set of 4 boundary points. Pick any 4 of the n points; they form a convex quadrilateral. There are three ways to pair up 4 points into two chords, but only *one* of them makes the chords cross inside — the pair of **diagonals**. The other two pairings join adjacent points (“opposite sides”) and stay apart:



This is a **bijection**: every crossing comes from a unique set of 4 endpoints (two crossing chords have 4 distinct ends — if they shared one, they would meet on the boundary, not inside), and every set of 4 points yields exactly one crossing. So the number of interior crossings equals the number of ways to choose 4 points, $\binom{n}{4}$.

$$R(n) = 1 + \binom{n}{2} + \binom{n}{4}.$$

Numeric examples. $n = 5$: $\binom{5}{2} = 10$ chords and $\binom{5}{4} = 5$ crossings, so $R = 1 + 10 + 5 = 16$. $n = 6$: $\binom{6}{2} = 15$ chords and $\binom{6}{4} = 15$ crossings, so $R = 1 + 15 + 15 = 31$.

Warning: the Answer Is *Not* 2^{n-1}

The values for $n = 1, 2, 3, 4, 5$ are 1, 2, 4, 8, 16 — exactly the powers of two, which tempts everyone to guess $R(n) = 2^{n-1}$. It is **wrong**. The doubling breaks at $n = 6$:

n	1	2	3	4	5	6	7
$R(n) = 1 + \binom{n}{2} + \binom{n}{4}$	1	2	4	8	16	31	57
2^{n-1} (the trap)	1	2	4	8	16	32	64

*Five data points agreeing with 2^{n-1} prove nothing: at $n = 6$ the true count is **31**, not 32. This is the textbook warning against extrapolating a pattern from small cases — the polynomial $1 + \binom{n}{2} + \binom{n}{4}$ just happens to coincide with 2^{n-1} for $n \leq 5$.*

Verifying with Euler's Formula

The incremental count is convincing but informal; Euler's formula pins it down rigorously. Treat the whole figure — boundary points, crossings, arcs, and chord-segments — as one connected planar graph and apply $V - E + F = 2$.

Vertices are the n points on the circle plus the $\binom{n}{4}$ interior crossings:

$$V = n + \binom{n}{4}.$$

Edges are the n boundary arcs between adjacent points, plus the chord pieces. Each of the $\binom{n}{2}$ chords is cut into segments by the crossings on it; since every crossing lies on exactly two chords, it creates 2 extra chord pieces, giving

$$E = n + \binom{n}{2} + 2\binom{n}{4}.$$

Solving Euler's formula for the face count:

$$F = 2 - V + E = 2 + \binom{n}{2} + \binom{n}{4}.$$

Finally — and this is the key last step — **discard the outer face**. Euler's F counts *every* region including the unbounded one outside the circle, but the problem asks only for regions *inside* the disk. Removing that single outer region gives

$$R(n) = F - 1 = 1 + \binom{n}{2} + \binom{n}{4},$$

matching the incremental count exactly.

Concept Summary and Applications

- **Two-tool method:** count regions either *incrementally* (initial region + one per edge + one per new crossing) or *rigorously* via $V - E + F = 2$ (then subtract the outer face); they must agree.
- **General position maximises** regions: forcing three chords to be concurrent merges crossings and *loses* regions.
- **Applications:** planar subdivision and arrangement counting in computational geometry, face counts in graph drawings, and the classic “don't trust 1, 2, 4, 8, ...” cautionary tale about pattern-matching.

19 Expected Length of the Longest Piece

Problem: A unit stick is broken at two independent uniform points. What is the expected length of the **longest** of the three pieces?

Setup

Order the break points as $0 \leq U \leq V \leq 1$; the pieces are

$$X = U, \quad Y = V - U, \quad Z = 1 - V, \quad M = \max\{X, Y, Z\}.$$

The ordered pair (U, V) is uniform on the triangle $0 < u < v < 1$ (area $\frac{1}{2}$), so (X, Y) is uniform on $\{x > 0, y > 0, x + y < 1\}$ with **density** 2 (the constant f with $f \cdot \frac{1}{2} = 1$). Hence $\mathbb{P}((X, Y) \in A) = 2 \text{area}(A)$.

Tail Probability of the Maximum

The longest piece is always at least the average, so $M \geq \frac{1}{3}$ and $\mathbb{P}(M > m) = 1$ for $m < \frac{1}{3}$. For larger m the geometry splits into two regimes.

Case $\frac{1}{3} \leq m \leq \frac{1}{2}$ — **compute** $\mathbb{P}(M \leq m)$ **directly**. Here $\{M \leq m\}$ means *all three* pieces are $\leq m$: $X \leq m$, $Y \leq m$, and the third piece $Z = 1 - X - Y \leq m$, i.e. $X + Y \geq 1 - m$. In the (X, Y) plane these three cuts carve out a small triangle with corners

$$(m, m), \quad (1-2m, m), \quad (m, 1-2m).$$

Why the side has length $3m - 1$. Take the horizontal leg: it runs at height $y = m$ from the corner $(1-2m, m)$ across to (m, m) , so its length is

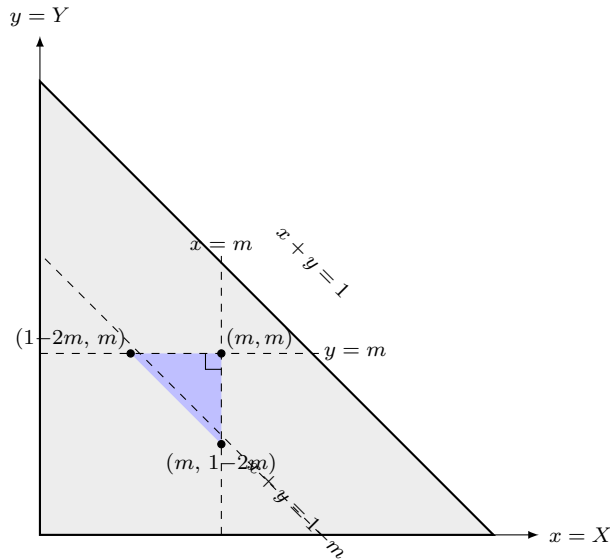
$$m - (1 - 2m) = 3m - 1.$$

By the $x \leftrightarrow y$ symmetry the vertical leg from $(m, 1-2m)$ up to (m, m) has the same length $3m - 1$, and the right angle sits at (m, m) . So the region is a right isosceles triangle with both legs equal to $3m - 1$ (equivalently $|1 - 3m|$), giving

$$\text{area} = \frac{1}{2}(3m - 1)^2.$$

Multiplying by the density 2,

$$\mathbb{P}(M \leq m) = 2 \cdot \frac{1}{2}(3m - 1)^2 = (3m - 1)^2, \quad \mathbb{P}(M > m) = 1 - (3m - 1)^2.$$



Drawn at $m = \frac{2}{5}$: the sample space is the large triangle (density 2), and $\{M \leq m\}$ is the shaded right-isosceles triangle — right angle at (m, m) , legs $3m - 1$; its area $\frac{1}{2}(3m - 1)^2$ times 2 gives $\mathbb{P}(M \leq m) = (3m - 1)^2$.

Case $\frac{1}{2} \leq m \leq 1$ — compute the tail $\mathbb{P}(M > m)$ directly. Now it is easier to go after the tail. The event $\{M > m\}$ means *at least one* piece exceeds m , i.e. the union $\{X > m\} \cup \{Y > m\} \cup \{Z > m\}$.

Why the factor 3 (disjointness). When $m \geq \frac{1}{2}$, two pieces **cannot both** exceed m : if two of them were each larger than $m \geq \frac{1}{2}$, their lengths alone would already sum to more than 1, which is impossible for pieces of a unit stick. So the three events are **mutually exclusive**, and the probability of their union is simply the sum of the three (no overlaps to subtract):

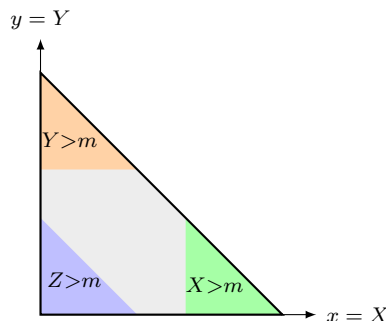
$$\mathbb{P}(M > m) = \mathbb{P}(X > m) + \mathbb{P}(Y > m) + \mathbb{P}(Z > m) = 3\mathbb{P}(X > m).$$

Why each term is the square $(1 - m)^2$. The event $\{X > m\}$ is the corner triangle $\{x > m, y > 0, x + y < 1\}$, with corners $(m, 0)$, $(1, 0)$, and $(m, 1 - m)$. Its horizontal leg (from $x = m$ to $x = 1$ along $y = 0$) has length $1 - m$, and its vertical leg (from $(m, 0)$ up to the hypotenuse $x + y = 1$ at $(m, 1 - m)$) also has length $1 - m$. So it is a right triangle with both legs $1 - m$, area $\frac{1}{2}(1 - m)^2$; times the density 2,

$$\mathbb{P}(X > m) = 2 \cdot \frac{1}{2}(1 - m)^2 = (1 - m)^2.$$

The **square** is just the area of a right triangle whose two legs both shrink like $1 - m$ as m grows toward 1. Hence

$$\mathbb{P}(M > m) = 3(1 - m)^2.$$



Drawn at $m = 0.6$ (so $1 - m = 0.4$): the three events are the three *disjoint* corner triangles, each a right triangle with legs $1 - m$, area $\frac{1}{2}(1 - m)^2$ and probability $(1 - m)^2$. They do not touch (because $m \geq \frac{1}{2}$), so the tail is their sum $3(1 - m)^2$.

Collecting,

$$\mathbb{P}(M > m) = \begin{cases} 1, & 0 \leq m < \frac{1}{3}, \\ 1 - (3m - 1)^2, & \frac{1}{3} \leq m \leq \frac{1}{2}, \\ 3(1 - m)^2, & \frac{1}{2} \leq m \leq 1. \end{cases}$$

Compute the Expectation

Using $\mathbb{E}[M] = \int_0^1 \mathbb{P}(M > m) dm$:

$$\underbrace{\int_0^{1/3} 1 dm}_{1/3} + \underbrace{\int_{1/3}^{1/2} (6m - 9m^2) dm}_{[3m^2 - 3m^3]_{1/3}^{1/2} = 11/72} + \underbrace{\int_{1/2}^1 3(1 - m)^2 dm}_{1/8} = \frac{1}{3} + \frac{11}{72} + \frac{1}{8} = \frac{44}{72} = \frac{11}{18}.$$

$$\mathbb{E}[\text{longest piece}] = \frac{11}{18} \approx 0.611.$$

The One-Line Shortcut (Uniform Spacings)

Breaking a unit stick at $n - 1$ uniform points produces n pieces called **uniform spacings**. Sort them from smallest to largest, $D_{(1)} \leq D_{(2)} \leq \dots \leq D_{(n)}$. There is a clean formula for the expected value of the j -th smallest,

$$\mathbb{E}[D_{(j)}] = \frac{1}{n} \sum_{i=1}^j \frac{1}{n - i + 1}.$$

Apply to $n = 3$. Plug $n = 3$ into the boxed formula. The inner terms $\frac{1}{n-i+1}$ run through $\frac{1}{3}, \frac{1}{2}, 1$, and each partial sum is scaled by $\frac{1}{n} = \frac{1}{3}$:

$$\underbrace{\mathbb{E}[D_{(1)}]}_{\text{shortest}} = \frac{1}{3} \left(\frac{1}{3} \right) = \frac{1}{9}, \quad \underbrace{\mathbb{E}[D_{(2)}]}_{\text{middle}} = \frac{1}{3} \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{5}{18}, \quad \underbrace{\mathbb{E}[D_{(3)}]}_{\text{longest}} = \frac{1}{3} \left(\frac{1}{3} + \frac{1}{2} + 1 \right) = \frac{11}{18}.$$

Reading the three sums: the shortest keeps only the first term; the middle adds the second; the longest adds the third as well. As a check they must cover the whole stick, and indeed $\frac{1}{9} + \frac{5}{18} + \frac{11}{18} = \frac{2+5+11}{18} = 1$. ✓

The boxed formula itself is derived next; the derivation can be skipped.

Explanation of the One-Line Formula (Can Skip If Too Complex)

Step 1 — spacings are normalised exponentials. A standard and useful fact: if $T_1, \dots, T_n$ are independent $\text{Exponential}(1)$ random variables and $S = T_1 + \dots + T_n$, then the scaled vector $(\frac{T_1}{S}, \dots, \frac{T_n}{S})$ has *exactly* the same distribution as the n uniform spacings. Intuitively, dropping n independent “interarrival” times and rescaling so they sum to 1 reproduces the stick breaks. In particular the pieces sorted by size correspond to the T_i sorted by size, and since the average spacing is $\frac{1}{n}$ while the average T_i is 1,

$$\mathbb{E}[D_{(j)}] = \frac{1}{n} \mathbb{E}[T_{(j)}],$$

where $T_{(j)}$ is the j -th smallest of the exponentials.

Step 2 — exponential order statistics via memorylessness. Here the magic of the exponential distribution does the work. Wait for the *first* (smallest) of n independent $\text{Exp}(1)$ clocks to ring. The minimum of n rate-1 exponentials is itself exponential with rate n , so its mean waiting time is $\frac{1}{n}$. Because the exponential is **memoryless**, once the first clock rings the remaining $n - 1$ clocks are “as good as new”; the wait for the *next* ring is the minimum of $n - 1$ fresh $\text{Exp}(1)$ clocks, mean $\frac{1}{n-1}$. Continuing, the gap between the $(i - 1)$ -th and i -th rings has mean $\frac{1}{n-i+1}$. Adding these gaps up to the j -th ring,

$$\mathbb{E}[T_{(j)}] = \sum_{i=1}^j \frac{1}{n-i+1}.$$

Dividing by n (Step 1) gives the boxed formula. (Substituting $n = 3$ recovers the shortest/middle/longest values found above.)

Concept Summary and Applications

- **Two routes:** geometry (uniform density on the simplex + tail integration) and the spacings formula. The longest piece averages about 61% of the stick — far above the naive $\frac{1}{3}$, because randomness usually produces one dominant piece.
- **Density weighting:** on the (X, Y) simplex, probability = $2 \times$ area; forgetting the factor 2 is the classic error.
- **Applications:** load balancing and max-bucket occupancy, spacing statistics for arrivals on an interval, and coverage/gap problems.

20 Continuous Order Statistics

Problem: A lottery ticket has 10 slots, each hiding an independent Uniform $[0, 1]$ number. Your payout is the **maximum** of the ten. What is its fair price (the expected payout)?

Distribution of the Maximum

Write $M = \max(U_1, \dots, U_{10})$ for the payout. To average it we first describe its distribution with two standard objects: the CDF and the density.

The CDF $F_M(x)$. The *cumulative distribution function* is just the probability that M does not exceed a level x , written $F_M(x) = \mathbb{P}(M \leq x)$. The maximum is $\leq x$ exactly when *every* one of the ten numbers is $\leq x$. Each U_i is Uniform $[0, 1]$, so $\mathbb{P}(U_i \leq x) = x$, and because the ten draws are independent the probabilities multiply:

$$F_M(x) = \mathbb{P}(U_1 \leq x) \cdots \mathbb{P}(U_{10} \leq x) = x^{10}, \quad 0 \leq x \leq 1.$$

The density $f_M(x)$, and why it is $F'_M(x)$. The *probability density function* f_M measures how fast probability accumulates near x : $f_M(x) dx$ is the chance that M lands in the tiny interval $[x, x + dx]$. That chance is the CDF's increase across the interval,

$$f_M(x) dx \approx \mathbb{P}(x < M \leq x + dx) = F_M(x + dx) - F_M(x),$$

and dividing by dx is precisely the definition of a derivative. So the density is the derivative of the CDF:

$$f_M(x) = F'_M(x) = \frac{d}{dx} x^{10} = 10x^9, \quad 0 \leq x \leq 1.$$

(Equivalently $F_M(x) = \int_0^x f_M(t) dt$ — the CDF accumulates the density.)

The expectation as $\int x f_M(x) dx$. An average is a probability-weighted sum of values. Slice $[0, 1]$ into slivers $[x, x + dx]$; the outcome $M \approx x$ carries probability $f_M(x) dx$, so weighting each value x by its probability and summing (integrating) gives

$$\mathbb{E}[M] = \int_0^1 x f_M(x) dx = \int_0^1 x \cdot 10x^9 dx = 10 \int_0^1 x^{10} dx = \frac{10}{11} \approx 0.909.$$

$$\mathbb{E}[\text{max of 10 } U[0, 1]] = \frac{10}{11} \approx \$0.91.$$

The General Result (and the Beta Distribution)

The same idea gives the mean of *any* order statistic, not just the maximum. Sort the n draws as $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$, so $X_{(k)}$ is the k -th smallest.

Density of the k -th smallest. For $X_{(k)}$ to sit right at x , three things must line up among the n draws at once:

- exactly one draw lands in the sliver $[x, x + dx]$ — there are n ways to pick which one, each with probability $f(x) dx = dx$ (a uniform has density 1);

- exactly $k - 1$ of the remaining draws fall *below* x — probability x each;
- the other $n - k$ draws fall *above* x — probability $1 - x$ each.

The number of ways to split the draws into these three groups is the multinomial coefficient $\frac{n!}{(k-1)!1!(n-k)!}$, so the density is

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}, \quad 0 \leq x \leq 1.$$

What “Beta” means. Any density shaped like $x^{a-1}(1-x)^{b-1}$ on $[0, 1]$, scaled to integrate to 1, is called the **Beta(a, b) distribution**. The scaling constant is the *Beta function*

$$B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx = \frac{(a-1)!(b-1)!}{(a+b-1)!} \quad (\text{integer } a, b),$$

and the Beta(a, b) density $x^{a-1}(1-x)^{b-1}/B(a, b)$ has mean $\frac{a}{a+b}$. Matching exponents, $f_{(k)}$ is exactly Beta($k, n - k + 1$).

The mean. Integrate x against this density; the Beta function evaluates the integral:

$$\mathbb{E}[X_{(k)}] = \frac{n!}{(k-1)!(n-k)!} \int_0^1 x^k(1-x)^{n-k} dx = \frac{n!}{(k-1)!(n-k)!} \cdot \frac{k!(n-k)!}{(n+1)!} = \frac{k}{n+1}.$$

For n i.i.d. Uniform $[0, 1]$ draws, the k -th smallest is Beta($k, n - k + 1$) with mean

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}.$$

So the n order-statistic means are $\frac{1}{n+1}, \frac{2}{n+1}, \dots, \frac{n}{n+1}$ — equally spaced, cutting $[0, 1]$ into $n + 1$ equal expected gaps. In particular the maximum ($k = n$) is $\frac{n}{n+1}$ (here $\frac{10}{11}$, matching the direct computation) and the minimum ($k = 1$) is $\frac{1}{n+1}$.

Concept Summary and Applications

- **Method:** for a maximum, work with the CDF ($F_M = F^n$); for a minimum, the survival function ($1 - (1 - F)^n$); differentiate for the density, or integrate the tail $\mathbb{E}[M] = \int_0^\infty \mathbb{P}(M > x) dx$.
- **Uniform mnemonic:** picture dropping n random points on $[0, 1]$. On average they slice the interval into $n + 1$ gaps of *equal* expected length $\frac{1}{n+1}$, so the k -th smallest point sits at expected position $\frac{k}{n+1}$. This single picture recalls every order-statistic mean with no integration — in this problem the maximum is the 10-th of 10 points, at $\frac{10}{11}$, exactly our answer. (It is the same “ $n + 1$ equal expected pieces” idea as the broken-stick spacings in the previous section.)
- **Applications:** second-price / highest-bid auctions, extreme-value and tail-risk estimation, and record/max-drawdown statistics.

21 Discrete Order Statistics

Problem: What is the expected value of the **maximum** of two fair six-sided dice?

Use the CDF (Cumulative Distribution Function)

The **CDF** (*cumulative distribution function*) of the maximum is $\mathbb{P}(\max \leq k)$, the probability that the max does not exceed k . The maximum is $\leq k$ exactly when *both* dice are $\leq k$; each die is $\leq k$ with probability $k/6$, and the two are independent, so the probabilities multiply:

$$\mathbb{P}(\max \leq k) = \left(\frac{k}{6}\right)^2.$$

To get the probability of *exactly* k , subtract consecutive CDF values — the event “ $\max = k$ ” is “ $\max \leq k$ ” with “ $\max \leq k - 1$ ” removed:

$$\mathbb{P}(\max = k) = \mathbb{P}(\max \leq k) - \mathbb{P}(\max \leq k - 1) = \frac{k^2 - (k - 1)^2}{36} = \frac{2k - 1}{36}.$$

Then, averaging k over its probabilities,

$$\mathbb{E}[\max] = \sum_{k=1}^6 k \cdot \frac{2k - 1}{36} = \frac{1 + 6 + 15 + 28 + 45 + 66}{36} = \frac{161}{36} \approx 4.47.$$

$$\mathbb{E}[\max \text{ of two d6}] = \frac{161}{36} \approx 4.47.$$

By the mirror symmetry $k \mapsto 7 - k$, $\mathbb{E}[\min] = \frac{91}{36} \approx 2.53$, and as a check $\mathbb{E}[\min] + \mathbb{E}[\max] = \frac{252}{36} = 7 = 2 \cdot 3.5$ (the two always sum to the two dice).

Concept Summary and Applications

- **Method:** for discrete maxima use $\mathbb{P}(\max \leq k) = F(k)^n$; the tail sum $\mathbb{E}[X] = \sum_{k \geq 1} \mathbb{P}(X \geq k)$ is often fastest.
- **Symmetry:** for a distribution symmetric about its mean μ , $\mathbb{E}[\min] + \mathbb{E}[\max] = 2\mu$, and range $= \mathbb{E}[\max] - \mathbb{E}[\min]$.
- **Applications:** best-of / worst-of payoffs, range and spread statistics, and tournament or team-formation problems.

Additional Problem 1 — Expected Absolute Difference of Two d6

Problem: Roll two fair six-sided dice, D_1 and D_2 . What is the expected value of the absolute difference $|D_1 - D_2|$ between the two rolls?

Why the answer is $\mathbb{E}[\max] - \mathbb{E}[\min]$. For any two numbers the absolute difference equals the larger minus the smaller: $|a - b| = \max(a, b) - \min(a, b)$. Applying this to the two dice and taking expectations (expectation is linear, so it splits across the subtraction),

$$\mathbb{E}|D_1 - D_2| = \mathbb{E}[\max] - \mathbb{E}[\min] = \frac{161}{36} - \frac{91}{36} = \frac{70}{36} = \frac{35}{18} \approx 1.94.$$

This is just the *range* of the two dice — the average gap between the higher and lower roll.

Verification by direct counting. We can confirm $\frac{35}{18}$ without the max/min values, by counting how many of the 36 ordered pairs realise each difference d . For $d \geq 1$, $|D_1 - D_2| = d$ occurs in $2(6 - d)$ ways — $6 - d$ pairs with $D_1 - D_2 = d$ and $6 - d$ pairs with $D_2 - D_1 = d$ — while $d = 0$ contributes nothing to the sum. Hence

$$\mathbb{E}|D_1 - D_2| = \frac{1}{36} \sum_{d=1}^5 d \cdot 2(6 - d) = \frac{2}{36} \sum_{d=1}^5 d(6 - d) = \frac{2}{36} (6 \cdot 15 - 55) = \frac{2}{36} \cdot 35 = \frac{35}{18}. \checkmark$$

(using $\sum_{d=1}^5 d = 15$ and $\sum_{d=1}^5 d^2 = 55$), matching the range computation.

Additional Problem 2 — Absolute Difference of Two d30

Problem: Roll two fair 30-sided dice, each uniform on $\{1, 2, \dots, 30\}$ and independent. What is the expected value of the absolute difference between the two outcomes?

We solve it for a general n -sided die and then set $n = 30$.

Step 1 — count pairs with a given difference. Let X, Y be the two rolls; there are n^2 equally likely ordered pairs. For a difference $d \geq 1$, $|X - Y| = d$ occurs when $X - Y = d$ (then $Y \in \{1, \dots, n - d\}$, giving $n - d$ pairs) or $Y - X = d$ (another $n - d$ pairs), so $2(n - d)$ ordered pairs in total. Therefore

$$\mathbb{E}|X - Y| = \frac{1}{n^2} \sum_{d=1}^{n-1} d \cdot 2(n - d) = \frac{2}{n^2} \sum_{d=1}^{n-1} d(n - d).$$

Step 2 — evaluate the sum. Write $d(n - d) = nd - d^2$ and use the standard sums $\sum_{d=1}^{n-1} d = \frac{(n-1)n}{2}$ and $\sum_{d=1}^{n-1} d^2 = \frac{(n-1)n(2n-1)}{6}$:

$$\sum_{d=1}^{n-1} d(n - d) = n \cdot \frac{(n-1)n}{2} - \frac{(n-1)n(2n-1)}{6} = (n-1)n \left(\frac{n}{2} - \frac{2n-1}{6} \right).$$

Put the bracket over 6: $\frac{3n-(2n-1)}{6} = \frac{n+1}{6}$, so the sum is $\frac{(n-1)n(n+1)}{6} = \frac{n(n^2-1)}{6}$.

Step 3 — assemble.

$$\mathbb{E}|X - Y| = \frac{2}{n^2} \cdot \frac{n(n^2-1)}{6} = \frac{n^2-1}{3n}.$$

For two i.i.d. uniforms on $\{1, \dots, n\}$: $\mathbb{E}|X - Y| = \frac{n^2-1}{3n}.$

For $n = 30$: $\frac{30^2-1}{3 \cdot 30} = \frac{899}{90} \approx 9.99$. (The d6 value $\frac{35}{18}$ is the same formula at $n = 6$.)

$$\mathbb{E}|X - Y| = \frac{899}{90} \approx 9.99 \text{ for two 30-sided dice.}$$

Additional Problem 3 — Larger of A/B and B/A

Problem: You roll two fair six-sided dice, die A and die B (uniform on $\{1, \dots, 6\}$, independent). After seeing both, you may choose to be paid either A/B or B/A dollars, whichever is larger. How much should you be willing to pay to play?

Since you always take the larger ratio, the payoff is

$$X = \max\left(\frac{A}{B}, \frac{B}{A}\right) = \frac{\max(A, B)}{\min(A, B)} \geq 1,$$

and the fair price is the expected value $\mathbb{E}[X]$.

Step 1 — organise the 36 outcomes. Of the 36 equally likely ordered pairs (A, B) : the 6 ties $A = B$ give $X = 1$; and each unordered pair $\{a, b\}$ with $a < b$ appears *twice* (as (a, b) and (b, a)), each time giving payoff b/a . So

$$\mathbb{E}[X] = \frac{1}{36} \left(6 \cdot 1 + 2 \sum_{1 \leq a < b \leq 6} \frac{b}{a} \right).$$

Step 2 — the double sum, grouped by the smaller value a .

$$\begin{aligned} a = 1 : \frac{2}{1} + \frac{3}{1} + \frac{4}{1} + \frac{5}{1} + \frac{6}{1} &= 20, & a = 2 : \frac{3}{2} + \frac{4}{2} + \frac{5}{2} + \frac{6}{2} &= 9, \\ a = 3 : \frac{4}{3} + \frac{5}{3} + \frac{6}{3} &= 5, & a = 4 : \frac{5}{4} + \frac{6}{4} &= \frac{11}{4}, \\ a = 5 : \frac{6}{5} &= \frac{6}{5}. \end{aligned}$$

Adding over the common denominator 20: $20 + 9 + 5 + \frac{11}{4} + \frac{6}{5} = \frac{400+180+100+55+24}{20} = \frac{759}{20}$.

Step 3 — take the expectation.

$$\mathbb{E}[X] = \frac{1}{36} \left(6 + 2 \cdot \frac{759}{20} \right) = \frac{1}{36} \left(6 + \frac{759}{10} \right) = \frac{1}{36} \cdot \frac{819}{10} = \frac{819}{360} = \frac{91}{40} = 2.275.$$

$$\mathbb{E}\left[\max\left(\frac{A}{B}, \frac{B}{A}\right)\right] = \frac{91}{40} = 2.275, \text{ so you should pay up to about \$2.28 per play.}$$

22 Optimal Stopping and Threshold Rules (Backward Induction)

Problem: You roll a fair six-sided die.

- With no choices, what is the expected value of a single roll?
- Now you may, after seeing the first roll, either *keep* it or *reroll once* and take that second roll. Playing optimally, what is your expected value?
- Now you may roll up to *three* times total: after each of the first two rolls you may keep the current value or roll again. What is your expected value under the best strategy?
- What if you may roll *infinitely many* times (keep or roll again after every roll, at no cost)?

This is an **optimal stopping** problem: you see a random outcome and must decide whether to *stop* and take it, or *continue* in the hope of something better. The winning idea is to compare the value in your hand against the **continuation value** — the expected payoff of playing on optimally — and to work **backward** from the last decision, where there is no more choice to make.

Part (a) — a single roll

With no decision to make, the value is just the average face of a fair die:

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

Part (b) — one optional reroll

Suppose your first roll shows n . You have two options:

- keep** it: you walk away with n ;
- reroll**: you give up n and take a fresh roll, worth on average 3.5 (the part-(a) value).

A rational player keeps whichever is larger, so you keep when n beats the continuation value 3.5 and reroll otherwise:

$$\text{keep if } n \geq 4, \quad \text{reroll if } n \leq 3.$$

This is a **threshold rule**: accept anything at or above the cutoff 4, throw back anything below it. The resulting payoff is 3.5 for each of the three low first-rolls (you reroll) and 4, 5, 6 for the three high ones (you keep), so

$$\mathbb{E} = \frac{1}{6} \left(\underbrace{3.5 + 3.5 + 3.5}_{n=1,2,3: \text{ reroll}} + \underbrace{4 + 5 + 6}_{n=4,5,6: \text{ keep}} \right) = \frac{25.5}{6} = 4.25.$$

This is not the same as “take the larger of two rolls.” In Section 21 (Discrete Order Statistics) you roll twice and keep whichever is bigger, $\max(\text{roll}_1, \text{roll}_2)$, worth $\frac{161}{36} \approx 4.47$. Here you must decide to reroll before seeing the second roll, and once you reroll you are **stuck with** that second value even if it is worse than the first. Giving up the safety net of hindsight costs you: $4.25 < 4.47$. The max-of-two only applies when you may look at both outcomes and then choose.

Part (c) — up to three rolls (backward induction)

With more rolls allowed we cannot just eyeball the answer; we compute it **backward**, one stage at a time. Let V_k denote the expected value when you are about to roll and are allowed *up to k rolls in total* from here on, playing optimally. Each stage obeys the same rule — keep the current roll if it beats the value of continuing:

$$V_k = \mathbb{E}[\max(X, V_{k-1})].$$

Base case $V_1 = 3.5$: with only one roll left you must accept it, so the value is the plain average. **One step back**, V_2 : keep the roll if it beats $V_1 = 3.5$ (that is, $n \geq 4$), otherwise reroll into a position worth 3.5. This is exactly part (b), so $V_2 = 4.25$. **One more step back**, V_3 : keep the roll if it beats $V_2 = 4.25$ (that is, $n \geq 5$), otherwise reroll into a position worth 4.25:

$$V_3 = \frac{1}{6} \left(\underbrace{4.25 + 4.25 + 4.25 + 4.25}_{n=1,2,3,4: \text{ reroll}} + \underbrace{5 + 6}_{n=5,6: \text{ keep}} \right) = \frac{28}{6} = \frac{14}{3} \approx 4.667.$$

Notice the threshold *rose* from 4 to 5: with an extra roll in reserve you can afford to be pickier.

Part (d) — unlimited rolls

Now suppose there is **no limit** on the number of rolls (and rolling is free): after every roll you may keep the value or roll again.

Step 1 — keep extending the backward induction. The recursion $V_k = \mathbb{E}[\max(X, V_{k-1})]$ still applies for any finite limit k ; the threshold is always “keep iff the roll is at least V_{k-1} ”:

Rolls allowed k	Continuation value V_{k-1}	Keep if roll $\geq$	Value V_k
1	—	(must keep)	3.5
2	3.5	4	4.25
3	4.25	5	4.667
4	4.667	5	$\frac{89}{18} \approx 4.944$
5	4.944	5	$\frac{277}{54} \approx 5.130$
6	5.130	6	5.275
10	5.580	6	5.650

Once $V_{k-1} > 5$ (from $k = 6$ on) only a 6 is worth keeping, so

$$V_k = \frac{5}{6} V_{k-1} + \frac{1}{6} \cdot 6 \implies 6 - V_k = \frac{5}{6} (6 - V_{k-1}) \implies V_k = 6 - \frac{47}{54} \left(\frac{5}{6} \right)^{k-5} \quad (k \geq 5).$$

The gap to 6 shrinks by a factor $\frac{5}{6}$ with every extra roll, so $V_k \rightarrow 6$ as $k \rightarrow \infty$.

Step 2 — the unlimited game directly. With no limit, the continuation value no longer depends on how many rolls are left: it is the same number V every time. So V must satisfy the *fixed-point* (stationary) equation

$$V = \mathbb{E}[\max(X, V)].$$

Subtract V from both sides: $\mathbb{E}[\max(X, V)] - V = \mathbb{E}[(X - V)^+]$, the expected amount by which the roll beats V . If $V < 6$, a 6 beats it with probability $\frac{1}{6}$, so this is strictly positive and the equation fails. The only solution is $V = 6$ (check: $\mathbb{E}[\max(X, 6)] = 6$).

Step 3 — a strategy that achieves it. “Roll until you get a 6, then stop.” Each roll is a 6 with probability $\frac{1}{6}$, so the number of rolls is geometric with mean 6 and is finite with probability 1; you always end with a 6.

***Why this is (almost) trivial — and how interviews make it interesting.** Waiting costs nothing here, so you can hold out for the best face. Real optimal-stopping problems add a price for waiting: a fee per reroll, discounting, or a deadline. With a cost c per reroll the stationary equation becomes $V = \mathbb{E}[\max(X, V - c)]$ and the threshold drops (e.g. $c = \$1$: keep any roll ≥ 3 , $V = 4$); see Section 13 (Optimal Stopping with a Reroll Cost).*

No reroll: 3.5. One optional reroll: 4.25. Up to three rolls: $\frac{14}{3} \approx 4.67$.
 Unlimited free rolls: 6 (roll until a 6).

The Backward-Induction Procedure

The same recipe solves any finite-horizon optimal-stopping problem:

1. **Set up the stages and horizon.** Identify the states and the *last* moment a decision is made.
2. **Solve the final stage first.** At the end there is no choice left, so its value is just the expected payoff of whatever you are forced to take.
3. **Step backward.** At each earlier stage the value is $V_{\text{now}} = \mathbb{E}[\max(\text{take it now}, V_{\text{next}})]$, where V_{next} is the already-computed value of continuing.
4. **Read off the threshold.** The optimal action is a cutoff: *stop* exactly when the current outcome is at least the continuation value V_{next} , otherwise *continue*.
5. **Repeat to the first stage.** That first-stage value is the answer.

$V_k = \mathbb{E}[\max(X, V_{k-1})]$, with the base case $V_1 = \mathbb{E}[X]$; stop iff the current draw $\geq V_{k-1}$.

A useful sanity check: **more attempts can only help**, so V_k increases with k , and the acceptance threshold rises with it (more tries in hand $\Rightarrow$ hold out for more).

Concept Summary and Applications

- **Method:** work *backward* from the final decision; at each stage compare “take it now” against the continuation value V_{next} .
- **Threshold rule:** the optimum is always a cutoff — accept iff the current outcome is at least the continuation value.
- **Monotonicity:** more rounds remaining raise both the value and the cutoff; fewer rounds make you less choosy.
- **Applications:** hiring / secretary problems, sequential search with the option to keep a past

offer, “keep or redraw” games, and early exercise of American options.

Additional Problem 1 — Four Coins with a Full Retoss

Problem: You toss four fair coins and are paid \$1 for each head (so \$0 to \$4). (a) With no options, what is a fair price to play? (b) Now, after seeing the first toss, you may either keep that result or *retoss all four coins* once and take the new result. What is a fair price now?

Part (a). The payoff equals the number of heads H , and the number of heads in four fair tosses averages

$$\mathbb{E}[H] = 4 \cdot \frac{1}{2} = 2 \implies \text{fair price } \$2.$$

Part (b) — one backward-induction step. Let H_1 be the heads on the first toss. Keeping pays H_1 ; retossing throws it away for a fresh toss worth $\mathbb{E}[H_2] = 2$ on average. So the continuation value is 2, and the threshold rule says

$$\text{keep if } H_1 \geq 2, \quad \text{retoss if } H_1 \leq 1.$$

The number of heads in four tosses follows the counts $\binom{4}{k} = 1, 4, 6, 4, 1$ out of 16:

$$\mathbb{P}(H_1 = 0, 1, 2, 3, 4) = \frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}.$$

When $H_1 \in \{0, 1\}$ (probability $\frac{5}{16}$) you retoss and collect the continuation value 2; otherwise you keep H_1 . Hence

$$\mathbb{E}[\text{payoff}] = \underbrace{2 \cdot \frac{5}{16}}_{H_1 \leq 1: \text{ retoss}} + \underbrace{2 \cdot \frac{6}{16} + 3 \cdot \frac{4}{16} + 4 \cdot \frac{1}{16}}_{H_1 \geq 2: \text{ keep}} = \frac{10 + 12 + 12 + 4}{16} = \frac{38}{16} = \frac{19}{8}.$$

Without the option: \$2. With one full retoss (keep if ≥ 2 heads): $\frac{19}{8} = \$2.375$.

Additional Problem 2 — Four Coins, Unlimited Reflips at \$1 Each (Q418)

Problem: You flip four fair coins and get \$1 for each head. You may reflip *all four* coins **as many times as you want**, but each reflip costs \$1. What is the optimal strategy, and what is the game worth?

Step 1 — the distribution. As in Additional Problem 1, the number of heads H takes the values 0, 1, 2, 3, 4 with probabilities $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$, and $\mathbb{E}[H] = 2$.

Step 2 — the common mistake. It is tempting to reason: “a reflip costs \$1 and a fresh flip is worth $\mathbb{E}[H] = 2$ on average, so reflipping is worth $2 - 1 = 1$; keep anything ≥ 1 and reflip only on 0.” This is **wrong**, because after a reflip you are *not* stuck with the new result — you may reflip again. The value of reflipping is not $\mathbb{E}[H] - 1$ but the value of *the whole game* minus \$1.

Step 3 — the stationary (fixed-point) equation. Let V be the expected payoff of the game (net of all future fees) just before a flip, under optimal play. Because reflips are unlimited, the situation after any reflip is identical to the start, so the value is the *same* V every time — exactly the idea of part (d) of the main problem. After seeing H :

- **keep:** receive H ;
- **reflip:** pay \$1 and face the same game again, worth $V - 1$.

Hence

$$V = \mathbb{E}[\max(H, V - 1)], \quad \text{keep iff } H \geq V - 1.$$

Step 4 — test each threshold for consistency. Since H is an integer, the rule is “keep iff $H \geq t$ ” for some t . For each t , solve for V , then check that the implied cutoff $V - 1$ gives back the same t .

Rule	Equation for V	V and $V - 1$	Consistent?
keep ≥ 1 (reflip 0)	$V = \frac{32}{16} + \frac{1}{16}(V - 1)$	$V = \frac{31}{15} \approx 2.07, V - 1 \approx 1.07$	no: $1 < 1.07$
keep ≥ 2 (reflip 0, 1)	$V = \frac{28}{16} + \frac{5}{16}(V - 1)$	$V = \frac{23}{11} \approx 2.09, V - 1 \approx 1.09$	yes
keep ≥ 3 (reflip 0, 1, 2)	$V = \frac{16}{16} + \frac{11}{16}(V - 1)$	$V = 1, V - 1 = 0$	no: keep all

For the consistent rule, the numerator $28 = 2 \cdot 6 + 3 \cdot 4 + 4 \cdot 1$ collects the kept outcomes and $\frac{5}{16} = \frac{1+4}{16}$ is the chance of reflipping. Multiplying by 16: $16V = 28 + 5V - 5$, so $11V = 23$ and $V = \frac{23}{11}$. The cutoff $V - 1 = \frac{12}{11}$ lies between 1 and 2, so indeed a 1 is reflipped and a 2 is kept.

Step 5 — sanity check by counting reflips. Each flip is reflipped with probability $\frac{5}{16}$, so the number of reflips is geometric with mean $\frac{5/16}{11/16} = \frac{5}{11}$. When you stop, $H \geq 2$, with $\mathbb{E}[H \mid H \geq 2] = \frac{28/16}{11/16} = \frac{28}{11}$. Net value = $\frac{28}{11} - \frac{5}{11} \cdot \$1 = \frac{23}{11}$. ✓

Reflip on 0 or 1 head; keep 2, 3 or 4. Value $V = \frac{23}{11} \approx \2.09 (the unlimited-reflip option is worth $\frac{1}{11} \approx \$0.09$ over playing once).

Takeaway. With repeated options, compare the current outcome with the *continuation value of the whole game* ($V - \text{cost}$), not with the one-shot average ($\mathbb{E}[H] - \text{cost}$). Here the continuation value $\frac{12}{11}$ is slightly above 1, which tips $H = 1$ into “reflip.” The same fixed-point method handles any per-attempt cost; see also Section 13 (Optimal Stopping with a Reroll Cost).

23 Backward Induction as Dynamic Programming: Multi-Path

Section 22 solved stopping problems by working *backward* along a simple line of stages. This section pushes the same tool one step further, to problems where the states branch and rejoin — the natural home of **dynamic programming**.

Backward Induction and Dynamic Programming Are One Idea

Backward induction and dynamic programming (DP) are two names for the same method. Both rest on four pieces:

1. a **state** that captures everything you need to know to act (here, the running total of points);
2. a **value function** $f(\text{state})$ — the best achievable value from that state onward;
3. a **recursion** (the *Bellman equation*) expressing f at a state in terms of the states you can move to, taking a max over your available actions;
4. **boundary values** at the terminal states, from which everything else is built.

You solve the terminal states first and work backward, so every subproblem is ready before the states that depend on it.

The only difference between the two names is the *shape* of the state space. In Section 22 the states formed a single chain (rolls remaining: $3 \rightarrow 2 \rightarrow 1$), one path from start to end — “backward induction” pictures marching straight back along that line. Here the states form a **branching graph**: from a total s you can move to $s+2$ or $s+3$, and the *same* state can be reached by several different paths (a total of 5 arises both as $2+3$ and as $3+2$). “Dynamic programming” stresses computing each state’s value *once* and reusing it for every path that passes through, instead of recomputing — this reuse (“memoization”) is what makes branching problems tractable.

Representative Problem — The Dartboard

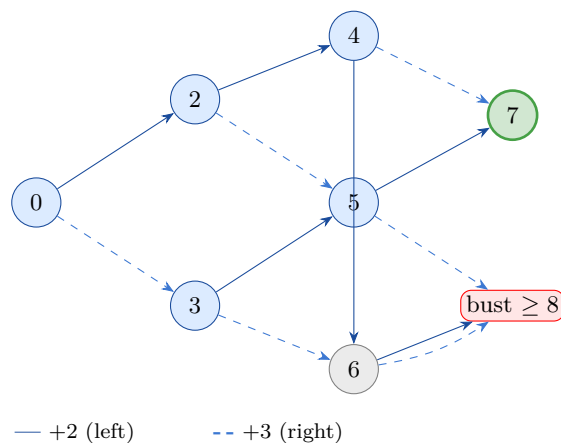
Problem: A dartboard is split in half: the left half is worth 2 points, the right half 3 points. On any throw you hit the board with probability 80%; when you hit, you land on the side you aimed for with probability 70% (and the other side with probability 30%). With unlimited throws, and choosing which side to aim for before each throw, what is the largest probability that your running total is *exactly* 7?

Three facts to see first.

- **7 has only one recipe.** The only way to build 7 from 2s and 3s is $2+2+3$ (since $2x+3y=7$ forces $x=2, y=1$): two left hits and one right hit, in some order.
- **The 80% hit rate is a red herring.** A miss scores 0; with unlimited throws you simply throw again, so misses only delay you, never change whether 7 is reachable. Conditioning on an eventual hit, each hit lands on the aimed side with probability 0.7 and the other side with 0.3.
- **Overshooting is the real enemy.** The total only ever rises in steps of +2 or +3, so if you ever jump *past* 7 you have lost it forever (e.g. $5 \xrightarrow{+3} 8$, or landing on 6, from which any hit gives 8 or 9).

Why the answer is not 1. It is tempting to say “aim left until a 2 comes, again until a second 2, then aim right for the 3 — each eventually happens.” This is wrong: while you wait for a 2, every throw *also* has a 0.3 chance of a stray 3, which can shove your total past 7. You cannot wait for a specific value without risking an overshoot, so the probability is strictly below 1 and depends on aiming cleverly.

The state graph (why it is “multi-path”). Track only the running total. Each total leads to two others (+2 or +3), and several totals are reachable by more than one route — e.g. 5 from both 2 and 3:



Because total 5 (and 6, and the goal 7) is reached by two different paths, this is a graph, not a chain — DP computes each $f(s)$ once and reuses it.

The Bellman recursion. Let $f(s)$ be the *maximum* probability of finishing on exactly 7 starting from total s . On each hit you choose a side: aiming *left* gives +2 with probability 0.7 and +3 with 0.3; aiming *right* gives +3 with 0.7 and +2 with 0.3. Taking the better aim,

$$f(s) = \max \left\{ \underbrace{0.7 f(s+2) + 0.3 f(s+3)}_{\text{aim left (for 2)}}, \underbrace{0.3 f(s+2) + 0.7 f(s+3)}_{\text{aim right (for 3)}} \right\},$$

with boundary $f(7) = 1$ and $f(s) = 0$ for $s \geq 8$ (overshot). Now work *downward* from the boundary.

$f(6)$: both $6+2=8$ and $6+3=9$ are busts, so $f(6) = 0$.

$f(5)$: $5+2=7$ ($f=1$), $5+3=8$ ($f=0$). Aim left = $0.7(1) + 0.3(0) = 0.7$; aim right = $0.3(1) = 0.3$. So $f(5) = 0.7$ (aim for 2).

$f(4)$: $4+2=6$ (0), $4+3=7$ (1). Aim left = $0.3(1) = 0.3$; aim right = $0.7(1) = 0.7$. So $f(4) = 0.7$ (aim for 3).

$f(3)$: $3+2=5$ (0.7), $3+3=6$ (0). Aim left = $0.7(0.7) = 0.49$; aim right = $0.3(0.7) = 0.21$. So $f(3) = 0.49$ (aim for 2).

$f(2)$: $2+2=4$ (0.7), $2+3=5$ (0.7). Aim left = $0.7(0.7) + 0.3(0.7) = 0.70$; aim right = 0.70 too. So $f(2) = 0.70$ (either).

$f(0)$: $0+2=2$ (0.70), $0+3=3$ (0.49).

aim left = $0.7(0.70) + 0.3(0.49) = 0.490 + 0.147 = 0.637$, aim right = $0.3(0.70) + 0.7(0.49) = 0.553$.

So $f(0) = \max(0.637, 0.553) = 0.637$ (aim for 2). (The total $s = 1$ never occurs, since every step from 0 is +2 or +3.)

Optimal policy and answer.

s	0	2	3	4	5	6
$f(s)$	0.637	0.70	0.49	0.70	0.70	0
aim for	2	either	2	3	2	—

$$P(\text{score exactly } 7) = \frac{637}{1000} = 0.637, \text{ aiming for 2 at totals 0, 3, 5 and for 3 at total 4.}$$

Strategy matters. Adapting the aim to the current total is the whole game. Naively *always* aiming for 2s scores exactly 7 with probability only $\binom{3}{2}(0.7)^2(0.3) = 0.441$, and always aiming for 3s gives about 0.189; steering toward 7 lifts this to 0.637 (a direct simulation confirms ≈ 0.637).

Concept Summary and Applications

- **Method:** define a state, a value function, and a Bellman recursion with a max over actions; fix the terminal values and evaluate backward. When states branch and rejoin, store each value once and reuse it (DP / memoization).
- **Policy:** the optimal action at each state falls out of which term achieves the max — here, which side to aim for at each total.
- **Watch the boundary:** terminal “win” states get value 1 and overshoot / “bust” states get value 0; getting these right is half the problem.
- **Applications:** shortest-path and reachability on graphs, optimal play in dice/board games, inventory and resource control, sequence alignment, and any “best expected outcome over a sequence of choices” problem.

24 Renewal Theory and the Inspection Paradox

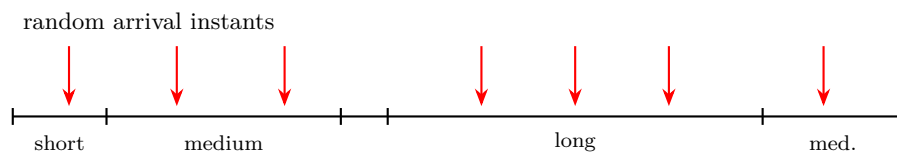
What Renewal Theory Is (in Plain Words)

Renewal theory studies things that happen over and over, separated by *random* waits. Buses arriving at a stop, light bulbs burning out and being replaced, machines failing, customers walking into a shop, earthquakes on a fault line — each is a stream of events with a random gap between one and the next. Call those gaps $L_1, L_2, L_3, \dots$ and assume they are independent and share the same distribution, with average length $\mathbb{E}[L]$. The word “renewal” captures the key idea: each time an event happens, the process *starts over* statistically — it forgets the past, and the waiting clock resets. Renewal theory answers questions like “how many events by time t ?” and, the one we care about, “if I show up at a random moment, how long until the next event?”

The Inspection Paradox (in Plain Words)

Here is the surprise. Suppose buses come *on average* every 10 minutes, and you arrive at the stop at a random time. How long do you wait? The tempting answer is “half the average gap,” i.e. 5 minutes. **But you typically wait longer than that.** This is the **inspection paradox**.

Why? Because a random arrival instant is *not* equally likely to fall in every gap — it is more likely to fall inside a *long* gap than a short one, simply because a long gap covers more of the timeline. Imagine throwing a dart at the time axis, which is chopped into intervals of different widths: the dart is more likely to land in a wide interval than a narrow one. So the gap you happen to land in is *not* a typical gap; it is biased toward being long. This is called **length-biasing** (or size-biased sampling).



Seven random arrivals dropped on a timeline of unequal gaps: the long gap catches three of them, the tiny gap none. The gap you experience tends to be longer than a typical gap — that is the paradox.

Everyday versions of the same effect. Ask students “how big is your class?” and the average answer exceeds the true average class size, because more students sit in the big classes. Your friends have, on average, more friends than you do (the “friendship paradox”), because popular people appear on many friend lists. Each is length-biasing in disguise: sampling *by a random member* over-weights the large groups.

Turning the Intuition into a Formula

Your waiting time is the *residual life*: the time from your random arrival to the next event. Two effects combine.

(i) *You land in long gaps more often.* The gap you fall into has a **length-biased** distribution — a gap of length ℓ is ℓ times more likely to be the one you hit — with density $\tilde{f}(\ell) = \ell f(\ell)/\mathbb{E}[L]$ and mean

$$\mathbb{E}[\tilde{L}] = \frac{\mathbb{E}[L^2]}{\mathbb{E}[L]} \geq \mathbb{E}[L].$$

(ii) *Within that gap you sit at a uniform spot.* Given the gap you are in has length ℓ , your arrival is uniformly placed inside it, so your wait averages $\ell/2$. Combining,

$$\mathbb{E}[W] = \frac{1}{2} \mathbb{E}[\tilde{L}] = \frac{\mathbb{E}[L^2]}{2 \mathbb{E}[L]}.$$

Splitting $\mathbb{E}[L^2] = \text{Var}(L) + \mathbb{E}[L]^2$ exposes exactly where the paradox lives:

$$\mathbb{E}[W] = \underbrace{\frac{\mathbb{E}[L]}{2}}_{\text{naive guess}} + \underbrace{\frac{\text{Var}(L)}{2 \mathbb{E}[L]}}_{\text{paradox penalty}} = \frac{\mathbb{E}[L]}{2} (1 + \text{CV}^2), \quad \text{CV}^2 = \frac{\text{Var}(L)}{\mathbb{E}[L]^2}.$$

Here CV is the **coefficient of variation** — the standard deviation divided by the mean, $\text{CV} = \sqrt{\text{Var}(L)}/\mathbb{E}[L]$ — a unitless measure of how spread out the gaps are relative to their average. The extra wait beyond the naive $\mathbb{E}[L]/2$ comes *entirely* from the variability of the gaps. Two extremes: **constant** gaps ($\text{CV} = 0$) give $\mathbb{E}[W] = \mathbb{E}[L]/2$ — no paradox; **exponential** gaps (a Poisson process, $\text{CV} = 1$) give $\mathbb{E}[W] = \mathbb{E}[L]$ — by memorylessness you wait a *whole* mean gap, no matter how long you have already stood there.

Representative Problem — The Bus and the Ice-Cream Stop

Problem: A bus runs a loop that takes exactly 10 minutes, restarting as soon as it finishes. (a) If you arrive at a uniformly random time, what is your expected wait? (b) Now suppose there is an ice-cream shop on the route; each time the bus passes it, the driver flips a fair coin and, on heads, stops for 10 minutes to eat ice cream before continuing. What is your expected wait now? (c) How does the answer change if the coin is biased?

Part (a) — a clockwork 10-minute loop. The bus visits your stop once every 10 minutes exactly. Arriving uniformly at random, the time *since* the last bus is uniform on $[0, 10]$, so the time *until* the next is also uniform on $[0, 10]$:

$$\mathbb{E}[W] = \frac{10}{2} = 5 \text{ minutes.}$$

Here every gap is the same length, so there is no length-biasing — the naive answer is correct.

Part (b) — the ice-cream stop makes gaps random. Between two successive visits to your stop, the bus passes the shop exactly once, so there is one coin flip per cycle. With stop-probability p (fair coin: $p = \frac{1}{2}$), the gap between visits is

$$L = \begin{cases} 10, & \text{probability } 1 - p, \\ 20, & \text{probability } p, \end{cases} \quad \mathbb{E}[L] = 10(1 - p) + 20p = 10(1 + p).$$

The tempting answer $\mathbb{E}[L]/2$ is **wrong** — this is exactly the inspection paradox. Use the residual-life formula. First the second moment:

$$\mathbb{E}[L^2] = 10^2(1-p) + 20^2p = 100 + 300p,$$

so

$$\mathbb{E}[W] = \frac{\mathbb{E}[L^2]}{2\mathbb{E}[L]} = \frac{100 + 300p}{2 \cdot 10(1+p)} = 5 \frac{1+3p}{1+p} \text{ minutes.}$$

For the fair coin $p = \frac{1}{2}$:

$$\mathbb{E}[W] = 5 \cdot \frac{1+3/2}{1+1/2} = 5 \cdot \frac{5/2}{3/2} = \frac{25}{3} \approx 8.33 \text{ minutes.}$$

Sanity check via the variance form. Here $\mathbb{E}[L] = 15$ and $\text{Var}(L) = \mathbb{E}[L^2] - \mathbb{E}[L]^2 = 250 - 225 = 25$, so $\text{CV}^2 = \frac{25}{225} = \frac{1}{9}$ and $\mathbb{E}[W] = \frac{15}{2}(1 + \frac{1}{9}) = \frac{25}{3}$. The 0.83 minutes above the naive 7.5 is precisely the paradox penalty.

Part (c) — biased coin. The general answer $\mathbb{E}[W] = 5 \frac{1+3p}{1+p}$ rises smoothly from 5 to 10 minutes as p goes from 0 to 1:

- $p = 0$ (never stops): every gap is 10, so $\mathbb{E}[W] = 5$ min. ✓
- $p = 1$ (always stops): every gap is 20, so $\mathbb{E}[W] = 10$ min. ✓

Both endpoints have *constant* gaps, so the paradox penalty vanishes there; it is largest for intermediate p , where the gap length is most uncertain.

Plain loop: $\mathbb{E}[W] = 5$ min. **Fair-coin stop:** $\mathbb{E}[W] = \frac{25}{3} \approx 8.33$ min. **Biased coin (stop prob. p):** $\mathbb{E}[W] = 5 \frac{1+3p}{1+p}$ min.

Concept Summary and Applications

- **Key formula:** a random observer's wait to the next event is $\mathbb{E}[W] = \frac{\mathbb{E}[L^2]}{2\mathbb{E}[L]} = \frac{\mathbb{E}[L]}{2}(1 + \text{CV}^2)$ — always at least the naive $\mathbb{E}[L]/2$, with the excess set by the gaps' variability.
- **Why:** sampling by a random time-point (or a random member) over-weights long gaps (large groups); this length-biasing raises the mean of the sampled gap to $\mathbb{E}[L^2]/\mathbb{E}[L]$.
- **Two anchors:** deterministic gaps $\Rightarrow$ no paradox; a Poisson process $\Rightarrow$ you wait a full mean gap ($\mathbb{E}[W] = \mathbb{E}[L]$).
- **Applications:** bus/subway waiting times, time-to-failure and maintenance planning, queueing and server load, network sampling (friendship paradox), and any “sampled-by-size” bias in data.

25 Sicherman Dice and Generating Functions

How Generating Functions Encode Dice

A **generating function** is a way to pack a whole list of counts into a single polynomial, so that ordinary algebra on the polynomial mirrors probability on the dice. The trick is to turn “face values” into *exponents* and “how many faces” into *coefficients*.

A single die. Encode a die by a polynomial in a placeholder variable x : each face showing value a contributes a term x^a . A standard six-sided die, with faces 1, 2, 3, 4, 5, 6, becomes

$$D(x) = x^1 + x^2 + x^3 + x^4 + x^5 + x^6.$$

More generally, if a die has c_a faces showing value a , its generating function is $\sum_a c_a x^a$: **the exponent records a face value, and the coefficient records how many faces carry that value**. Two handy readings: evaluating at $x = 1$ counts the faces, $D(1) = 6$ (divide by this to turn counts into probabilities); and having only *positive* labels means there is no x^0 term.

The sum of several dice — multiply the polynomials. This is the whole point. Roll two dice with generating functions $A(x) = \sum_a p_a x^a$ and $B(x) = \sum_b q_b x^b$ and multiply:

$$A(x) B(x) = \sum_a \sum_b p_a q_b x^{a+b} \implies [x^s] A(x) B(x) = \sum_{a+b=s} p_a q_b.$$

The coefficient of x^s in the product is exactly the number of ordered face pairs (a, b) that sum to s . In other words, **multiplying generating functions performs the “convolution” that produces the distribution of a sum** — adding independent dice corresponds to multiplying their polynomials. For two standard dice,

$$D(x)^2 = x^2 + 2x^3 + 3x^4 + 4x^5 + 5x^6 + 6x^7 + 5x^8 + 4x^9 + 3x^{10} + 2x^{11} + x^{12},$$

and the coefficients 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 are precisely the familiar frequencies of the sums 2 through 12 (divide by $D(1)^2 = 36$ for probabilities). More dice simply multiply in more factors. Crucially, *the sum distribution depends only on this product, not on how its factors are divided between the two dice* — which is exactly the door the Sicherman puzzle walks through.

Representative Problem — Sicherman Dice (Q220)

Problem: You have two ordinary six-sided dice, whose sums 2 through 12 occur with the usual frequencies. Erase all the faces and rewrite them with a new set of *positive integers*. Can you choose numbers so that the distribution of the sum of the two dice is unchanged? If so, how?

Answer. Yes — and there is exactly one nonstandard way, the **Sicherman dice**

$$\{1, 2, 2, 3, 3, 4\} \quad \text{and} \quad \{1, 3, 4, 5, 6, 8\}.$$

The generating-function machinery both finds them and proves they are unique.

Step 1 — reduce the goal to a polynomial equation. Relabelling the two dice means choosing new die polynomials $Q_1(x), Q_2(x)$. Since the sum distribution is the coefficient list of the product, “same sum distribution as two standard dice” means precisely

$$Q_1(x) Q_2(x) = D(x)^2.$$

A polynomial Q_i is a *legal die* exactly when it has non-negative integer coefficients, no constant term (positive labels), and $Q_i(1) = 6$ (six faces). So the whole problem is: split the factors of $D(x)^2$ into two legal dice.

Step 2 — factor the standard die. Using $1 + x + \cdots + x^5 = \frac{x^6 - 1}{x - 1}$ and $x^6 - 1 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$,

$$D(x) = x(1 + x + x^2 + x^3 + x^4 + x^5) = x(1 + x)(1 + x + x^2)(1 - x + x^2).$$

Squaring, two standard dice carry *two copies each* of the four irreducible factors:

$$D(x)^2 = [x(1 + x)(1 + x + x^2)(1 - x + x^2)]^2.$$

Because integer polynomials factor *uniquely*, relabelling can only *redistribute* these eight factors between Q_1 and Q_2 — which makes the search finite.

Step 3 — use the constraints to pin the split. Evaluate each irreducible at $x = 1$ to get its “face-weight”:

$$x \mapsto 1, \quad (1 + x) \mapsto 2, \quad (1 + x + x^2) \mapsto 3, \quad (1 - x + x^2) \mapsto 1.$$

- **Six faces each** needs $Q_i(1) = 6 = 2 \cdot 3$, so each die must get exactly one $(1 + x)$ and one $(1 + x + x^2)$. (The $(1 - x + x^2)$ factors have weight 1 and do not change the face count, so they may go anywhere.)
- **Positive labels** need a zero constant term, so each die must get one of the two x factors.

The *only* remaining freedom is where the two $(1 - x + x^2)$ factors land:

- **one to each die** $\Rightarrow$ both dice equal $x(1 + x)(1 + x + x^2)(1 - x + x^2) = D(x)$ — the ordinary dice;
- **both to one die** $\Rightarrow$ the Sicherman dice.

The positive-label rule is what makes the answer unique. If a face value of 0 (or negative labels) were allowed, the “each die needs an x ” constraint disappears, and uniqueness collapses. For instance, moving one x factor from one die to the other gives $\{0, 1, 2, 3, 4, 5\}$ and $\{2, 3, 4, 5, 6, 7\}$ — shifting one die down by 1 and the other up by 1 leaves every sum unchanged. More generally you may shift the two dice by $\pm k$ for any k , and shift the Sicherman pair too (e.g. $\{0, 1, 1, 2, 2, 3\}$ and $\{2, 4, 5, 6, 7, 9\}$), so infinitely many relabellings work once 0 is permitted. Insisting on positive labels forces each die to be divisible by x , which pins the split and singles out the Sicherman dice.

Step 4 — read off the Sicherman pair. The die that gets *neither* extra factor is

$$Q_1(x) = x(1 + x)(1 + x + x^2) = x + 2x^2 + 2x^3 + x^4 \Rightarrow \{1, 2, 2, 3, 3, 4\},$$

(coefficients 1, 2, 2, 1 on x^1, x^2, x^3, x^4 : one 1, two 2s, two 3s, one 4), and the die that gets *both* is

$$Q_2(x) = x(1 + x)(1 + x + x^2)(1 - x + x^2)^2 = x + x^3 + x^4 + x^5 + x^6 + x^8 \Rightarrow \{1, 3, 4, 5, 6, 8\}.$$

By construction $Q_1(x)Q_2(x) = D(x)^2$, so this pair yields the identical sum distribution — sums $2, \dots, 12$ with counts $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$.

Step 5 — uniqueness. The constraints forced one x , one $(1 + x)$, and one $(1 + x + x^2)$ onto each die, leaving only the placement of the two $(1 - x + x^2)$ factors; “one each” just rebuilds the standard dice, so “both together” is the *only* nonstandard option. Hence the Sicherman pair is the unique positive-integer relabelling of two d6 preserving the sum distribution.

Yes: the unique nonstandard relabelling is $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$ (the Sicherman dice), found by splitting the factors of $D(x)^2 = [x(1 + x)(1 + x + x^2)(1 - x + x^2)]^2$ into two legal dice.

Concept Summary and Applications

- **Encoding:** a die is a polynomial $\sum_a c_a x^a$ — exponent = face value, coefficient = number of such faces; $D(1)$ = number of faces.
- **Sums multiply:** the generating function of a sum of independent dice is the *product* of their polynomials; the coefficient of x^s counts the ways to reach total s .
- **Relabelling = refactoring:** because the sum only sees the product, redistributing the (uniquely determined) irreducible factors gives every equivalent labelling — the Sicherman dice are one such split.
- **Applications:** counting problems via “coefficient extraction,” convolutions of distributions, coin/partition problems, and any question of the form “in how many ways can these pieces sum to a target?”

26 The Frobenius (Chicken McNugget) Formula

What the Frobenius Formula Says

Suppose you have coins (or stamps, or McNugget boxes) in a few fixed positive denominations and an unlimited supply of each. Using non-negative whole numbers of each coin, which total amounts can you make? If the denominations share no common factor, then *every* large enough amount is makeable, but some small amounts are not. The **Frobenius number** is the **largest amount that cannot be made**. (It is also called the **Chicken McNugget** problem: with boxes of 6, 9, and 20 nuggets, the largest un-orderable number is 43.)

For **two** coprime denominations m and n (i.e. $\gcd(m, n) = 1$), the largest integer that *cannot* be written as $mx + ny$ with $x, y \geq 0$ is

$$g(m, n) = mn - m - n,$$

and the number of non-makeable amounts is $\frac{(m-1)(n-1)}{2}$.

Two things are worth stressing. First, the **coprimality condition is essential**: if $\gcd > 1$, only multiples of that common factor are ever reachable, so there is *no* largest un-makeable number (infinitely many amounts fail). Second, for **three or more** denominations there is no simple closed formula in general — but a denomination that is itself makeable from the others is *redundant* and can be dropped, sometimes collapsing the problem back to the two-coin formula.

Representative Problem — Coins of 3, 8, and 20

Problem: With an unlimited supply of coins worth 3, 8, and 20, what is the largest positive integer that cannot be formed as a non-negative combination $3a + 8b + 20c$?

Step 1 — a largest un-makeable number exists. A finite Frobenius number exists exactly when the denominations are collectively coprime. Here $\gcd(3, 8, 20) = 1$, so all sufficiently large integers are makeable and there is a well-defined largest exception to find.

Step 2 — drop the redundant 20. Notice 20 is itself makeable from the others:

$$20 = 4 \cdot 3 + 1 \cdot 8.$$

So any total that uses a 20-coin could instead use four 3s and one 8 — the 20 adds nothing new. The amounts makeable from $\{3, 8, 20\}$ are exactly those makeable from $\{3, 8\}$, and the problem reduces to the two-coin case $3a + 8b$.

Step 3 — the two-coin formula, and why it holds. For coprime m, n the answer is $g(m, n) = mn - m - n$. Here is the reasoning, which also shows *why* coprimality is needed. Sort every non-negative integer by its remainder modulo m , and study each remainder class separately.

1. **An m -coin keeps you in the same class.** Adding m does not change a number's remainder mod m . So m -coins let you climb *within* a remainder class in steps of m , but never switch classes.

2. **Each class has a single threshold.** Within one class, once some value T is makeable, so are $T + m, T + 2m, \dots$ (just add m -coins), while everything in the class below T is not. The largest un-makeable number in that class is therefore $T - m$.
3. **Only the n -coins choose the class.** Since m -coins do not affect the remainder, the class of $mx + ny$ is decided by ny . To reach a class you must use just enough n -coins to land on its remainder; the cheapest way uses the smallest count k with $kn \equiv r \pmod{m}$, costing kn .
4. **Coprimality reaches every class exactly once.** Because $\gcd(m, n) = 1$, the multiples $0, n, 2n, \dots, (m-1)n$ hit all m remainders, each exactly once — so every class gets a finite threshold. (If $\gcd > 1$ this fails: some classes are never reached, and no finite Frobenius number exists.)
5. **Worst class wins.** The class reached with k n -coins has threshold kn , hence largest gap $kn - m$. The biggest gap over all classes is at the largest $k = m - 1$:

$$g(m, n) = (m - 1)n - m = mn - m - n.$$

For $m = 3, n = 8$, list the multiples of 8 and their remainders mod 3:

k	$8k$	$8k \bmod 3$	largest unreachable = $8k - 3$
0	0	0	—
1	8	2	5
2	16	1	13

The class $\equiv 1 \pmod{3}$ is the bottleneck — it needs two 8-coins (16) before it is reachable — so $16 - 3 = 13$ is the largest unreachable value:

$$g(3, 8) = 3 \cdot 8 - 3 - 8 = 24 - 11 = 13.$$

Step 4 — verify. 13 is not $3a + 8b$: with $b = 0$, 13 is not a multiple of 3; with $b = 1$, $13 - 8 = 5$ is not a multiple of 3; with $b \geq 2$, $8b > 13$. And every integer from 14 up is makeable:

$$14 = 8 + 3 + 3, \quad 15 = 3 \cdot 5, \quad 16 = 8 + 8, \quad 17 = 8 + 3 + 3 + 3,$$

after which adding 3s covers all larger integers. The complete list of un-makeable amounts is 1, 2, 4, 5, 7, 10, 13 (seven of them, matching $\frac{(3-1)(8-1)}{2} = 7$).

The largest integer not formable from 3, 8, 20 is 13 (the 20 is redundant since $20 = 4 \cdot 3 + 8$, and $g(3, 8) = 3 \cdot 8 - 3 - 8 = 13$).

Additional Problem — The Chicken McNugget Numbers (6, 9, 20)

Problem: Nuggets come in boxes of 6, 9, and 20. What is the largest number of nuggets you *cannot* order exactly?

Here no coin is redundant (none of 6, 9, 20 is makeable from the other two) and there is no closed formula, so we exploit the structure. The key observation: **6 and 9 are both multiples of 3**, so only the 20-boxes can change the total's remainder modulo 3. Handle each remainder class separately.

What 6 and 9 alone make. Since $6a + 9b = 3(2a + 3b)$ and $2a + 3b$ takes every value ≥ 2 (plus 0, missing only 1), the reachable totals are

$$\{0, 6, 9, 12, 15, 18, \dots\} = 0 \text{ and every multiple of 3 that is } \geq 6.$$

So the *only* multiple of 3 you cannot make from 6s and 9s is 3 itself.

Split by remainder mod 3 (note $20 \equiv 2$, $40 \equiv 1$, $60 \equiv 0 \pmod{3}$):

- **Totals $\equiv 0 \pmod{3}$:** use no 20-boxes; reachable set is $\{0, 6, 9, 12, \dots\}$, so the largest unreachable value is just **3**.
- **Totals $\equiv 2 \pmod{3}$:** you need at least one 20 (as $20 \equiv 2$). Reachable = $20 + \{0, 6, 9, \dots\} = \{20, 26, 29, 32, \dots\}$. The value $20 + 3 = 23$ falls in the “missing 3” gap and is unreachable, while $20, 26, 29, \dots$ are fine — largest unreachable = **23**.
- **Totals $\equiv 1 \pmod{3}$:** you need at least *two* 20s (since $40 \equiv 1$; one 20 gives the wrong remainder). Reachable = $40 + \{0, 6, 9, \dots\} = \{40, 46, 49, \dots\}$, and $40 + 3 = 43$ is the unreachable one — largest unreachable = **43**.

Taking the worst class, the Frobenius number is $\max(3, 23, 43) = 43$. Every total from 44 up is orderable ($44 = 6 + 9 + 9 + 20$, $45 = 9 \cdot 5$, $46 = 6 + 20 + 20$, $47 = 9 + 9 + 9 + 20$, $48 = 6 \cdot 8$, then add 6s).

Cross-check (Apéry-set method). For any set of denominations, find the smallest orderable amount in each residue class modulo one coin, then subtract that coin. Modulo 6, the smallest orderable amounts are

$$r = 0: 0, \quad r = 1: 49, \quad r = 2: 20, \quad r = 3: 9, \quad r = 4: 40, \quad r = 5: 29,$$

and the Frobenius number is the largest of these minus 6: $49 - 6 = 43$. ✓

The largest un-orderable number of nuggets from boxes of 6, 9, 20 is 43 (the $\equiv 1 \pmod{3}$ class needs two 20-boxes, and $43 = 40 + 3$ lands on the one gap 6s and 9s cannot fill).

Concept Summary and Applications

- **Two-coin formula:** for coprime m, n , the largest un-makeable amount is $mn - m - n$, with $\frac{(m-1)(n-1)}{2}$ amounts un-makeable in total.
- **Check coprimality first:** a finite answer exists only when the denominations are collectively coprime; otherwise infinitely many amounts fail.
- **Reduce redundant coins:** drop any denomination expressible from the others; with luck the problem falls back to two coins (no closed form exists for three or more in general).
- **Applications:** postage/stamp and coin-change problems, the McNugget puzzle, packaging and bundling questions, and modular-reachability arguments in number theory.

27 Number Theory: Digit Counting

Representative Problem — How Many Digits (Q21)

Problem: How many digits does 99^{10} have (in base 10)?

The key fact. A positive integer N with exactly d digits sits between the smallest and next-smallest powers of ten:

$$10^{d-1} \leq N < 10^d \quad (\text{e.g. a 3-digit number satisfies } 100 \leq N < 1000).$$

Taking $\log_{10}$ of all three parts (the log is increasing, so inequalities are preserved),

$$d - 1 \leq \log_{10} N < d \implies d - 1 = \lfloor \log_{10} N \rfloor,$$

so the digit count is

$$d = \lfloor \log_{10} N \rfloor + 1.$$

The idea: $\log_{10} N$ measures “how many powers of ten fit into N ,” and its integer part is one less than the number of digits.

Apply it. Using $\log_{10}(ab) = \log_{10} a + \log_{10} b$ to pull the exponent out,

$$\log_{10}(99^{10}) = 10 \log_{10} 99.$$

Since $99 = 100 \cdot 0.99$,

$$\log_{10} 99 = \log_{10} 100 + \log_{10} 0.99 = 2 + \log_{10} 0.99 \approx 2 - 0.00436 = 1.99564,$$

so $10 \log_{10} 99 \approx 19.956$. Therefore

$$d = \lfloor 19.956 \rfloor + 1 = 19 + 1 = 20.$$

99^{10} has $\lfloor 10 \log_{10} 99 \rfloor + 1 = \boxed{20}$ digits.

Additional Problem 1 — How Often a Digit Appears (Q29)

Problem: Among all the integers from 1 to 1,000,000, how many times does the digit 2 appear in total (counting every occurrence, so 22 counts twice)?

The padding trick. Write every number from 0 to 999,999 as a *six-digit string*, padding with leading zeros ($0 = 000000$, $37 = 000037$, etc.). There are exactly $10^6 = 1,000,000$ such strings, and this padding does not change how many 2s appear (leading zeros are never 2s).

Now look at *one* digit position — say the hundreds place. As the number runs over all 10^6 strings, that single position cycles through $0, 1, 2, \dots, 9$ equally often, so each digit value shows up

$$\frac{10^6}{10} = 100,000$$

times in that position. The same holds for every one of the 6 positions, so the digit 2 appears

$$6 \times 100,000 = 600,000$$

times in total across 0 to 999,999. Finally adjust the range: 0 contributes no 2, and 1,000,000 (digits 1, 0, 0, 0, 0, 0, 0) contributes no 2, so the count for 1 to 1,000,000 is unchanged.

The digit 2 appears $6 \times 100,000 = \boxed{600,000}$ times.

(General rule: over 0 to $10^k - 1$, written as k -digit strings, each digit value appears $k \cdot 10^{k-1}$ times.)

Additional Problem 2 — Whether a Digit Appears (Q73)

Problem: A whole number is chosen uniformly at random from 1 to 100. What is the probability that it does *not* contain the digit 7?

Count the “avoiders” position by position. Again write the numbers as two-digit strings 00 to 99. For a string to contain no 7, each of its two digits may be any of $\{0, 1, 2, 3, 4, 5, 6, 8, 9\}$ — that is 9 allowed values per position. Since the positions are independent choices,

$$\#\{\text{two-digit strings with no 7}\} = 9 \times 9 = 81.$$

These are the numbers 00 through 99 avoiding 7. Drop 00 (it is not in 1–100), leaving 80 such numbers in 1–99; then add 100 (its digits 1, 0, 0 contain no 7), giving 81 avoiders in 1–100. Hence

$$P(\text{no digit 7}) = \frac{81}{100} = 0.81.$$

$$P(\text{no digit 7 in 1–100}) = \frac{81}{100} = 81\% \text{ (so } P(\text{contains a 7}) = 19\%).$$

(General rule: among k -digit strings, a fraction $(9/10)^k$ avoid any one chosen digit; its complement gives the probability the digit *does* appear.)

Additional Problem 3 — Sum of All Digits over a Range (Q195)

Problem: Add up the digits of *every* integer from 1 to 1,000,000. (This is the sum of the *digits*, not of the numbers — e.g. from 1 to 10 the numbers sum to 55 but the digits sum to 46.) What is the total?

Same padding idea, now summing instead of counting. Write every number from 0 to 999,999 as a six-digit string with leading zeros; the padding adds only 0-digits, so it does not change any digit sum. Fix one position: as the number runs over all 10^6 strings, that position shows each value $0, 1, \dots, 9$ exactly 10^5 times, so the digits in that one position add up to

$$\underbrace{(0 + 1 + 2 + \dots + 9)}_{=45} \times 10^5 = 45 \times 10^5.$$

With 6 independent positions, the grand total over 0 to 999,999 is

$$6 \times 45 \times 10^5 = 270 \times 10^5 = 27,000,000.$$

Since 0 adds nothing, this is also the total for 1 to 999,999. Finally add the digit sum of 1,000,000 itself, which is 1:

$$27,000,000 + 1 = 27,000,001.$$

The digits of all numbers from 1 to 1,000,000 sum to $6 \times 45 \times 10^5 + 1 = 27,000,001$.

(General rule: over 0 to $10^k - 1$, the digits sum to $k \times 45 \times 10^{k-1}$ — each of the k positions contributes the average digit 4.5 across 10^k numbers.)

Concept Summary and Applications

- **Number of digits:** N has $\lfloor \log_{10} N \rfloor + 1$ digits, because $10^{d-1} \leq N < 10^d$; pull out exponents with $\log_{10}(a^b) = b \log_{10} a$ to size huge powers quickly.
- **How often a digit appears:** pad numbers to a fixed length; each position is uniform over 0–9, so a digit shows up $k \cdot 10^{k-1}$ times over 0 to $10^k - 1$.
- **Whether a digit appears:** count the strings that *avoid* it (9 choices per position, fraction $(9/10)^k$), then take the complement.
- **Sum of the digits over a range:** each padded position averages the digit 4.5, so over 0 to $10^k - 1$ the digits sum to $k \times 45 \times 10^{k-1}$ (the same position-by-position idea, summed rather than counted).
- **Applications:** estimating the magnitude of large numbers, log-scale sizing and Fermi estimates, combinatorial digit counts, and quick market-making on “probability a number looks like...” questions.

28 Fibonacci and Tiling Recurrences

Representative Problem — Ten Light Bulbs, No Two Adjacent (Q61)

Problem: Ten light bulbs stand in a row, each either on or off, but *no two adjacent bulbs may be on at the same time*. How many valid on/off patterns are there?

Set up a recurrence by looking at the last bulb. Let a_n be the number of valid patterns for a row of n bulbs. The trick that unlocks these problems is to condition on the *last* bulb and reduce to smaller rows:

- **Last bulb off.** Then the first $n - 1$ bulbs may be any valid pattern — no new restriction. That contributes a_{n-1} patterns.
- **Last bulb on.** Then its neighbour, bulb $n - 1$, is forced *off* (no two adjacent on); the first $n - 2$ bulbs may be any valid pattern. That contributes a_{n-2} patterns.

These two cases are mutually exclusive and cover everything, so

$$a_n = a_{n-1} + a_{n-2} \quad (\text{the Fibonacci recurrence}).$$

Pin down the starting values by hand. For one bulb, the patterns are 0 and 1, so $a_1 = 2$. For two bulbs, 00, 01, 10 are valid (only 11 is banned), so $a_2 = 3$. Now climb the recurrence:

n	1	2	3	4	5	6	7	8	9	10
a_n	2	3	5	8	13	21	34	55	89	144

Each entry is the sum of the two before it ($5 = 3 + 2$, $8 = 5 + 3$, $\dots$, $144 = 89 + 55$).

There are $a_{10} = \boxed{144}$ valid patterns. (These are the Fibonacci numbers: $a_n = F_{n+2}$, so $a_{10} = F_{12} = 144$.)

Additional Problem — Climbing a Staircase

Problem: A staircase has n steps. On each move you may climb either **one** step or **two** steps. How many distinct sequences of moves take you from the bottom to the top?

Same idea — condition on the last move. Let $f(n)$ be the number of ways to reach step n . Your final move landed you there either from step $n - 1$ (a 1-step) or from step $n - 2$ (a 2-step), and these two possibilities are distinct and exhaustive:

$$f(n) = f(n - 1) + f(n - 2)$$

— the Fibonacci recurrence again. Only the starting values differ from the light-bulb problem: there is exactly $f(1) = 1$ way to climb one step (a single 1-step) and $f(2) = 2$ ways to climb two

(1+1 or 2). Building up,

n	1	2	3	4	5	6	7	8	9	10
$f(n)$	1	2	3	5	8	13	21	34	55	89

so, for example, a 5-step staircase has $f(5) = 8$ ways and a 10-step staircase has $f(10) = 89$.

The number of ways to climb n steps (by 1s and 2s) is the Fibonacci number $f(n) = f(n-1) + f(n-2)$ with $f(1) = 1, f(2) = 2$; here $f(n) = F_{n+1}$, e.g. $f(10) = 89$.

Concept Summary and Applications

- **The signature:** whenever a valid configuration is built one item at a time and each item's choice depends only on the *immediately previous* one — “no two adjacent,” “take 1 or 2,” “no 11” — the count obeys $a_n = a_{n-1} + a_{n-2}$, the Fibonacci recurrence.
- **Derivation recipe:** condition on the last element; split into the cases it allows; each case leaves a smaller instance of the *same* problem, giving the recurrence directly.
- **Only the seeds differ:** light bulbs ($a_1 = 2, a_2 = 3$), staircases ($f_1 = 1, f_2 = 2$), 1×2 domino tilings of a $1 \times n$ strip, and binary strings with no 11 are all the *same* recurrence, landing on different offsets of the Fibonacci sequence $1, 1, 2, 3, 5, 8, 13, \dots$.
- **Applications:** tiling and packing counts, restricted binary strings and codes, path-counting on a line, and any “build it up step by step with a local constraint” combinatorics problem.

29 Rational Numbers: Fractions in an Interval

Representative Problem — Which Denominators Fit (Q305)

Problem: Two positive integers a and b satisfy $\frac{a}{b} \in [0.48, 0.52]$. For which values of b is this possible — i.e. for which b does *some* positive integer a make a/b land in that interval?

Step 1 — turn the decimals into exact fractions. Since $0.48 = \frac{12}{25}$ and $0.52 = \frac{13}{25}$, the condition is $\frac{12}{25} \leq \frac{a}{b} \leq \frac{13}{25}$. Multiplying through by $b > 0$ turns it into a statement about a :

$$\frac{12b}{25} \leq a \leq \frac{13b}{25}.$$

So b works exactly when the interval $[\frac{12b}{25}, \frac{13b}{25}]$ contains at least one whole number a .

Step 2 — long intervals always contain an integer. The interval has length

$$\frac{13b}{25} - \frac{12b}{25} = \frac{b}{25}.$$

Any interval of length ≥ 1 must contain an integer, so as soon as $\frac{b}{25} \geq 1$, i.e. $b \geq 25$, the value b automatically works. That leaves only $1 \leq b \leq 24$ to examine by hand.

Step 3 — the small cases, and a clean pattern. For $1 \leq b \leq 24$ the interval is shorter than 1, so it may or may not catch an integer. The interval is centred at $\frac{12.5b}{25} = \frac{b}{2}$ with half-width $\frac{b}{50}$. Two cases:

- **b even:** take $a = \frac{b}{2}$, giving $\frac{a}{b} = \frac{1}{2} = 0.5$, which is inside $[0.48, 0.52]$. So *every* even b works — in particular $b = 2, 4, 6, \dots, 24$.
- **b odd:** the centre $\frac{b}{2}$ is a half-integer, so the nearest integer sits a distance $\frac{1}{2}$ away. The interval only reaches it when its half-width $\frac{b}{50} \geq \frac{1}{2}$, i.e. $b \geq 25$. So no odd $b \leq 24$ works, as the intervals $[\frac{12b}{25}, \frac{13b}{25}]$ for $b = 1, 3, 5, \dots$ confirm — $[0.48, 0.52]$, $[1.44, 1.56]$, $[2.4, 2.6]$, ... each falls strictly between two integers and catches none.

b works iff b is even or $b \geq 25$:

$$b \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24\} \cup \{25, 26, 27, \dots\}.$$

The smallest is $b = 2$ (with $a = 1$, giving exactly $\frac{1}{2} = 0.5$).

The Bigger Picture — Farey Sequences and the Stern–Brocot Tree

The problem is really asking: *what is the simplest fraction (smallest denominator) sitting inside a given interval?* Two classical structures answer exactly this, and both are worth knowing in layman's terms.

Farey sequences. The *Farey sequence* F_n is simply the list of all fractions between 0 and 1 whose denominator is at most n , written in lowest terms and in increasing order. The first few are

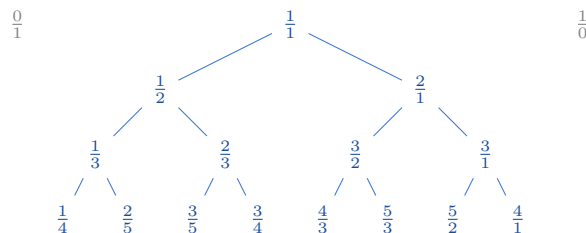
$$\begin{aligned} F_1 &: \frac{0}{1}, \frac{1}{1}. \\ F_2 &: \frac{0}{1}, \frac{1}{2}, \frac{1}{1}. \\ F_3 &: \frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1}. \\ F_4 &: \frac{0}{1}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{1}. \end{aligned}$$

(Going from F_3 to F_4 only adds $\frac{1}{4}$ and $\frac{3}{4}$, since $\frac{2}{4}$ reduces to the already-listed $\frac{1}{2}$.) As you raise n , new fractions squeeze in between the old ones. The magic is *how* they appear: the first fraction to show up between two neighbours $\frac{p}{q}$ and $\frac{r}{s}$ is their **mediant**

$$\frac{p+r}{q+s} \quad (\text{add the tops, add the bottoms}),$$

which always lies strictly between them and, remarkably, is the fraction with the *smallest possible denominator* in that gap. (Neighbours in a Farey sequence also satisfy the tidy identity $ps - qr = \pm 1$.) Note the mediant is *not* the same as averaging: the mediant of $\frac{0}{1}$ and $\frac{1}{1}$ is $\frac{1}{2}$, not $\frac{1}{1}$ -ish nonsense — it is a genuinely different, and very useful, operation.

The Stern–Brocot tree. Start with the two “endpoints” $\frac{0}{1}$ and $\frac{1}{0}$ and repeatedly insert mediants. This grows a binary tree that contains *every* positive fraction in lowest terms exactly once, ordered left-to-right by size. To find the simplest fraction in an interval, you **binary-search** the tree: form the mediant of your current bounds; if it lands in the target interval you are done, otherwise it tells you to go left (mediant too big) or right (too small), and you repeat. Because each step is a mediant, the first fraction you land inside the interval is guaranteed to have the smallest denominator.



The Stern–Brocot tree. The root $\frac{1}{1}$ is the mediant of the boundary values $\frac{0}{1}$ and $\frac{1}{0}$; every node is the mediant of its nearest left and right ancestors, and each fraction in lowest terms appears exactly once.

Back to the problem. Run this search on $[0.48, 0.52]$ starting from $\frac{0}{1}$ and $\frac{1}{1}$: their mediant is $\frac{0+1}{1+1} = \frac{1}{2} = 0.5$, which is already inside the interval. So $\frac{1}{2}$ is the simplest fraction in $[0.48, 0.52]$ — denominator 2 — which is exactly why $b = 2$ is the smallest denominator that works, and why every even b (a scaled-up copy of $\frac{1}{2}$) works too.

Concept Summary and Applications

- **Existence test:** a fraction a/b lands in $[\ell, u]$ for some integer a iff the interval $[\ell b, ub]$ contains an integer; if its length $(u - \ell)b \geq 1$ this is automatic.
- **Mediant, not average:** the mediant $\frac{p+r}{q+s}$ of two fractions lies between them and is the smallest-denominator fraction in the gap — the engine behind Farey sequences and the Stern–Brocot tree.

- **Simplest fraction in an interval:** binary-search the Stern–Brocot tree with mediants; the first one inside the interval has the least denominator.
- **Applications:** best rational approximations (gear ratios, calendar and timing fractions), continued fractions, reducing/deriving tolerances, and clock/alignment problems.

30 Region Counting in 2-Dimensional and 3-Dimensional Space

Both problems below share one idea: **add the cuts one at a time and count how many new regions each cut creates**. In 2D a new line is chopped into pieces by the lines it crosses; in 3D a new plane is chopped into flat pieces by the planes it crosses — and each piece splits exactly one region in two.

Representative Problem 1 — n Lines Dividing the Plane

Problem: Into how many regions do n straight lines divide the plane, when the lines are in “general position” (no two parallel, no three through a common point, which maximises the count)?

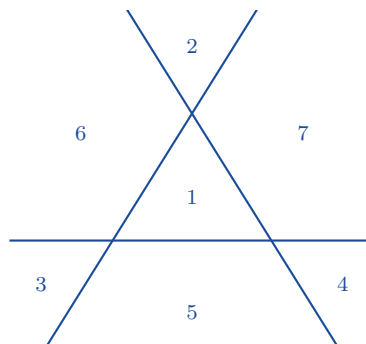
Add one line at a time. Start with the empty plane: 1 region. Now suppose $k - 1$ lines are already drawn and you add the k -th. In general position it crosses each of the existing $k - 1$ lines at a distinct point, so those $k - 1$ crossing points chop the new line into k pieces ($k - 2$ interior segments plus 2 infinite rays). *Each piece slices one existing region into two*, so the new line adds exactly k regions. Summing from the first line to the n -th,

$$L(n) = 1 + \underbrace{(1 + 2 + 3 + \cdots + n)}_{\text{one term per line}} = 1 + \frac{n(n+1)}{2}.$$

Written with binomial coefficients this is the memorable form

$$L(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} = 1 + n + \frac{n(n-1)}{2}.$$

These are the **lazy-caterer** (pancake) numbers 1, 2, 4, 7, 11, 16, 22, ... The picture below shows $n = 3$: three lines make a central triangle plus six unbounded pieces, $L(3) = 1 + 3 + 3 = 7$.



n lines in general position make $L(n) = 1 + n + \binom{n}{2} = \frac{n^2 + n + 2}{2}$ regions (the lazy-caterer numbers); e.g. $L(3) = 7$, $L(4) = 11$.

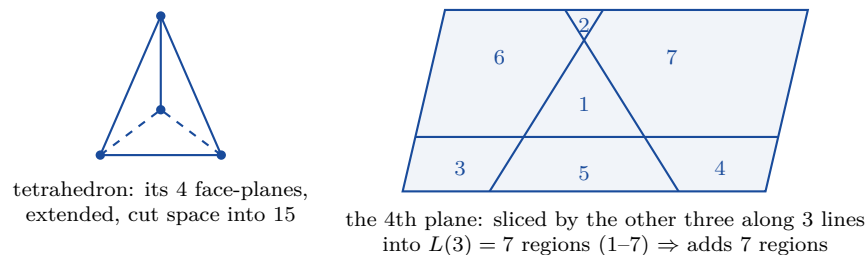
Representative Problem 2 — Tetrahedron Planes (Q38)

Problem: Extend the four faces of a tetrahedron into full infinite planes. Into how many regions do these four planes divide three-dimensional space?

The four face-planes are in general position (no two parallel, no three sharing a line, no four sharing a point). Add them one at a time; the new idea is that the k -th plane is cut by the $k - 1$ existing planes into flat pieces, and *each flat piece splits one 3D region in two*. On the new plane those $k - 1$ cutting lines are themselves in general position, so they carve it into $L(k - 1)$ pieces — exactly the 2D lazy-caterer count from Problem 1. Hence each plane adds $L(k - 1)$ regions:

planes so far	0	1	2	3	4
new regions added	1	1	2	4	7
running total	1	2	4	8	15

The fourth plane is cut by 3 lines into $L(3) = 7$ pieces (drawn in the right-hand figure below), so it adds 7 regions, giving $1 + 1 + 2 + 4 + 7 = 15$.



General formula. The same “add one term per dimension” pattern extends the lazy-caterer numbers by one binomial term:

$$R(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = \frac{n^3 + 5n + 6}{6}.$$

These are the **cake numbers** 1, 2, 4, 8, 15, 26, 42, $\dots$, and $R(4) = 15$ recovers the answer.

Why (recurrence). Add the planes one at a time. The n -th plane is sliced by the $n - 1$ existing planes into flat pieces, and *each piece splits one 3D region in two*, so it adds exactly that many regions. On that plane the $n - 1$ slicing lines are in general position, so they create

$$L(n - 1) = \binom{n - 1}{0} + \binom{n - 1}{1} + \binom{n - 1}{2}$$

pieces — the 2D lazy-caterer count for $n - 1$ lines. Hence

$$R(n) = R(n - 1) + L(n - 1),$$

and summing this telescoping recurrence gives the boxed formula. Our step table $1 + 1 + 2 + 4 + 7 = 15$ is exactly $R(3) + L(3)$ with $L(3) = 1 + 3 + 3 = 7$.

Four planes in general position (the tetrahedron’s faces) cut space into $R(4) = 1 + 4 + 6 + 4 = \boxed{15}$ regions.

Concept Summary and Applications

- **One recipe, every dimension:** add each cut and count the pieces it is chopped into; each piece splits one region. This gives “sum the first $d + 1$ binomial coefficients” in dimension d :

$$\text{points (1D)} : 1 + n, \quad \text{lines (2D)} : 1 + n + \binom{n}{2}, \quad \text{planes (3D)} : 1 + n + \binom{n}{2} + \binom{n}{3}.$$

- **General position maximises:** making cuts parallel or concurrent merges intersections and *loses* regions.
- **Name-check:** 2D gives the lazy-caterer/pancake numbers, 3D the cake numbers.
- **Applications:** arrangement counting in computational geometry, hyperplane arrangements, and “maximum pieces from n cuts” puzzles.

31 Painted Cube and 3D Geometry

The whole family of “painted cube” problems rests on one classification. Paint the outside of an $m \times m \times m$ cube and slice it into m^3 unit cubes; a unit cube gets painted only on the faces that lay on the surface, and *where* it sat determines how many painted faces it has:

Position	Painted faces	Count	
Corner	3	8	(a cube has 8 corners)
Edge (non-corner)	2	$12(m-2)$	(12 edges, $m-2$ each)
Face (non-edge)	1	$6(m-2)^2$	(6 faces, $(m-2)^2$ each)
Interior	0	$(m-2)^3$	(the hidden inner block)

Everything else is bookkeeping on these four groups.

Representative Problem — The $3 \times 3 \times 3$ Cube (Q61)

Problem: You dip a cube in paint and let it dry, then cut it into 27 unit sub-cubes and drop them in a bag. You pick one at random and roll it on the table. What is the probability that *no paint shows on top*?

Step 1 — classify the 27 sub-cubes. With $m = 3$ the table above gives

$$\underbrace{8}_{3 \text{ faces}} \text{ (corners), } \underbrace{12(1) = 12}_{2 \text{ faces}} \text{ (edges), } \underbrace{6(1)^2 = 6}_{1 \text{ face}} \text{ (face-centres), } \underbrace{1^3 = 1}_{0 \text{ faces}} \text{ (the centre).}$$

(Check: $8 + 12 + 6 + 1 = 27$.)

Step 2 — probability of a blank top for each type. After a random roll, each of the sub-cube’s 6 faces is equally likely to land on top, so the chance the top is unpainted is $\frac{6 - (\text{painted faces})}{6}$:

$$0 \text{ painted} \rightarrow 1, \quad 1 \rightarrow \frac{5}{6}, \quad 2 \rightarrow \frac{4}{6}, \quad 3 \rightarrow \frac{3}{6}.$$

Step 3 — average over the bag. Weight each type by how many sub-cubes it has and divide by 27:

$$P = \frac{1}{27} \left(1 \cdot 1 + 6 \cdot \frac{5}{6} + 12 \cdot \frac{4}{6} + 8 \cdot \frac{3}{6} \right) = \frac{1}{27} (1 + 5 + 8 + 4) = \frac{18}{27} = \frac{2}{3}.$$

$$P(\text{no paint on top}) = \frac{18}{27} = \frac{2}{3}.$$

(If instead you just asked for a sub-cube with *no painted face at all*, that is only the single centre cube: probability $\frac{1}{27}$.)

Additional Problem 1 — The $4 \times 4 \times 4$ Cube (Q62)

Problem: Paint the outside of a $4 \times 4 \times 4$ cube and cut it into 64 unit cubes. (1) How many unit cubes have at least one painted face? (2) If you pick one of the 64 at random and roll it, what is the probability the top face is painted?

Part (1) — count by the complement. The unpainted cubes are exactly the hidden interior block, of size $(4 - 2)^3 = 2^3 = 8$. So the number with *at least one* painted face is

$$64 - 8 = 56.$$

Part (2) — classify, then average. With $m = 4$ the table gives

corners 8 (3 faces), edges $12(2) = 24$ (2), faces $6(2)^2 = 24$ (1), interior 8 (0).

(Check: $8 + 24 + 24 + 8 = 64$.) After a random roll the chance the top *is* painted is $\frac{\text{painted faces}}{6}$, so

$$P = \frac{1}{64} \left(8 \cdot \frac{3}{6} + 24 \cdot \frac{2}{6} + 24 \cdot \frac{1}{6} + 8 \cdot 0 \right) = \frac{1}{64} \cdot \frac{24 + 48 + 24}{6} = \frac{1}{64} \cdot \frac{96}{6} = \frac{16}{64} = \frac{1}{4}.$$

56 of the 64 cubes have at least one painted face; and $P(\text{top face painted}) = \frac{16}{64} = \frac{1}{4}$.

A quick sanity check for part (2): the total painted area is 6 faces of side 4, i.e. $6 \cdot 16 = 96$ unit faces, spread over $64 \cdot 6 = 384$ unit faces in all; a uniformly random unit face is painted with probability $96/384 = \frac{1}{4}$ — matching the roll calculation, since rolling makes every unit face equally likely to be on top.

Additional Problem 2 — Red Edges Instead of Faces (Q407)

Problem: A grey $3 \times 3 \times 3$ cube is painted red on all its *edges* (only the 12 edge lines of the big cube, not the faces). It is cut into 27 unit cubes; one is chosen uniformly at random and placed randomly on a table. What is the probability that *no red edge is visible*?

Step 1 — classify by red edges (not faces). A unit cube gets a red edge only where it touches one of the big cube's 12 edges:

Position	Count	Red edges
Corner	8	3 (the 3 edges meeting at the big cube's corner)
Edge-middle	12	1 (lies along one big edge)
Face-centre	6	0 (touches no big edge)
Centre	1	0

(Check: $8 + 12 + 6 + 1 = 27$.) Compare the face-painting table: the edge-middle cubes now carry 1 red edge (not 2 painted faces), and the face-centres are clean.

Step 2 — which edges are hidden? A cube has 6 faces and 12 edges, and each edge borders exactly 2 faces. On the table, one face (equally likely any of the 6) is on the bottom. The natural convention is that the 4 edges of the bottom face, which lie on the table, are hidden; the other 8 are visible. So no red shows iff *every* red edge lies on the bottom face.

Step 3 — probability for each type.

- **0 red edges (7 cubes):** always clean, probability 1.

- **1 red edge (12 cubes):** that edge belongs to exactly 2 of the 6 faces; it is hidden iff one of those 2 is on the bottom: $\frac{2}{6} = \frac{1}{3}$.
- **3 red edges at a corner (8 cubes):** no single face contains all 3 (each face through that corner holds only 2 of them), so at least one always shows: probability 0.

Step 4 — average over the bag.

$$P = \frac{1}{27} \left(7 \cdot 1 + 12 \cdot \frac{1}{3} + 8 \cdot 0 \right) = \frac{7+4}{27} = \frac{11}{27} \approx 40.7\%.$$

$$P(\text{no red edge visible}) = \frac{11}{27} \text{ (edges lying on the table count as hidden).}$$

How to read “visible”. Unlike faces, an edge is shared by two faces, so the answer depends on the convention:

Convention	Per-type probabilities (0 / 1 / 3 red edges)	Answer
Edges touching the table are hidden (used above)	1, $\frac{1}{3}$, 0	$\frac{11}{27}$
Strict: viewed from sides and top, a bottom edge still shows as the lower border of a side face	1, 0, 0	$\frac{7}{27}$
Top view only: just the 4 edges of the top face	1, $\frac{2}{3}$, $\frac{1}{2}$	$\frac{19}{27}$

For the top view: a single red edge lies on the top face with probability $\frac{2}{6}$, so it is hidden with probability $\frac{2}{3}$. For a corner cube, if the top face contains that corner (probability $\frac{3}{6}$) two red edges show; otherwise the top face contains none of the three, so it is clean with probability $\frac{1}{2}$. Then

$$P = \frac{7 + 12 \cdot \frac{2}{3} + 8 \cdot \frac{1}{2}}{27} = \frac{7 + 8 + 4}{27} = \frac{19}{27}.$$

In an interview, state the convention first; the intended answer is $\frac{11}{27}$.

Takeaway. Same recipe as face painting: *classify by position, find the per-type probability, average over the bag*. Only the per-type rule changes: for faces it is a face-count ratio ($\frac{\text{painted faces}}{6}$); for edges you must ask which *faces* put an edge out of sight (an edge belongs to 2 faces).

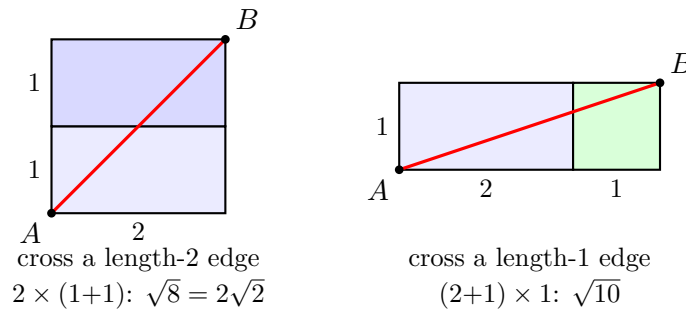
Additional Problem 3 — Shortest and Longest Paths on the Surface of a Box (Q408, Part 3)

Problem: (a) What is the shortest path between opposite corners of a $2 \times 1 \times 1$ rectangular box, if the path must stay on the *surface* of the box? (b) What is the maximum surface distance between *any* two points of the box?

Key idea — unfold the box. An ant walking on the box cannot cut through the inside, so the straight 3D diagonal ($\sqrt{2^2 + 1^2 + 1^2} = \sqrt{6}$) is not allowed. The trick is to *unfold* (flatten) the faces

the path crosses into one plane, like opening a cardboard box. Unfolding does not change lengths on the faces, and in the plane the shortest route between two points is a straight line. So: *list the possible unfoldings, draw a straight line in each, and take the shortest.*

Step 1 — which faces can a corner-to-corner path cross? Call the starting corner A and the opposite corner B . Each face touches either A or B (three faces each), so a shortest path leaves A on one face and arrives at B on an adjacent face, crossing the single edge they share. There are three edge directions to cross (length 2, 1, or 1), giving three unfoldings. When two adjacent faces are laid flat they form one rectangle: one side is the length of the shared edge, the other is the sum of the two remaining dimensions, and A, B sit at opposite ends of its diagonal.



Step 2 — compare the three lengths.

Path crosses an edge of length	Unfolded rectangle	Straight-line length
2	$2 \times (1 + 1)$	$\sqrt{2^2 + 2^2} = 2\sqrt{2} \approx 2.828$
1	$1 \times (2 + 1)$	$\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.162$
1 (other)	$1 \times (1 + 2)$	$\sqrt{10} \approx 3.162$

(a) Shortest surface path between opposite corners = $2\sqrt{2} \approx 2.83$: cross two of the long 2×1 faces, which unfold into a 2×2 square, and go along its diagonal.

General rule. For an $a \times b \times c$ box with $a \geq b \geq c$, the three candidates are $\sqrt{a^2 + (b+c)^2}$, $\sqrt{b^2 + (a+c)^2}$, $\sqrt{c^2 + (a+b)^2}$. Expanding, each equals $\sqrt{a^2 + b^2 + c^2 + 2 \cdot (\text{product of the two summed sides})}$, so the smallest is the one whose summed sides are the two *shortest*:

$$d_{\min}(\text{opposite corners}) = \sqrt{a^2 + (b+c)^2}$$

— pair the **longest** side with the **sum of the other two** (the cross term $2bc$ is the smallest of $2bc, 2ac, 2ab$).

Part (b) — is the farthest pair the opposite corners? It is tempting to say yes, but it is **false** for this box.

Counterexample: the centres of the two 1×1 end faces. Any surface path between them must

1. walk from the centre of the first end face to its edge: at least $\frac{1}{2}$ (all of this happens on the end face, so it makes no progress along the long axis);
2. cross the long faces from one end to the other: at least 2 (the change in the long coordinate);

3. walk from the edge of the second end face to its centre: at least $\frac{1}{2}$.

So their distance is at least $\frac{1}{2} + 2 + \frac{1}{2} = 3$, and the straight route over one long face achieves exactly 3. Since $3 > 2\sqrt{2} \approx 2.83$, the opposite corners are *not* the farthest pair.

Slightly farther still. Put coordinates $(y, z) \in [0, 1]^2$ on each end face and move the two points a little off-centre in opposite directions along a diagonal:

$$P = (a, 1 - a), \quad Q = (1 - a, a), \quad a = \frac{3 - \sqrt{3}}{2} \approx 0.634.$$

Unfolding over a long face, the straight segment has horizontal run $(1 - a) + 2 + a = 3$ and vertical offset $2a - 1 = 2 - \sqrt{3}$, so

$$\sqrt{3^2 + (2 - \sqrt{3})^2} = \sqrt{16 - 4\sqrt{3}} = 2\sqrt{4 - \sqrt{3}} \approx 3.012,$$

and for this pair every other route (over a different long face, or across two or three long faces) is equally long or longer. A numerical search over all pairs of end-face points finds nothing larger.

Why $a = \frac{3 - \sqrt{3}}{2}$? The surface distance is the *shortest* of all routes, and moving the points off-centre pulls two kinds of route in opposite directions:

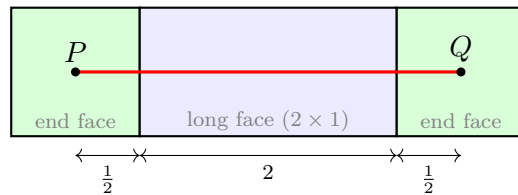
- **Over one long face** (all four long faces give the same length for this pair): run 3, offset $2a - 1$, length $\sqrt{9 + (2a - 1)^2}$ — this *grows* as a moves away from $\frac{1}{2}$.
- **Around a long edge** (end face $\rightarrow$ long face $\rightarrow$ adjacent long face $\rightarrow$ end face): unfolded, the run is $(1 - a) + 2 + (1 - a) = 4 - 2a$ and the offset is $a + a = 2a$, length $\sqrt{(4 - 2a)^2 + (2a)^2}$ — this *shrinks* as a grows, because the points get closer to that shared edge.

The distance is the smaller of the two, so it is largest where they are equal:

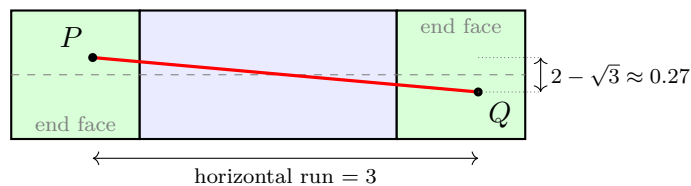
$$9 + (2a - 1)^2 = (4 - 2a)^2 + 4a^2 \iff 10 - 4a + 4a^2 = 16 - 16a + 8a^2 \iff 2a^2 - 6a + 3 = 0 \iff a = \frac{6 - \sqrt{12}}{4} = \frac{3 - \sqrt{3}}{2}$$

(the other root exceeds 1). Beyond this point the corner route becomes the shorter one and the distance starts to fall.

(i) **End-face centres: distance 3**



(ii) **Off-centre pair: distance $2\sqrt{4 - \sqrt{3}} \approx 3.012$**



Each panel shows the box unfolded along one long face (blue) with the two square end faces (green) folded down flat on either side. In (i) the path is $\frac{1}{2} + 2 + \frac{1}{2} = 3$. In (ii) the points are shifted off-centre in opposite

directions (by $a - \frac{1}{2} = \frac{2-\sqrt{3}}{2} \approx 0.134$ each), so the straight line is slightly tilted and longer; the shift is chosen so that the routes over the other long faces become no shorter than this one.

(b) The maximum surface distance is about 3.01 (at least 3, from the end-face centres) — larger than the $2\sqrt{2} \approx 2.83$ between opposite corners.

Takeaway. “Unfold, then draw a straight line” turns any “shortest path on a surface” problem into plane geometry; just remember to check *every* unfolding. And do not assume that the farthest points on a solid are its corners: on a long box, points in the middle of the small end faces are farther apart.

Concept Summary and Applications

- **Four groups by location:** corners (3 painted, always 8), edges (2, $12(m-2)$), faces (1, $6(m-2)^2$), interior (0, $(m-2)^3$); they sum to m^3 .
- **Count the interior by shrinking:** the unpainted block is $(m-2)^3$, so “at least one painted face” $= m^3 - (m-2)^3$.
- **Roll = average over faces:** a random roll makes each of a cube’s 6 faces equally likely on top, so “top painted” has probability $\frac{\text{painted faces}}{6}$; averaging over the bag equals (total painted unit-faces)/(total unit-faces).
- **Painted edges (Q407):** classify by red *edges* instead (corners 3, edge-middles 1, others 0); an edge is hidden only if a face containing it is out of sight, so state the visibility convention.
- **Surface paths (Q408):** unfold the faces a path crosses and draw a straight line; opposite corners of an $a \times b \times c$ box ($a \geq b \geq c$) are $\sqrt{a^2 + (b+c)^2}$ apart. The farthest surface points need not be corners ($2 \times 1 \times 1$: corners $2\sqrt{2}$, end-face centres 3).
- **Applications:** surface-area vs. volume scaling, expected-value counting by category, and Bayesian “I see a painted face, which cube is it?” variants (weight cubes by painted-face count).

32 Clock Hands, Integer Constraints, and Digit-Date Puzzles

Two staples of the mental-math round live here: *clock-hand* problems, solved by comparing the two hands' steady rates, and “*minimum starting number*” problems, solved by chasing a **divisibility** (modular-arithmetic) constraint backward.

Representative Problem 1 — When the Hands Next Align (Q46)

Problem: At 12:00 the hour and minute hands point the same way. Assuming the hands sweep continuously, at what time do they next align?

Compare the two rates. The minute hand sweeps a full circle each hour, and the hour hand one-twelfth of a circle:

$$\text{minute: } 360^\circ/\text{hr}, \quad \text{hour: } 30^\circ/\text{hr}.$$

So the minute hand *gains* on the hour hand at the relative rate

$$360^\circ - 30^\circ = 330^\circ \text{ per hour}.$$

Starting aligned, the hands align again the instant the minute hand has gained a whole lap of 360° on the hour hand:

$$t = \frac{360^\circ}{330^\circ/\text{hr}} = \frac{12}{11} \text{ hr} = \frac{12}{11} \cdot 60 = \frac{720}{11} = 65\frac{5}{11} \text{ min}.$$

That is $65\frac{5}{11}$ minutes after 12:00, i.e. 1:05 $\frac{5}{11}$ (about 1:05:27).

The hands next align at 1:05 $\frac{5}{11}$ (= 65 $\frac{5}{11}$ minutes after 12:00). In general they align every $\frac{12}{11}$ hour — 11 times in each 12-hour span.

Representative Problem 2 — Minimum Starting Passengers (Q79)

Problem: Suppose there are n people on a bus. At each of three stops, $\frac{2}{3}$ of the people on the bus get off and 7 people get on. What is the minimum number of people that were initially on the bus before the first stop, given that the number of people on the bus remains an integer after each stop?

Let the initial number of people on the bus be n . At each stop $\frac{2}{3}$ of the people get off, so $\frac{1}{3}$ remain, and then 7 people get on. If the number of people after each stop must remain an integer, then at each stage the number of people **before** the stop must be divisible by 3.

Let the numbers of people just before the first, second, and third stops be n , n_1 , n_2 , and after the third stop be n_3 . Then

$$n_1 = \frac{n}{3} + 7, \quad n_2 = \frac{n_1}{3} + 7, \quad n_3 = \frac{n_2}{3} + 7.$$

We seek the smallest positive integer n such that n , n_1 , and n_2 are all divisible by 3 before the corresponding stop.

Work backward. Since n_2 must be divisible by 3 and $n_2 = \frac{n_1}{3} + 7$,

$$\frac{n_1}{3} + 7 \equiv 0 \pmod{3}.$$

Because $7 \equiv 1 \pmod{3}$, this gives $\frac{n_1}{3} \equiv -1 \equiv 2 \pmod{3}$. The smallest such value is $\frac{n_1}{3} = 2$, i.e. $n_1 = 6$. But then

$$6 = \frac{n}{3} + 7 \implies \frac{n}{3} = -1,$$

which is impossible. So take the next value, $\frac{n_1}{3} = 5$, i.e. $n_1 = 15$. Then

$$15 = \frac{n}{3} + 7 \implies \frac{n}{3} = 8 \implies n = 24.$$

The same via a closed form. Composing the recurrence three times expresses the final count n_3 directly in terms of the start n :

$$n_3 = \frac{n}{27} + 7\left(1 + \frac{1}{3} + \frac{1}{9}\right) = \frac{n}{27} + \frac{91}{9},$$

which rearranges to

$$\boxed{n = 27n_3 - 273}.$$

A valid start needs $n > 0$, so $27n_3 > 273$, i.e. $n_3 > 10.11$; the smallest integer final count is therefore $n_3 = 11$, giving $n = 27 \cdot 11 - 273 = 24$. (Taking $n_3 = 10$ would force $n = -3 < 0$, which is impossible.)

Check.

- First stop: from 24, $\frac{2}{3} \cdot 24 = 16$ get off, leaving 8; then 7 get on, so 15 remain.
- Second stop: from 15, $\frac{2}{3} \cdot 15 = 10$ get off, leaving 5; then 7 get on, so 12 remain.
- Third stop: from 12, $\frac{2}{3} \cdot 12 = 8$ get off, leaving 4; then 7 get on, so 11 remain.

All counts are integers.

The minimum starting number is 24 (giving $24 \rightarrow 15 \rightarrow 12 \rightarrow 11$).

Representative Problem 3 — Next Date with All Distinct Digits (Q99)

Problem: Starting from today's date, written DD/MM/YYYY, what is the next date in the future whose eight digits are all different?

Distinctness depends only on the *set* of digits, so the DD/MM/YYYY vs YYYY-MM-DD layout does not matter. Two facts pin the search: a two-digit *month* has tens digit 0 (Jan–Sep) or 1 (Oct–Dec), and a two-digit *day* has tens digit in $\{0, 1, 2, 3\}$. Remarkably the answer is over three centuries away, because whole centuries near us are impossible.

No date in 2000–2099. A year 20YY always contains a 2 and a 0. Every month then collides: Jan–Sep are 0X (the tens 0 clashes with the year's 0); Oct = 10 has a 0; Nov = 11 repeats a 1; Dec = 12 has a 2. *No month fits at all.*

No date in 2100–2199. A year 21YY contains 2 and 1, so Oct–Dec (which all use a 1) are out, forcing a Jan–Sep month 0X — which needs the digit 0. The day’s tens digit must come from $\{0, 1, 2, 3\}$, but 0 (month), 1 and 2 (year) are taken, forcing tens digit 3, i.e. day 30 or 31 — whose units 0, 1 are already used. *No day fits.*

No date in 2200–2299. The year “22YY” already repeats the 2.

The 2300s pin everything down. A year 23YZ owns 2 and 3; Oct–Dec are blocked, so the month is Jan–Sep with tens digit 0. The day’s tens digit lies in $\{0, 1, 2, 3\}$, but 0 (month), 2 and 3 (year) are used, so it must be 1 (day 10–19). Thus $\{0, 1, 2, 3\}$ are forced into fixed roles (month-tens 0, day-tens 1, year-lead 2, 3), and the four remaining slots — the year’s last two digits, the month’s unit, and the day’s unit — must be drawn *distinct* from $\{4, 5, 6, 7, 8, 9\}$.

Minimise the date (year first, then month, then day):

- **Year** 23YZ with $Y, Z \in \{4, \dots, 9\}$: smallest is 2345.
- **Month** 0m with m from the leftovers $\{6, 7, 8, 9\}$: smallest is 06 (June).
- **Day** 1d with d from $\{7, 8, 9\}$: smallest is 17.

The next all-distinct-digit date is 17/06/2345 — digits 1, 7, 0, 6, 2, 3, 4, 5 (everything except 8 and 9). No date from 2000 through 2344 qualifies, so the answer does not depend on the exact start date.

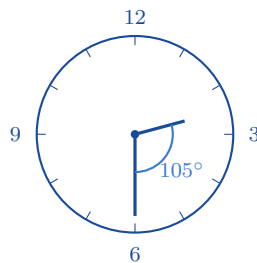
Additional Problem 1 — Angle Between the Hands (Q35)

Problem: What is the angle between the hour and minute hands at 2:30?

Place each hand in degrees, measured clockwise from the 12.

- **Minute hand:** 30 minutes $\times 6^\circ$ per minute (a full 360° in 60 minutes) = 180° — pointing straight down at the 6.
- **Hour hand:** it moves 30° per hour *plus* a bit for the minutes, 0.5° per minute. At 2:30 it is at $2 \times 30^\circ + 30 \times 0.5^\circ = 60^\circ + 15^\circ = 75^\circ$ (halfway between the 2 and the 3).

The angle between them is $|180^\circ - 75^\circ| = 105^\circ$.



At 2:30 the hands are 105° apart. In general the angle at $h:m$ is $|30h - 5.5m|$ degrees (take 360° minus this if it exceeds 180°).

Additional Problem 2 — Next Date with at Most One Repeated Digit (Q329)

Problem: Relax the previous puzzle: starting from today (DD/MM/YYYY), find the next date whose eight digits repeat *at most once* — that is, at most one digit value may appear twice, and every other digit is distinct (so at least seven of the eight digits are different).

Even with this looser rule, the near future is surprisingly barren, for the same tens-digit reasons as Q99. Recall a month's tens digit is 0 or 1, and a near-term year $20YZ/203Z$ carries the digits 2 and 0.

The 2020s are out. A year $202Z$ already contains a *pair* of 2s — that single repeat uses up the whole allowance — plus a 0. Any month then adds a *second* repeat: Jan–Sep (0X) and Oct = 10 repeat the 0; Nov = 11 repeats a 1; Dec = 12 makes a third 2. So no date in the 2020s works.

2030–2033 are out too. The years 2030, 2032, 2033 each repeat a digit inside the year itself (00, 22, 33), using the allowance; the forced month-0 then collides again. The year $2031 = \{2, 0, 3, 1\}$ has distinct digits, but now the four tens-slots values $\{0, 1, 2, 3\}$ are all occupied (month-tens 0, and the day-tens must be one of 0, 1, 2, 3, all taken) — so the day forces a second repeat. None qualifies.

2034 frees a digit. The year $2034 = \{2, 0, 3, 4\}$ has four distinct digits, which frees the digit 1. Now the one allowed repeat is the pair of 0s (the year's 0 plus a Jan–Sep month's leading 0), the day's tens digit can be 1, and the remaining slots take leftover digits. Minimising month then day: month 05 (May — months 01–04 either strand the day or reuse 2, 3, 4), and day 16, giving

16/05/2034, digits 1, 6, 0, 5, 2, 0, 3, 4 (only the 0 repeats).

A day-by-day scan from today confirms nothing earlier qualifies.

The next date with at most one repeated digit is 16/05/2034 (the single repeat is the pair of 0s). Every year from the 2020s through 2033 is impossible.

Concept Summary and Applications

- **Clock hands are a rate race:** the minute hand gains $330^\circ/\text{hr}$ on the hour hand, so they align every $\frac{12}{11}$ hr; the instantaneous angle is $|30h - 5.5m|$ degrees (30° per hour-mark, 5.5° of net gain per minute).
- **“Minimum integer” problems are divisibility problems:** write the step as a recurrence, demand each intermediate value be a whole number, and work *backward* using congruences (mod) the divisor.
- **Watch for impossible early roots:** the smallest congruence solution can force a negative count; step to the next valid one.
- **Digit-date puzzles:** treat a date as its eight-digit multiset and use the tens-digit constraints (month tens $\in \{0, 1\}$, day tens $\in \{0, 1, 2, 3\}$) to eliminate whole centuries fast; then minimise year, then month, then day.

- **Applications:** calendar/day-of-week arithmetic, gear and orbital alignment (LCM of periods), cycle lengths, and any “keep it a whole number” constraint.

33 Apple Transport Optimization (Relay / Depot Problems)

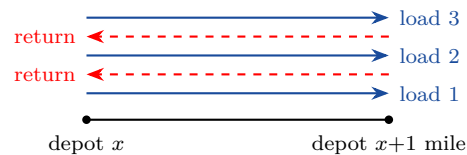
The Set-Up and the One Idea That Solves It

A truck must move a big pile of apples across a long road. It can carry at most C apples at a time and it “pays” for travel in apples. Because the pile is bigger than one truckload, the truck cannot drive straight through: it must **shuttle** — carry a load forward, drop it at a roadside depot, drive back for the next load, and so on — moving the whole pile forward bit by bit. The question is always: how many apples survive to the destination?

The one idea: cost per mile depends on how many loads are left. Suppose the pile currently needs k truckloads ($k = \lceil \text{pile}/C \rceil$). To advance the *entire* pile by one mile, the truck must make k loaded forward trips and, between them, $k - 1$ return trips (it does not return after the last load). How many apples that mile costs depends on whether an *empty* return trip costs anything — which gives two different models:

Model A (loss on *every* mile driven, including empty returns — the classic camel/fuel convention): a mile of progress costs k (forward) + $(k - 1)$ (return) = $2k - 1$ apples.

Model B (loss *only while carrying* apples — empty returns are free): a mile of progress costs k apples.



Moving a 3-load pile ($k = 3$) forward one mile: 3 loaded trips (solid) and 2 empty returns (dashed). Model A charges all 5 legs ($2k - 1 = 5$ apples); Model B charges only the 3 loaded legs ($k = 3$ apples).

The strategy (same for both models). Shuttle forward at the current rate until the pile shrinks to *exactly* the next multiple of C — at that moment one whole load disappears and the rate drops ($k \rightarrow k - 1$). Stopping anywhere else would leave a partial load that still needs a full round trip, wasting apples. So the journey splits into **phases**, one per value of k , and in each phase we lose exactly C apples (except the last phase, which just drives to the end).

Representative Problem 1 — 2000 Apples, Model A (Q43)

Problem: A truck at a depot starts with 2000 apples and can carry at most 1000 at a time. For every mile it travels, it loses 1 apple. What is the maximum number of apples that can be transported across a 1000-mile desert? (Use **Model A**: every mile driven costs an apple, including empty returns.)

Step 1 — the first phase, while the pile is 2000 ($k = 2$). Moving all 2000 apples forward needs two loads, so each mile of progress takes

- one forward trip carrying 1000 apples,

- one return trip (to fetch the rest),
- one forward trip carrying the other 1000 apples.

That is 3 miles of driving per mile of progress, so under Model A the rate is $2k - 1 = 3$ apples per mile. We keep this up until the pile falls to 1000 (one load), i.e. until 1000 apples have been lost:

$$3x = 1000 \implies x = \frac{1000}{3} \approx 333.3 \text{ miles.}$$

So after $\frac{1000}{3}$ miles there are exactly 1000 apples at the depot.

Step 2 — the final phase ($k = 1$). Now everything fits in one load, so the truck simply drives the rest of the way at 1 apple per mile. The remaining distance is $1000 - \frac{1000}{3} = \frac{2000}{3}$ miles, costing $\frac{2000}{3}$ apples:

$$1000 - \frac{2000}{3} = \frac{1000}{3} = 333\frac{1}{3}.$$

Under Model A the truck delivers at most $\frac{1000}{3} = \boxed{333\frac{1}{3}}$ apples.

For comparison (Model B): the first phase costs only $k = 2$ apples per mile, lasting $1000/2 = 500$ miles; the last 500 miles cost 1 per mile, so $1000 - 500 = \mathbf{500}$ apples arrive. The wording “the truck loses an apple per mile” is ambiguous — state which model you assume.

Representative Problem 2 — 3000 Apples, Model B (Q266)

Problem: You have 3000 apples in Edinburgh and want to move as many as possible to London, 1000 miles away. The truck holds at most 1000 apples, and *when the truck is carrying apples* it consumes one apple per mile. What is the maximum number you can deliver? (The phrase “when carrying apples” makes this **Model B**: empty return trips are free.)

Step 1 — 3000 apples, three loads ($k = 3$). Each mile of progress needs 3 loaded forward trips (each losing 1 apple) and 2 empty returns (free), so the rate is $k = 3$ apples per mile. Run until the pile drops to 2000:

$$3000 - 3x = 2000 \implies x = \frac{1000}{3} \approx 333.3 \text{ miles.}$$

Step 2 — 2000 apples, two loads ($k = 2$). Now 2 loaded trips per mile: 2 apples per mile. Run until the pile drops to 1000:

$$2000 - 2y = 1000 \implies y = 500 \text{ miles.}$$

Distance covered so far: $\frac{1000}{3} + 500 = \frac{2500}{3} \approx 833.3$ miles, leaving $1000 - \frac{2500}{3} = \frac{500}{3}$ miles to go.

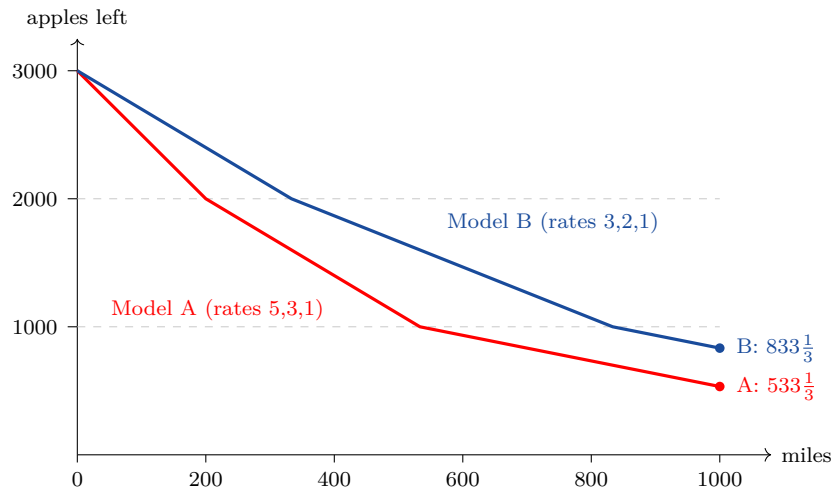
Step 3 — the last 1000 apples, one load ($k = 1$). Drive the final $\frac{500}{3}$ miles at 1 apple per mile:

$$1000 - \frac{500}{3} = \frac{2500}{3} = 833\frac{1}{3}.$$

Under Model B the truck delivers at most $\frac{2500}{3} = \boxed{833\frac{1}{3}}$ apples.

For the same Representative Problem if under Model A. Now empty returns cost too, so the phase rates are $2k - 1 = 5, 3, 1$ apples per mile:

phase 1 ($k = 3$) : 5 per mile, $1000/5 = 200$ mi $\Rightarrow$ 2000 left at mile 200;
 phase 2 ($k = 2$) : 3 per mile, $1000/3 \approx 333.3$ mi $\Rightarrow$ 1000 left at mile $533\frac{1}{3}$;
 phase 3 ($k = 1$) : 1 per mile, $1000 - 533\frac{1}{3} = 466\frac{2}{3}$ mi $\Rightarrow$ $1000 - 466\frac{2}{3} = 533\frac{1}{3}$ delivered.



Apples remaining versus distance for 3000 apples over 1000 miles. Each kink is where the pile hits a multiple of 1000 and one load drops away. Model A (red) burns apples faster because empty returns cost, so it arrives with fewer.

Model A vs. Model B — Key Insights

	Model A	Model B
What costs an apple	every mile driven	only miles driven <i>loaded</i>
Empty return trip	costs 1/mile	free
Rate with k loads left	$2k - 1$	k
Typical wording	“camel eats a banana per km”, fuel	“drops an apple while carrying”
2000 apples, 1000 mi (Q43)	$333\frac{1}{3}$	500
3000 apples, 1000 mi (Q266)	$533\frac{1}{3}$	$833\frac{1}{3}$

- **Read the wording first.** “Consumes an apple *per mile traveled*” (like a camel eating bananas, or a vehicle burning fuel) points to Model A; “loses an apple *while carrying* apples” points to Model B. When in doubt, say which you assume — the answers differ a lot.
- **Phases by load count.** In both models, the pile needs $k = \lceil \text{pile}/C \rceil$ loads and the per-mile cost depends only on k . Advance until the pile is an exact multiple of C ; each full phase burns exactly C apples and lasts $C/(\text{rate})$ miles.
- **Model A phases are shorter** ($C/(2k - 1)$ miles vs. C/k), because the returns eat into the pile — so Model A always delivers less.

- **Early miles are the expensive ones.** The first phase (most loads) has the highest rate, which is why a big starting pile yields surprisingly few extra apples at the end; extra loads beyond a point are pure waste.
- **Applications:** the jeep/desert-crossing and camel-and-bananas problems, fuel depot planning, supply-chain relay costs, and any “carry capacity + per-distance consumption” optimization.

34 Resource Rotation and Equal Wear

Representative Problem — Tricycle with Two Spare Tyres (Q189)

Problem: You have a tricycle (three tyres on the ground) and two spare tyres, so five tyres in total. You will ride 1000 miles and may swap tyres at stops along the way. (a) If every tyre is to be worn *equally* by the end, how many miles does each tyre spend on the ground? (b) What is the minimum number of stops needed to achieve this?

Label the tyres $T_1, \dots, T_5$. At every moment exactly 3 of them are on the ground and 2 are resting.

(a) Count the total work, then share it equally. Think of “tyre-miles”: one tyre rolling one mile is one tyre-mile of wear. Every mile of the trip uses 3 tyres, so the trip produces

$$3 \times 1000 = 3000 \text{ tyre-miles}$$

of wear in total, *no matter how* you rotate. Spreading it equally over 5 tyres gives each

$$\frac{3000}{5} = 600 \text{ miles on the ground}$$

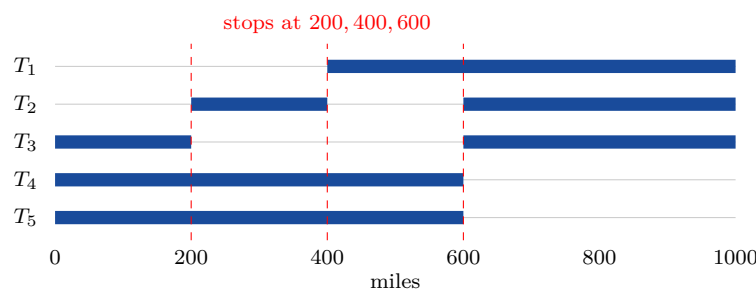
— and therefore $1000 - 600 = 400$ miles resting.

(b) Minimum stops. Split the trip into *segments* between stops; within a segment a fixed pair of tyres rests. Each tyre must rest a total of exactly 400 miles. (Number of stops = number of segments $- 1$.)

Three stops are enough. Group the tyres as a trio $\{T_1, T_2, T_3\}$ and a pair $\{T_4, T_5\}$. Let the trio take turns resting two at a time for 200 miles each, then let the pair rest for the final 400:

miles	resting	on the ground
0–200	T_1, T_2	T_3, T_4, T_5
200–400	T_1, T_3	T_2, T_4, T_5
400–600	T_2, T_3	T_1, T_4, T_5
600–1000	T_4, T_5	T_1, T_2, T_3

Each of T_1, T_2, T_3 rests in two 200-mile segments (400 total); T_4, T_5 rest in the single 400-mile segment. Every tyre is on the ground exactly 600 miles, using four segments — swaps at miles 200, 400, 600, i.e. **3 stops**.



Blue = on the ground. Every row has exactly 600 miles of blue, and every vertical slice has exactly 3 blue tyres.

Two stops are impossible. Two stops give three segments with lengths $x + y + z = 1000$, each resting a pair — $3 \times 2 = 6$ “rest slots.” Every tyre needs at least one slot (otherwise it rests $0 \neq 400$ miles), and 6 slots over 5 tyres forces one tyre (call it D) to rest twice and the other four to rest once. A tyre that rests in only one segment rests that segment’s whole length, so *every segment containing a single-rester must be exactly 400 miles*. With D in two segments, the four single-resters fill the remaining slots: one in each of those two segments and two in the third — so all three segments contain a single-rester and must each be 400 miles. Then $x + y + z = 1200 \neq 1000$ (and D would rest 800 miles). Contradiction.

(a) Each tyre is on the ground $\frac{3 \times 1000}{5} = \boxed{600}$ miles. (b) The minimum is $\boxed{3}$ stops (at miles 200, 400, 600 with the schedule above); 2 stops cannot work.

The Key Concept, the Strategy, and the General Formula

Key concept — conserve the total, then divide. However cleverly you rotate, the *total* amount of wear is fixed by the task: with k items in use at every moment over a duration D , the total usage is $k \cdot D$. Equal wear simply means splitting that fixed total evenly over all n items. This “count the total work first” step is the entire trick for part (a) — no schedule is needed.

With n interchangeable items, k in use at all times, over a total duration D :

$$\text{each item is used } \frac{kD}{n} \quad \text{and rests } \frac{(n-k)D}{n}.$$

Here $n = 5$, $k = 3$, $D = 1000$: use 600, rest 400.

Strategy — a rotation that always works. Cut the trip into n equal segments of length D/n and let the resting set “slide” around the circle: in segment i , rest items $i, i+1, \dots, i+(n-k)-1$ (counting cyclically). Each item then rests in exactly $n-k$ of the n segments, i.e. $(n-k)D/n$ in total — equal wear guaranteed, with $n-1$ stops. For the tricycle this is 5 segments of 200 miles and 4 stops (at 200, 400, 600, 800).

Refinement — fewer stops by grouping. The simple rotation is not always the fewest stops. When the resting requirement can be met by grouping items (as with the trio taking turns in 200-mile blocks while the pair rests together for 400), you can merge segments; the tricycle needs only 3 stops. To prove a stop count is minimal, count the “rest slots” (segments $\times (n-k)$) against the number of items and the lengths each must rest, as in the two-stop impossibility argument.

Concept Summary and Applications

- **Total first:** the total usage $k \cdot D$ is fixed by the task; equal wear means each of n items gets kD/n .
- **Cyclic rotation:** n equal segments with a sliding resting set always equalizes wear, using $n-1$ stops.
- **Minimality needs a counting argument:** compare available rest slots with what each item

must receive; grouping can beat the plain rotation.

- **Applications:** tyre rotation, load-balancing servers or workers, shift scheduling so everyone works equal hours, wear-levelling in flash memory, and rotating inventory or equipment.

35 Optimal Partition: Maximise the Product (Integer and Continuous)

Representative Problem — Split Coins to Maximise the Product (Q148)

Problem: You have x coins. Into how many stacks, and with how many coins in each stack, should you divide them so that the *product* of the stack sizes is as large as possible?

Warm-up example ($x = 10$). A few ways to split 10:

10 (product 10), $5+5$ (25), $3+3+3+1$ (27), $3+3+4$ (36), $3+3+2+2$ (36), $2+2+2+2+2$ (32).

The best is 36 — mostly 3s. The rest of the solution explains why.

Step 1 — never use a stack of 1. A stack of 1 multiplies the product by 1, i.e. contributes nothing, yet uses up a coin. Merging that coin into any other stack ($k \rightarrow k+1$) strictly increases the product. So every stack has at least 2 coins.

Step 2 — never use a stack of 4 or more. For $n \geq 4$, splitting it as $n = 2 + (n-2)$ gives

$$2(n-2) - n = n-4 \geq 0,$$

so the split is never worse: equal at $n = 4$ ($2 \cdot 2 = 4$) and strictly better for $n \geq 5$ (e.g. $5 \rightarrow 2 \cdot 3 = 6 > 5$, $6 \rightarrow 2 \cdot 4 = 8 > 6$). So any stack of 4 or more can be broken up without losing anything. Hence the optimal stacks can be taken to be only 2s and 3s.

Step 3 — prefer 3 over 2. Compare how much each size multiplies the product *per coin*: a stack of 3 multiplies by 3 using 3 coins, i.e. by $3^{1/3}$ per coin; a stack of 2 gives $2^{1/2}$ per coin. Raise both to the 6th power to compare: $(3^{1/3})^6 = 9 > 8 = (2^{1/2})^6$. So $3^{1/3} > 2^{1/2}$ and 3s win. In particular never keep three 2s: $2+2+2 = 3+3$ but $2^3 = 8 < 9 = 3^2$ — so at most *two* 2s ever remain.

Rule: use as many 3s as possible, absorbing the remainder with 2s:

$$\begin{array}{lll} x \equiv 0 \pmod{3}: & \text{all 3s} & \text{product } 3^{x/3} \\ x \equiv 2 \pmod{3}: & \frac{x-2}{3} \text{ threes and one 2} & \text{product } 2 \cdot 3^{(x-2)/3} \\ x \equiv 1 \pmod{3}: & \frac{x-4}{3} \text{ threes and two 2s} & \text{product } 4 \cdot 3^{(x-4)/3} \end{array}$$

(For $x \equiv 1$, never leave a lone 1: turn one $3+1$ into $2+2$, since $4 > 3$.)

Integer examples. $x = 10 \equiv 1$: $3+3+2+2$, product 36 (beats $3+3+3+1 = 27$). $x = 8 \equiv 2$: $3+3+2$, product 18. $x = 9 \equiv 0$: $3+3+3$, product 27. The full table of best products:

x	2	3	4	5	6	7	8	9	10	11	12
best	2	3	4	6	9	12	18	27	36	54	81

Use about $x/3$ stacks, almost all of size 3, with one 2 (if $x \equiv 2$) or two 2s (if $x \equiv 1$).
E.g. 10 coins $\rightarrow 3+3+2+2$, maximum product 36.

Extending to the Continuous Case — Why e Is Optimal

The continuous problem. Now let the pieces be any positive real numbers: split a total S into parts of *any* size to maximise their product. (Think of cutting a rope of length S into pieces and multiplying the lengths.)

Step 1 — for a fixed number of parts, make them equal. By the AM–GM inequality (explained in the next subsection), n parts with sum S have product at most $(S/n)^n$, reached when every part equals S/n . So only the *size* $t = S/n$ of the equal parts remains to be chosen.

Step 2 — choose the best part size (the derivation from Q148). With parts of size t there are S/t of them, so the product is

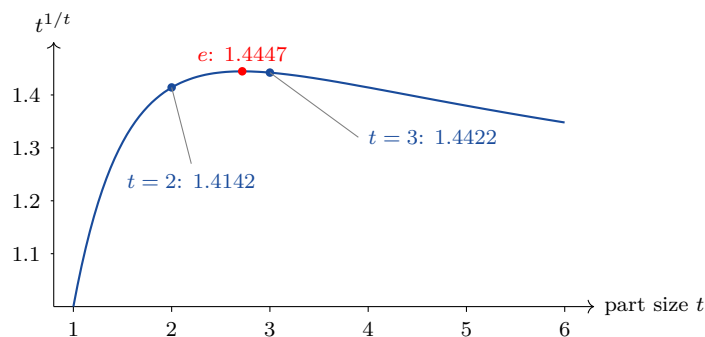
$$t^{S/t} = (t^{1/t})^S.$$

Since S is fixed, we maximise $f(t) = t^{1/t}$. Take logarithms, $g(t) = \ln f(t) = \frac{\ln t}{t}$, and differentiate with the quotient rule:

$$g'(t) = \frac{\frac{1}{t} \cdot t - \ln t \cdot 1}{t^2} = \frac{1 - \ln t}{t^2}.$$

Because $t^2 > 0$, the sign of g' is the sign of $1 - \ln t$: positive for $t < e$ and negative for $t > e$. So g (and hence f) *rises up to $t = e$ and falls after*, and the unique maximiser is

$$1 - \ln t = 0 \implies \boxed{t = e \approx 2.718}.$$



The “growth per unit” $t^{1/t}$ peaks at $t = e$. Among whole numbers, 3 sits closest to the peak (1.4422 vs. 1.4142 for 2) — which is exactly why the integer rule is “use 3s.”

Step 3 — respect that the number of parts is a whole number. The ideal $n = S/e$ is usually not an integer, so compare the two nearest whole numbers of parts, $\lfloor S/e \rfloor$ and $\lceil S/e \rceil$, each with equal parts. For $S = 10$: $S/e \approx 3.68$, and

$$n = 3 : \left(\frac{10}{3}\right)^3 \approx 37.04, \quad n = 4 : 2.5^4 = 39.0625, \quad n = 5 : 2^5 = 32.$$

So the continuous optimum is four pieces of 2.5, product 39.06 — larger than the integer answer 36, because the pieces are now free to sit closer to e .

Continuous case: cut into $n \approx S/e$ equal pieces, each close to $e \approx 2.718$ (pick whichever of $\lfloor S/e \rfloor, \lceil S/e \rceil$ gives the larger $(S/n)^n$). **Integer case:** the nearest usable size to e is 3, hence “as many 3s as possible.”

The Connection to Arithmetic and Geometric Means

For positive numbers $a_1, \dots, a_n$, the **arithmetic mean** (AM) is $\frac{a_1 + \dots + a_n}{n}$ and the **geometric mean** (GM) is $\sqrt[n]{a_1 a_2 \dots a_n}$. The **AM–GM inequality** says

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}, \quad \text{with equality iff } a_1 = a_2 = \dots = a_n.$$

Why it is true for two numbers. For $a, b > 0$,

$$\left(\frac{a+b}{2}\right)^2 - ab = \frac{a^2 + 2ab + b^2 - 4ab}{4} = \frac{(a-b)^2}{4} \geq 0,$$

so $\frac{a+b}{2} \geq \sqrt{ab}$, with equality only when $a = b$. (Picture a rectangle with fixed perimeter: its area ab is largest when it is a square.)

How it drives the partition problem. Raise AM–GM to the n -th power: if the parts add up to S , then

$$a_1 a_2 \dots a_n \leq \left(\frac{S}{n}\right)^n.$$

So, with the *sum fixed*, the *product* is capped by the equal-parts value and reaches it only when all parts are equal. This gives two lessons:

- **Equalise the parts.** Any two unequal parts can be moved toward each other (keeping their sum) and the product goes up — e.g. $5 + 1$ (5) $\rightarrow 3 + 3$ (9). In the integer case, this is why the optimal stacks differ by at most 1 (only 2s and 3s together).
- **AM–GM fixes the shape, calculus fixes the size.** AM–GM says “make them equal” for a *given* number of parts; it does not say how many parts. The $t^{1/t}$ calculation answers that: equal parts of size e . Integers force the size to 3.

Concept Summary and Applications

- **Integer rule:** split into as many 3s as possible, using one or two 2s for the remainder; never 1s, never parts ≥ 4 .
- **Continuous rule:** equal parts of size $\approx e$; the number of parts is $\lfloor S/e \rfloor$ or $\lceil S/e \rceil$, whichever gives the larger $(S/n)^n$.
- **Why e :** maximising the product per unit, $t^{1/t}$, gives $\frac{d}{dt} \frac{\ln t}{t} = \frac{1 - \ln t}{t^2} = 0 \Rightarrow t = e$.
- **AM–GM:** for a fixed sum, the product is maximised by equal parts — the principle behind “balance your resources.”
- **Applications:** maximising volumes/areas with fixed perimeter or budget, splitting work or capital into balanced chunks, radix economy (base 3 is the most “efficient” number base), and the related urn/ball allocation puzzles.

36 Auction Theory and the Winner's Curse

In an auction the thing being sold often has a value you cannot see. The trap is to bid as if you will receive an *average* item. But you only get the item when your bid wins — and *when* your bid wins tells you something about the item. That hidden information is the **winner's curse**: winning is itself news, and usually bad news.

Representative Problem — Bidding for a Car (Q28)

Problem: A seller offers you a car whose true value V is uniformly distributed between \$0 and \$1000; you do not know V . You make one bid b . If your bid is at least the car's value ($V \leq b$), you buy the car at price b and can resell it for 1.5 times its true value, $1.5V$. Otherwise no trade happens. What bid maximises your expected profit?

The naive (wrong) reasoning. “On average the car is worth \$500, I can resell it for $1.5 \times 500 = \$750$, so any bid below \$750 is profitable.” This ignores *which* cars you actually end up buying.

Step 1 — write down the profit. If you win ($V \leq b$) your profit is the resale value minus what you paid, $1.5V - b$. If you lose, the profit is 0.

Step 2 — average over the unknown value. V is uniform on $[0, 1000]$, so its density is $\frac{1}{1000}$. You win exactly when $V \leq b$, so for a bid $0 \leq b \leq 1000$

$$\mathbb{E}[\Pi(b)] = \int_0^b (1.5v - b) \frac{1}{1000} dv = \frac{1}{1000} \left[\frac{1.5}{2} v^2 - bv \right]_0^b = \frac{0.75b^2 - b^2}{1000} = -\frac{b^2}{4000}.$$

Step 3 — optimise. $-\frac{b^2}{4000}$ is negative for every $b > 0$ and equals 0 only at $b = 0$. Bids above \$1000 win for sure but earn $1.5 \mathbb{E}[V] - b = 750 - b \leq -250$, even worse. So the best bid is \$0 — do not bid at all.

Step 4 — why: condition on winning. You get the car only when $V \leq b$. Given that, V is uniform on $[0, b]$, so

$$\mathbb{E}[V \mid \text{you win}] = \frac{b}{2}.$$

Winning tells you the car is worth only *half your bid* on average. Reselling at $1.5 \times$ returns $1.5 \cdot \frac{b}{2} = 0.75b$, which is less than the b you paid — you lose about $0.25b$ on every car you win. The “average car” worth \$500 is irrelevant: you never get to buy the good ones cheaply, because a low bid only wins when the car is worth even less.

Optimal bid = \$0. Conditional on winning, the car is worth $b/2$ on average, and $1.5 \cdot \frac{b}{2} = 0.75b < b$, so every positive bid loses money in expectation.

The Profit Threshold — When Does Bidding Pay?

Replace the resale multiple 1.5 by a general k (e.g. a mechanic who can raise the car's value, or a buyer who values it more). The same calculation gives

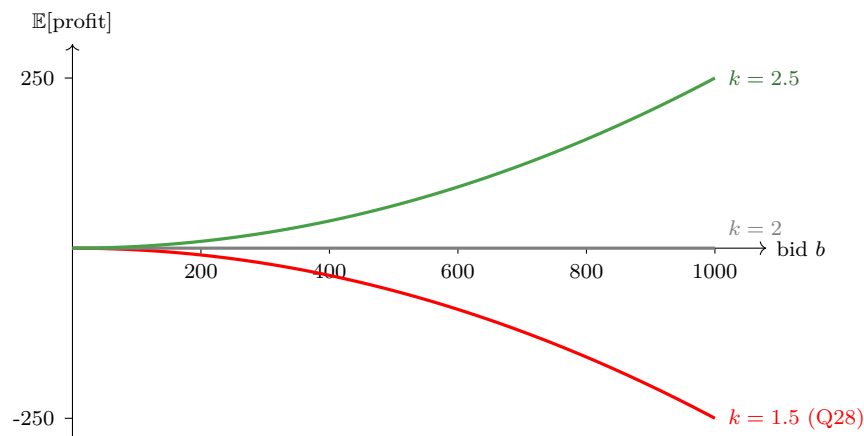
$$\mathbb{E}[\Pi(b)] = \int_0^b (kv - b) \frac{1}{1000} dv = \frac{1}{1000} \left(\frac{k}{2} - 1 \right) b^2.$$

Everything hinges on the sign of $\frac{k}{2} - 1$:

- $k < 2$: every positive bid loses $\Rightarrow$ bid 0;
- $k = 2$: break-even for any bid;
- $k > 2$: profit grows with $b \Rightarrow$ bid as high as possible.

Threshold: bidding is profitable *only if* the resale multiple exceeds **2**, because winning with bid b means the item is worth $b/2$ on average, and you need $k \cdot \frac{b}{2} > b$.

When $k > 2$ the profit keeps rising up to $b = 1000$, where you win every time and earn $k \cdot 500 - 1000$ (e.g. $k = 2.5$ gives \$250). The threshold 2 does not depend on the \$1000 ceiling: for V uniform on $[0, M]$ it is still $k > 2$.



Expected profit versus bid. Below the threshold ($k = 1.5$) it only goes down as you bid more, so bid nothing; at $k = 2$ it is zero for every bid; above the threshold ($k = 2.5$) it rises, so bid the maximum.

Concept Summary and Applications

- **The winner's curse:** in a *common-value* setting (the item is worth the same to everyone, but nobody knows exactly how much), winning selects the cases where the item is worth *less* than you bid. Always value the item *conditional on winning*, $\mathbb{E}[V \mid \text{win}]$, not the unconditional average $\mathbb{E}[V]$.
- **Uniform benchmark:** with $V \sim U[0, M]$ and a winning rule $V \leq b$, $\mathbb{E}[V \mid \text{win}] = b/2$; expected profit is $\frac{1}{M}(\frac{k}{2} - 1)b^2$.
- **Profit threshold:** bid only if the value multiplier $k > 2$; for $k < 2$ the optimal bid is 0, for $k > 2$ bid as high as the range allows.
- **Adverse selection:** the same logic appears whenever the other side accepts *only* when it is good for them — your quote gets hit when you are wrong. Ask “what does it mean that this trade happened?”
- **Both sides informed (wallet game):** when each bidder holds part of the value, the ascending-auction equilibrium is to quit at the price that breaks even against an opponent *with your own signal* — $2x$ in the problem below.

- **Applications:** market making (getting filled is informative), oil-lease and spectrum auctions, IPO allocations, used-car (“lemons”) markets, and takeover bids.

Additional Problem — Bidding for a Shared Pot (Q139)

Problem: You and another player each independently draw a number uniformly from $[0, 1]$; each number is private (you see yours, not theirs). Both numbers go into a pot, and you take turns bidding for the pot. What is your optimal bidding strategy?

Rules of the game. Your number is x , the opponent’s is y ; the pot is worth $x + y$, but each player sees only their own number. Bids go back and forth with a rising price (an ascending, or “English,” auction). A player may drop out at any time; **the last player still bidding wins the pot and pays the price at which the other dropped out.** The winner’s payoff is $(x + y) - \text{price}$; the loser’s payoff is 0.

Step 1 — the naive answer, $x + \frac{1}{2}$, is wrong. The tempting reasoning: you know x , and y averages $\frac{1}{2}$, so the pot is “worth” $x + \frac{1}{2}$ and you should pay up to that. But this uses the *unconditional* average of y . You only win when the opponent drops out first, and players drop out early precisely when their number is *low*. So conditional on winning, y is below average — exactly the winner’s curse from the car problem. Valuing the pot at $x + \frac{1}{2}$ makes you overpay.

Step 2 — guess a symmetric strategy and check it. Suppose each player stays in until the price reaches $2 \times (\text{their own number})$, and test whether that is a best response. If the opponent quits at price $2y$ and you plan to quit at price T , you win exactly when the opponent quits first,

$$2y < T \iff y < \frac{T}{2},$$

and in that case you pay $2y$. Your expected profit is

$$\int_0^{T/2} [(x + y) - 2y] dy = \int_0^{T/2} (x - y) dy = x \cdot \frac{T}{2} - \frac{1}{2} \left(\frac{T}{2} \right)^2.$$

Differentiate in T : $\frac{x}{2} - \frac{T}{4} = 0 \Rightarrow T = 2x$. (Equivalently, the integrand $x - y$ is positive while $y < x$ and negative after, so you want to keep winning exactly up to $y = x$, i.e. $T/2 = x$.) Your best response to “ $2y$ ” is “ $2x$ ”, so

stay in until the price reaches $2x$

is a symmetric equilibrium.

Step 3 — intuition: think about the marginal opponent. At the moment you are about to drop out at price p , ask: “what if my opponent were about to drop out at this same price?” With both players using the same increasing rule, that opponent has the *same* number, $y = x$. Then the pot is worth $x + x = 2x$, so you should be willing to pay up to exactly $2x$ and no more. Conditioning on this “tie” case builds the winner’s-curse correction in automatically.

Step 4 — check the profit. With both players using $2 \times$, you win exactly when $y < x$, paying $2y$ for a pot worth $x + y$, so you keep $x - y > 0$ on every win. Your expected profit is

$$\int_0^x (x - y) dy = \frac{x^2}{2} \geq 0,$$

and, conditional on winning, $\mathbb{E}[y \mid \text{win}] = \frac{x}{2}$ — well below the naive $\frac{1}{2}$.

Stay in until the price reaches twice your own number, $2x$ (not $x + \frac{1}{2}$). Expected profit $x^2/2$.

Sealed-bid variant. If instead each player submits one sealed bid and the higher bidder wins the pot and pays their own bid, the equilibrium bid is $b(x) = x$. Check: if the opponent bids $b(y) = y$ and you bid z , you win when $y < z$ and your expected profit is $\int_0^z (x + y - z) dy = xz - \frac{z^2}{2}$, maximised at $z = x$. Conditional on winning, the pot is worth $x + \frac{x}{2} = 1.5x$ on average; you pay x and keep $\frac{x}{2}$. Expected profit is again $\int_0^x (x + y - x) dy = \frac{x^2}{2}$ — the same as the ascending auction (revenue equivalence: the ascending winner pays $2y$, which averages x given $y < x$). The rule “bid half your value” belongs to a different problem (private values, where you know the item’s worth to you exactly) and does not apply here.

37 Market Making and Optimal Bidding

What “Making a Market” Means

To **make a market** on an uncertain quantity X is to quote two prices at once:

- a **bid** — the price at which *you will buy* X ;
- an **ask** (or offer) — the price at which *you will sell* X ;

together with a **size** — how many units you will trade at those prices. It is written “bid @ ask, size,” e.g. “6.5 at 7.5, 10 up” means “I buy up to 10 at 6.5 and sell up to 10 at 7.5.” The difference ask – bid is the **spread**. The counterparty chooses whether to buy from you or sell to you (or neither) *after* hearing your quote.

Representative Problem — Market on the Sum or Product of Two Dice (Q149)

Problem: Two fair six-sided dice will be rolled. Make a market — bid, ask, and size — on (a) the **sum** of the two dice, and (b) the **product** of the two dice. Justify your prices and quantities.

The recipe. (1) Find the *fair value* $\mathbb{E}[X]$ — the centre of the market. (2) Measure the *risk* with the variance / standard deviation — this drives the spread and the size. (3) Quote a spread around fair value and a size that reflects that risk.

Part (a) — the sum $S = D_1 + D_2$.

Fair value. One die averages $\mathbb{E}[D] = \frac{1+2+\dots+6}{6} = 3.5$, and expectation adds, so

$$\mathbb{E}[S] = \mathbb{E}[D_1] + \mathbb{E}[D_2] = 3.5 + 3.5 = 7.$$

Distribution. The 36 equally likely outcomes give

S	2	3	4	5	6	7	8	9	10	11	12
#ways	1	2	3	4	5	6	5	4	3	2	1

— symmetric around 7 and fairly concentrated.

Risk. For one die, $\mathbb{E}[D^2] = \frac{1^2+2^2+\dots+6^2}{6} = \frac{91}{6}$, so

$$\text{Var}(D) = \frac{91}{6} - 3.5^2 = \frac{35}{12}.$$

Variances of independent variables add, so

$$\text{Var}(S) = 2 \cdot \frac{35}{12} = \frac{35}{6} \approx 5.83, \quad \text{SD}(S) = \sqrt{35/6} \approx 2.42.$$

Market. Centre on 7 with a modest spread and decent size:

6.5 bid, 7.5 ask, size 10.

Buying at 6.5 or selling at 7.5 earns 0.5 of *edge* per unit on average (the distance from fair value). A tighter 6.8–7.2 would be more competitive; a wider quote is justified if the counterparty might be informed or you already hold inventory.

Part (b) — the product $P = D_1 D_2$.

Fair value. For independent variables, the expectation of a product is the product of expectations:

$$\mathbb{E}[P] = \mathbb{E}[D_1] \mathbb{E}[D_2] = 3.5 \times 3.5 = 12.25.$$

Distribution. The product takes 18 different values:

P	1	2	3	4	5	6	8	9	10
#ways	1	2	2	3	2	4	2	1	2

P	12	15	16	18	20	24	25	30	36
#ways	4	2	1	2	2	2	1	2	1

It is **right-skewed**: most outcomes are small, but a long tail reaches 36. In fact $P \leq 11$ about 53% of the time, so the *median* sits below the mean 12.25.

Risk. Independence also gives $\mathbb{E}[P^2] = \mathbb{E}[D_1^2] \mathbb{E}[D_2^2] = \left(\frac{91}{6}\right)^2 = \frac{8281}{36} \approx 230.03$, so

$$\text{Var}(P) = 230.03 - 12.25^2 = 230.03 - 150.06 \approx 79.97, \quad \text{SD}(P) \approx 8.94.$$

That is almost *four times* the standard deviation of the sum.

Market. Centre on 12.25 but go wider and smaller:

11 bid, 13.5 ask, size 5.

(A rounder 10.5–14 for size 10 is also defensible.)

Sum: fair value 7, SD ≈ 2.4 — quote about 6.5 at 7.5, size 10. Product: fair value 12.25, SD ≈ 8.9 and right-skewed — quote wider and smaller, about 11 at 13.5, size 5.

Concept Summary — Expected Value, Variance, Spread, and Size

	Fair value	SD	Quote	Size
Sum of two dice	7	2.42	6.5 – 7.5	10
Product of two dice	12.25	8.94	11 – 13.5	5

- **Expected value sets the centre.** If you bought and sold many times at a price p , your average profit per unit would be $\mathbb{E}[X] - p$ (when buying) or $p - \mathbb{E}[X]$ (when selling). Centring the market on $\mathbb{E}[X]$ makes both sides profitable by half the spread.
- **Variance sets the spread.** The spread is your compensation for risk and for being wrong. The more X can move (SD), the more a single trade can hurt, and the more room the other side has to know something you don't — so quote wider. A useful habit is to think of the spread as a fraction of the SD.
- **Variance also sets the size.** Risk grows with size $\times$ SD. To keep the risk of a fill roughly constant, size should go *down* as SD goes up — big size on the tame sum, small size on the volatile product.
- **Adverse selection and skew.** The counterparty picks the side *after* seeing your quote, so they only trade when your price is wrong in their favour. Quote wider if they may be informed; with a skewed payoff (like the product), remember the median differs from the mean.

- **Inventory.** If you are already long, shade both prices down (to encourage buyers and discourage sellers); if short, shade them up.

Additional Problem — The Optional Bet (Q140)

Problem: Two players, A and B , each roll a fair die privately. Whoever has the higher number wins a unit bet (a tie is a push), but nobody is forced to bet. A looks at their roll and chooses whether to offer the bet; if A declines, the round ends and nobody pays anything. B knows whether A offered, looks at their own roll, and then accepts or declines; if B declines, again nothing is paid. Which player has the higher expected payout?

Step 1 — the offer is information. A only offers when A 's roll is good. So when B sees an offer, B should update: " A 's number is probably high." B then accepts only if, *given that A offered*, B is at least as likely to win as to lose.

Step 2 — try " A offers with 5 or 6." If an offer means $a \in \{5, 6\}$ (each equally likely):

- B with $b = 6$ beats 5 and ties 6 — positive for B , so B accepts;
- B with $b = 5$ ties 5 and loses to 6 — negative, so B declines;
- smaller b are even worse, so they decline.

So B only ever accepts holding a 6. But then A holding a 5 *loses* every accepted bet — A would rather not offer with a 5. The strategy falls apart.

Step 3 — the unravelling argument (any strategy). Suppose A offers with some set of values, and let $a_{\min}$ be the smallest of them, with $a_{\min} < 6$.

- B never accepts with $b < a_{\min}$: such a b loses to *every* value A offers with.
- B always accepts with $b = 6$: a 6 never loses and beats $a_{\min}$.

So whenever the bet is accepted, B holds $b \geq a_{\min}$: A 's $a_{\min}$ never wins and sometimes loses (to the 6). A would do better by *not* offering with $a_{\min}$. Remove it and repeat — the offering set shrinks until only $a = 6$ is left. But if A offers only with 6, every B declines (B can at best tie). So no bet is ever accepted.

Neither player has an advantage: in equilibrium no bet is ever accepted, so both expected payouts are 0. (B 's information makes B impossible to exploit, but it does not give B a positive payout — there is nothing to win when bets are declined.)

The no-trade / adverse-selection theorem. This is a textbook case of the **no-trade theorem** (Milgrom–Stokey): when two rational players share the same beliefs about the world and a bet is purely speculative and zero-sum, *no* bet can be profitable for the person proposing it once the other side is allowed to read the proposal as information. The engine is **adverse selection**: the counterparty accepts *only* in the cases that are good for them — which are exactly the cases bad for you. Every bet B takes is one A expects to lose, and every bet A would win, B declines. Since each player can always guarantee 0 by declining, the result is 0 for both. In market-making language: *if your quote gets hit, ask why* — the willingness to trade is itself a signal.

***Caveat — it depends on the folding rule.** The 0–0 result relies on the rule that declining is free: a declined bet simply ends the round with no money moving. Change that and the answer changes. In a **poker-style** game — say both players first put an ante into a pot, and a player who folds (declines) gives up that pot — declining now costs something. Then A can profit by bluffing (offering with a weak roll to make B fold and hand over the pot), B must sometimes call weak-looking offers to prevent this, and the equilibrium involves mixed betting/calling ranges with a non-zero edge for one side. So “nobody wins” is a property of free, voidable bets, not of betting games in general.*

Applications

- **Trading interviews:** quote a two-sided market on any random quantity — centre on the mean, widen and shrink size with the variance and skew.
- **Real markets:** bid–ask spreads on stocks and options widen with volatility and with the risk of informed traders (adverse selection).
- **Negotiation and betting:** a counterparty eager to take your bet or hit your quote is information — update before you trade.

38 The Doubling Cube: When to Raise the Stakes

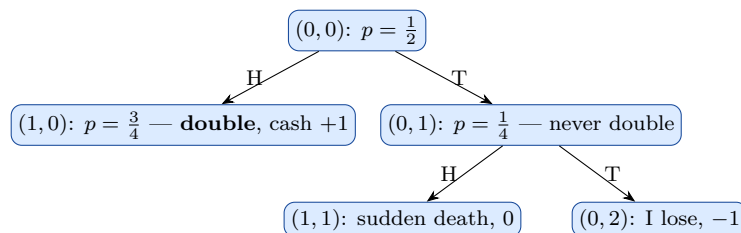
In backgammon (and many trading-style games) a player may offer to **double** the stake mid-game. The opponent then either **takes** (play continues for twice the stake) or **drops** (resigns now and pays the current stake). The right to double is an *option*: it has value because you can wait and use it only when the game has turned in your favour.

Representative Problem — A Coin Race with a Doubling Option (Q395)

Problem: A coin is flipped repeatedly. Heads gives me 1 point, tails gives you 1 point. The first to reach 2 points wins the game, and the loser pays the winner \$1. However, I have a one-time option to raise the stake to \$2. What is the value of this option?

Setup. Track the score as $(a, b) = (\text{my points}, \text{your points})$; each flip adds a point to one side with probability $\frac{1}{2}$. Without the option the game is symmetric and worth \$0 to me, so *the option's value is exactly the extra value it creates*. I may raise the stake from \$1 to \$2 once, at any point.

Warning — the score difference is not enough. It is tempting to track only the difference $b - a \in \{-1, 0, 1\}$. But the tied states $(0, 0)$ and $(1, 1)$ are *not* interchangeable: from $(1, 1)$ the very next flip ends the game (sudden death), so there is no later chance to double from ahead, while from $(0, 0)$ there is. Merging them gives the wrong answer $\frac{1}{3}$, so we keep the full score.



Step 1 — win probabilities and what a double is worth. Let $p(a, b) = \Pr(\text{I win} \mid (a, b))$. One flip at a time,

$$p(1, 1) = \frac{1}{2}, \quad p(1, 0) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot p(1, 1) = \frac{3}{4}, \quad p(0, 1) = \frac{1}{2} p(1, 1) + \frac{1}{2} \cdot 0 = \frac{1}{4}, \quad p(0, 0) = \frac{1}{2}.$$

If the stake is already \$2, my value in a state is $2(2p - 1)$ (win \$2 with probability p , lose \$2 otherwise):

$$(1, 0): 2(2 \cdot \frac{3}{4} - 1) = 1, \quad (0, 0): 0, \quad (1, 1): 0, \quad (0, 1): 2(2 \cdot \frac{1}{4} - 1) = -1.$$

So a double is worth at most \$1, reached only when I lead 1–0 (win chance $\frac{3}{4}$). At $\frac{3}{4}$ this is exactly the *take/drop point*: if you play on at \$2 my value is $2 \cdot \frac{1}{2} = \$1$; if you resign you pay the \$1 stake. Either way I collect \$1, so the answer is the same whether or not you may decline the double.

Step 2 — value with the option, state by state. Let $V(a, b)$ be my value while I still hold the option (stake \$1). In each state I take the better of *doubling now* or *flipping on with the option kept*.

- $(1, 1)$ — **sudden death**. Doubling is worth 0; flipping gives $\frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$. So $V(1, 1) = 0$.
- $(1, 0)$ — **I lead**. Doubling locks in \$1. Flipping on instead gives $\frac{1}{2}(+1) + \frac{1}{2}V(1, 1) = \frac{1}{2}$. Since $1 > \frac{1}{2}$, **double here**: $V(1, 0) = 1$.

- $(0, 1)$ — **I trail**. Doubling is worth $-\$1$, so never; flip on:

$$V(0, 1) = \frac{1}{2}V(1, 1) + \frac{1}{2}(-1) = \frac{1}{2} \cdot 0 - \frac{1}{2} = -\frac{1}{2}.$$

- $(0, 0)$ — **the start**. Doubling is worth 0; flip on, keeping the option:

$$V(0, 0) = \frac{1}{2}V(1, 0) + \frac{1}{2}V(0, 1) = \frac{1}{2}(1) + \frac{1}{2}\left(-\frac{1}{2}\right) = \frac{1}{4}.$$

The game without the option is worth \$0, so the option is worth $V(0, 0) = \frac{1}{4} = \0.25 .

Strategy: do nothing at the start; if the first flip puts me ahead 1–0, double immediately (banking \$1); if it puts me behind, never double.

Why not $\frac{1}{3}$? That answer comes from writing $V(\text{behind}) = \frac{1}{2}V(\text{tied}) + \frac{1}{2}(-1)$ with a single “tied” value. But from behind (0–1) a winning flip leads to (1, 1) — sudden death, value 0 — not back to the start (0, 0), value $\frac{1}{4}$. Using the correct $V(1, 1) = 0$ gives $V(0, 1) = -\frac{1}{2}$ and hence $\frac{1}{4}$.

The General Principle — Double at the Opponent’s Take Point

When should the opponent take? Suppose you double when the opponent’s chance of winning is q (yours is $p = 1 - q$), with stake 1 and no further doubling. If the opponent *drops*, they lose 1 for sure. If they *take*, their value is $2(q \cdot 1 - (1 - q) \cdot 1) = 2(2q - 1)$. Taking is at least as good as dropping when

$$2(2q - 1) \geq -1 \iff q \geq \frac{1}{4}.$$

So the opponent should take while they win at least 25% of the time, and drop below that.

What a double is worth to you. The opponent picks whichever response is worse for you, so a double at win probability p is worth

$$\min\left\{\underbrace{2(2p - 1)}_{\text{they take}}, \underbrace{1}_{\text{they drop}}\right\}.$$

This is capped at 1 (the current stake): once the opponent can simply resign, doubling never earns more than cashing the game.

Threshold rule (one double, final stake). The opponent’s take point is $q = \frac{1}{4}$, i.e. **offer the double when your win probability reaches $p = \frac{3}{4}$** . Below $p = \frac{1}{2}$ never double (it multiplies a losing position); between $\frac{1}{2}$ and $\frac{3}{4}$ doubling raises your immediate value but throws away the option; at $p = \frac{3}{4}$ doubling *cashes* the game for the full stake.

Why wait until $\frac{3}{4}$ rather than double as soon as you are ahead? Keep the option and double exactly when p first reaches $\frac{3}{4}$ (worth +1), or lose at 0 (worth −1). Because p is a fair (martingale) process, the chance of reaching $\frac{3}{4}$ before 0 is $\frac{p}{3/4} = \frac{4p}{3}$, so waiting is worth

$$\frac{4p}{3}(1) + \left(1 - \frac{4p}{3}\right)(-1) = \frac{8p}{3} - 1,$$

whereas doubling now (they take) is worth $4p - 2$. Waiting is better exactly when $\frac{8p}{3} - 1 \geq 4p - 2$, i.e. $p \leq \frac{3}{4}$ — and the two agree at $p = \frac{3}{4}$. The right moment is the last moment your opponent can still accept.

Discrete games: don't "lose your market." When p moves in jumps (as with coin flips), you may leap from below $\frac{3}{4}$ to well above it in one step. Then the opponent drops and you collect only 1, whereas an earlier double they had accepted would have been worth $2(2p - 1) > 1$ in that branch. So in jumpy games you sometimes double slightly *before* $\frac{3}{4}$. In Q395 the lead jumps from $p = \frac{1}{2}$ exactly to $\frac{3}{4}$, so doubling at 1–0 is exactly right.

Concept Summary and Applications

- **The double is an option:** its value comes from waiting and exercising only in good states; value it by backward induction over the *full* state (not just the score difference).
- **Take point $\frac{1}{4}$:** with one final double, the opponent takes iff their win chance is at least 25%; a double is worth $\min\{2(2p - 1), 1\}$ to you.
- **Doubling point $\frac{3}{4}$:** offer when your win probability reaches $\frac{3}{4}$ (earlier only if the next move could jump you past it). With re-doubles allowed (a “live” cube, as in backgammon) the take point drops to about 20% because the taker gains the right to redouble.
- **Applications:** backgammon cube decisions, raising bets in poker-style games, early exercise of options, and any “commit more capital once the odds are clearly in your favour” decision.

39 The Pirate Game (Backward Induction with Voting)

Representative Problem — Five Pirates, 100 Coins (Q249)

Problem: Five pirates of different ages share 100 gold coins. The oldest proposes how to split the coins, and *all* pirates (including the proposer) vote. If 50% **or more** vote for it, the coins are shared that way. Otherwise the proposer is thrown overboard, and the next-oldest pirate proposes, and so on. A pirate who would get the *same* number of coins either way votes *against* (he enjoys seeing the proposer thrown overboard). All pirates are rational, greedy, want to survive, and are perfect at mathematics. What happens?

Label the pirates from oldest to youngest P_1, P_2, P_3, P_4, P_5 ; P_1 proposes first.

The voting rule in one line. With n pirates alive, a proposal passes with at least $\lceil n/2 \rceil$ votes, and the proposer votes for himself. Each other pirate compares the offer with what he would get *if the proposer died* (the outcome of the game with one fewer pirate). He votes

- **for** if the offer is *more* than that,
- **against** if it is *less* or *the same* (the tie-breaking rule).

So to buy a pirate's vote, the proposer must give him **strictly more** than he would get in the next round. That is why we must first know the next round — we reason **backward**, starting from the smallest game.

Step 1 — 1 pirate (P_5). He keeps everything: $P_5 = 100$.

Step 2 — 2 pirates (P_4, P_5). P_4 needs $\lceil 2/2 \rceil = 1$ vote — his own. So he takes everything:

$$P_4 = 100, \quad P_5 = 0.$$

Step 3 — 3 pirates (P_3, P_4, P_5). P_3 needs 2 votes: his own plus one more. If P_3 dies, the Step 2 outcome is $(P_4, P_5) = (100, 0)$. So:

- P_4 would need 101 coins — impossible;
- P_5 gets 0 if P_3 dies, so *one* coin buys his vote.

P_3 proposes $(P_3, P_4, P_5) = (99, 0, 1)$: P_3 and P_5 vote for, P_4 against — it passes 2–1.

Step 4 — 4 pirates ($P_2, \dots, P_5$). P_2 needs $\lceil 4/2 \rceil = 2$ votes: his own plus one. If P_2 dies, the Step 3 outcome is $(P_3, P_4, P_5) = (99, 0, 1)$. The prices of the votes are:

- P_3 : would need 100 — too expensive;
- P_4 : gets 0 otherwise, so 1 coin suffices;
- P_5 : gets 1 otherwise, so he would need 2.

P_2 buys the cheapest vote, P_4 's: $(P_2, P_3, P_4, P_5) = (99, 0, 1, 0)$. Votes: P_2 and P_4 for, P_3 and P_5 against — 2 of 4, exactly 50%, so it passes.

Step 5 — all 5 pirates. P_1 needs $\lceil 5/2 \rceil = 3$ votes: his own plus two. If P_1 dies, the Step 4 outcome is $(P_2, P_3, P_4, P_5) = (99, 0, 1, 0)$. The prices:

- P_2 : would need 100 — too expensive;
- P_3 : gets 0 otherwise, so 1 coin;
- P_4 : gets 1 otherwise, so 2 coins;
- P_5 : gets 0 otherwise, so 1 coin.

P_1 buys the two cheapest votes, P_3 and P_5 :

$$(P_1, P_2, P_3, P_4, P_5) = (98, 0, 1, 0, 1).$$

Votes: P_1, P_3, P_5 for; P_2, P_4 against — it passes 3–2.

The oldest pirate survives and proposes $(98, 0, 1, 0, 1)$ from oldest to youngest: he keeps 98 coins and gives 1 coin each to the third and fifth pirates.

The General Principle — Build Up from the Smallest Game

pirates	P_1	P_2	P_3	P_4	P_5	votes needed
1					100	1
2				100	0	1
3			99	0	1	2
4		99	0	1	0	2
5	98	0	1	0	1	3

1. **Start with the fewest players.** With one (or two) pirates the answer is obvious. Each larger game only needs the answer to the game one size smaller, because that is what happens if the current proposer is thrown overboard.
2. **Each voter's price is "next round +1".** A pirate votes yes only if the offer beats what he would get in the smaller game (strictly, because of the tie-breaking rule). So his vote costs (his next-round payoff) + 1.
3. **Buy the cheapest votes.** The proposer needs $\lceil n/2 \rceil$ votes, i.e. $\lceil n/2 \rceil - 1$ besides his own. The cheapest are the pirates who get *nothing* in the next round — each costs just 1 coin. Everyone else gets 0.
4. **Read off the pattern.** The pirates who get 0 in one round are exactly the ones bought for 1 coin in the next, so the 1-coin gifts **alternate**: they go to every other pirate counting from the proposer. The proposer keeps

$$100 - (\lceil n/2 \rceil - 1) \text{ coins.}$$

For $n = 5$: $100 - 2 = 98$. This keeps working until the bribes use up all 100 coins (roughly 200 pirates); beyond that the analysis changes because the proposer can no longer afford enough votes.

5. **Check the rules.** Small changes flip the answer: if a tie (exactly 50%) *fails*, the proposer needs one more vote; if indifferent pirates vote *for*, votes cost 0 instead of 1. Always redo the smallest cases under the stated rules.

Concept Summary and Applications

- **Backward induction:** solve the smallest version first; each bigger version depends only on the next smaller one.
- **Price of a vote = outside option +1:** a rational voter compares your offer with what he gets if your proposal fails.
- **Cheapest coalition:** buy exactly the votes you need from those with the worst outside option; give everyone else nothing.
- **Applications:** bargaining and coalition-building (legislatures, shareholder votes), ultimatum-style negotiations, and any sequential game where a failed offer passes the move to the next player.

40 The Kelly Criterion: How Much to Bet

Knowing that a bet is favourable tells you *whether* to bet; it does not tell you *how much*. Bet too little and you waste your edge; bet too much and a few losses can wipe you out. The **Kelly criterion** answers “how much” for a bettor who plays many times and reinvests the winnings.

Representative Problem — Betting on a Die (Q234)

Problem: What is the optimal way to bet on the outcomes of a six-sided die roll?

The question is deliberately open: “optimal” depends on what the bets *pay* and on what you are trying to maximise. A good answer sets up the payoff, finds when a bet is worth taking at all, and then sizes it.

Step 1 — the model. Each face $1, \dots, 6$ has probability $\frac{1}{6}$. You can bet on an event A (for example “the roll is a 4” or “the roll is even”) with probability $p = P(A)$. Staking 1 unit, you receive r units of *net profit* (“ r to 1”) if A happens and lose your 1 unit if it does not. The expected profit per unit staked is

$$\mathbb{E}[\text{profit}] = pr - (1 - p),$$

so the bet is favourable exactly when

$$pr - (1 - p) > 0 \iff r > \frac{1 - p}{p}.$$

(Equivalently, if $b = r + 1$ is the total return including your stake, the bet is favourable when $pb > 1$.)

Step 2 — fair odds. For a single face, $p = \frac{1}{6}$, so the *fair* payout is $r = \frac{1-p}{p} = 5$, i.e. 5 : 1. At exactly 5 : 1 the expected profit is

$$\frac{1}{6} \cdot 5 - \frac{5}{6} \cdot 1 = 0.$$

The same holds for any correctly priced bet on a fair die. Hence:

- at **fair odds**, every betting strategy has expected profit 0;
- at **worse than fair odds** (e.g. a casino paying 4 : 1 on a face), every strategy loses on average — the optimal bet is *no bet*;
- only if some bet pays **better than fair odds** is there anything to gain.

Step 3 — two different meanings of “optimal.”

- **One roll, maximise expected money.** Bet nothing if no bet has positive expected value; otherwise put *everything* on the bet with the highest expected return. (This is reckless over many rounds, as shown below.)
- **Many rolls, maximise long-run growth.** Bet a *fraction* of your current wealth each time — the Kelly fraction. For a bet with win probability p and net odds r ,

$$f^* = \frac{pr - (1 - p)}{r} \quad (\text{bet nothing if this is } \leq 0).$$

Step 4 — apply it to one face. With $p = \frac{1}{6}$,

$$f^* = \frac{\frac{r}{6} - \frac{5}{6}}{r} = \frac{r - 5}{6r}.$$

So: if $r < 5$, do not bet; if $r = 5$, the edge is zero, so do not bet; if $r > 5$, bet the fraction $\frac{r-5}{6r}$ of your current wealth on that face. For example, if a face paid 7 : 1, you would stake $\frac{7-5}{42} = \frac{1}{21} \approx 4.8\%$ of your wealth each roll.

There is no single “optimal” bet until the payouts are given. Bet only when the offered odds beat the fair odds implied by $\frac{1}{6}$ (for one face: net payout above 5 : 1); at fair or worse odds, don’t bet. When a bet is favourable and you play repeatedly, stake the Kelly fraction $f^* = \frac{pr - (1-p)}{r}$ of your wealth — $\frac{r-5}{6r}$ for a single face.

Kelly Criterion — The Setup

You start with wealth W_0 and face the same favourable bet over and over:

- it wins with probability p and loses with probability $q = 1 - p$;
- it pays net odds $b : 1$ — stake 1, win b more, or lose the 1;
- each round you bet a fixed *fraction* f of your *current* wealth.

After a win your wealth is multiplied by $(1 + bf)$; after a loss, by $(1 - f)$. After n rounds with k wins,

$$W_n = W_0 (1 + bf)^k (1 - f)^{n-k}.$$

Wealth *multiplies*; it does not add. That single fact drives everything below.

Why Maximise the Logarithm of Wealth

Maximising the average wealth is a trap. Take $p = 0.6$ at even money ($b = 1$). Betting *everything* ($f = 1$) every round maximises expected wealth: it grows like $E[W_n] = W_0 (1.2)^n$. But you survive n rounds only if you win all of them, with probability $0.6^n \rightarrow 0$. The average is huge only because of an astronomically rare lucky path; you are almost certainly broke.

What matters is the growth rate of a typical path. Take logarithms — they turn products into sums:

$$\ln \frac{W_n}{W_0} = k \ln(1 + bf) + (n - k) \ln(1 - f).$$

Divide by n . By the law of large numbers, $k/n \rightarrow p$, so

$$\frac{1}{n} \ln \frac{W_n}{W_0} \rightarrow G(f) = p \ln(1 + bf) + q \ln(1 - f) \quad (\text{with probability } 1).$$

So $W_n \approx W_0 e^{nG(f)}$: $G(f)$ is the **long-run growth rate per bet**. Maximising it is the same as maximising the expected *log* of wealth, $\mathbb{E}[\ln W_n] = \ln W_0 + nG(f)$. This is why Kelly uses **log utility**:

- it maximises the compound growth rate of a typical path — over enough rounds, the Kelly bettor ends up ahead of any other fixed-fraction strategy with probability approaching 1;
- $\ln W \rightarrow -\infty$ as $W \rightarrow 0$, so it never risks total ruin (it keeps $f < 1$ whenever a loss is possible);
- it has *diminishing marginal utility*: an extra dollar matters less when you are rich, which matches how most people and firms feel about money;
- it is scale-free: the right bet is a *fraction* of wealth, whatever the wealth.

Deriving the Kelly Fraction

Maximise $G(f) = p \ln(1 + bf) + q \ln(1 - f)$. Differentiate:

$$G'(f) = \frac{pb}{1 + bf} - \frac{q}{1 - f}.$$

Set $G'(f) = 0$ and cross-multiply:

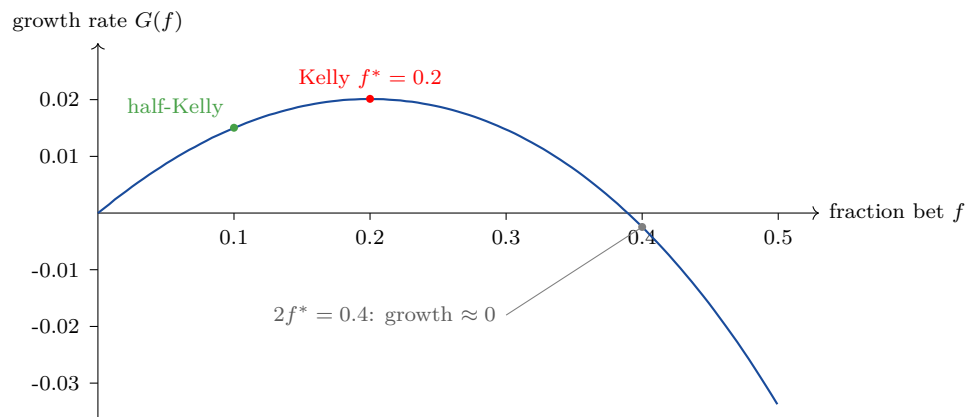
$$pb(1 - f) = q(1 + bf) \implies pb - pbf = q + qbf \implies pb - q = bf(p + q) = bf.$$

$$f^* = \frac{pb - q}{b} = p - \frac{q}{b} \quad \left(= \frac{\text{expected profit per unit}}{\text{profit if you win}} \right).$$

For even money ($b = 1$): $f^* = p - q = 2p - 1$, the size of your edge.

The second derivative $G''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{q}{(1-f)^2} < 0$, so G is concave and f^* is the maximum.

Example. $p = 0.6$, even money: $f^* = 0.6 - 0.4 = 0.20$. Betting 20% each round gives growth $G(0.2) = 0.6 \ln 1.2 + 0.4 \ln 0.8 \approx 0.0201$ per bet — about 2% compound growth per round.



Growth rate per bet for $p = 0.6$ at even money. It peaks at $f^* = 0.2$; half-Kelly (0.1) keeps 75% of the peak growth; betting $2f^* = 0.4$ gives roughly zero growth, and anything more makes your wealth shrink over time.

Intuition and Properties

- **Bet your edge over your odds.** $f^* = \frac{pb-q}{b}$: the numerator is the expected profit per unit bet (your edge); dividing by b scales it down when the payout is long (long odds $\Rightarrow$ more variance per unit bet).
- **No edge, no bet.** $f^* > 0$ exactly when $pb > q$, i.e. the bet has positive expected value. With no edge, Kelly says bet nothing, however tempting the odds.
- **Never bet it all.** $f^* < 1$ whenever losing is possible. On a fair coin ($p = \frac{1}{2}$), even at huge odds $f^* = \frac{1}{2} - \frac{1}{2b} < \frac{1}{2}$: you never stake more than half, because you still lose half the time.
- **Proportional betting.** After wins you bet more dollars, after losses fewer — the bet size automatically shrinks as you lose, which is what prevents ruin.
- **Overbetting is worse than underbetting.** Near the peak, G is roughly a parabola: betting $c f^*$ gives about $(2c - c^2)$ of the maximum growth. So half Kelly keeps 75% of the growth, while double Kelly ($c = 2$) gives about *zero* growth, and more than that gives negative growth even though every bet is favourable.
- **Continuous version.** For an investment with expected excess return μ and volatility σ , the Kelly fraction of wealth to hold is $f^* = \mu/\sigma^2$.

Risks, Limitations, and Why Fractional Kelly

Full Kelly maximises long-run growth, but the ride is rough and the result depends on inputs you rarely know exactly.

- **Large drawdowns.** Even at full Kelly, deep losing streaks are normal. In the continuous-time approximation, the chance that your wealth *ever* falls to half of what you started with is about 50%; to a third, about 33%. Many people and funds cannot tolerate that.
- **You don't know p exactly.** Kelly assumes the true probabilities. If you overestimate your edge, you overbet — and overbetting is costly (at twice the true f^* , growth is about zero). Estimation error makes the *effective* bet too big.
- **Long-run guarantee, short-run noise.** Kelly's optimality holds “in the long run”; over a short horizon, a Kelly bettor can easily do badly.
- **Model assumptions.** Independent, repeatable bets; infinitely divisible stakes; no transaction costs; known, stable odds; no fat tails. Real markets break some of these.
- **Log utility may not be yours.** Someone more risk-averse than log utility should bet less.

Fractional Kelly. Bet $f = c f^*$ with $0 < c < 1$ (commonly $c = \frac{1}{2}$).

- Growth: about $(2c - c^2)$ of the maximum — 75% at half Kelly.
- Volatility: scales with c — half Kelly has half the standard deviation (a quarter of the variance).
- Drawdowns: the chance of ever falling to a fraction x of your starting wealth drops from about x to about $x^{2/c-1}$ — for half Kelly, the chance of ever being halved falls from $\frac{1}{2}$ to $\frac{1}{8}$.
- Safety margin: it protects you if your estimate of p is too optimistic.

Giving up a quarter of the growth to halve the volatility and sharply cut the risk of big drawdowns is why practitioners usually bet half Kelly or less.

Concept Summary and Applications

- **First decide whether to bet:** only if the odds beat the fair odds ($pb > q$, or $r > \frac{1-p}{p}$).
- **Then decide how much:** maximise the growth rate $G(f) = p \ln(1 + bf) + q \ln(1 - f)$, giving $f^* = p - \frac{q}{b}$ ($= 2p - 1$ at even money).
- **Log utility** targets the growth of a typical wealth path, not the average (which a rare lucky path dominates) — and never risks ruin.
- **In practice, bet a fraction of Kelly:** half Kelly keeps 75% of the growth with far smaller drawdowns and protects against overestimating your edge.
- **Applications:** position sizing in trading and portfolio management, sports and casino betting with an edge, capital allocation across strategies, and interview questions of the form “you have an edge — how much do you bet?”

41 Self-Referential Logic Puzzles

A statement is **self-referential** when it talks about itself — for example, “this sentence is false.” Such statements can create a **paradox**: if the sentence is true, then it is false; if it is false, then it is true. Interviewers use self-reference to test whether you can step outside the usual “look up the fact” mode and reason carefully about what makes an answer right or wrong.

Representative Problem — Minimise the Correct Answers (Q17)

Problem: You may ask *one* true-or-false question to every American, and you want as *few* people as possible to answer correctly. What question do you ask?

Goal. The smallest possible number of correct answers is **zero**. Can one question *guarantee* zero, no matter how people respond? Only if *every* possible answer is wrong.

Step 1 — why an ordinary question fails. Any factual true/false question (“Is Canberra the capital of Australia?”) has a fixed correct answer. Even people who know nothing and simply guess get it right about half the time, so you should expect *at least* around 50% of guessers correct — nowhere near zero. To do better, the correct answer must *depend on what the person says*.

Step 2 — a question that makes every answer wrong. Ask:

“Will you answer *no* to this question?”

(In true/false form: “*True or false: you will answer ‘false’ to this question.*”) Each person answers, and we grade the answer against what they actually did:

- **If they answer “no”:** it is now a plain fact that they *did* answer “no,” so the truthful answer to “Will you answer no?” is *yes*. They said “no” — **wrong**.
- **If they answer “yes”:** they did *not* answer “no,” so the truthful answer is *no*. They said “yes” — **wrong**.

Whatever a person says, the correct answer is the opposite. So **every respondent is wrong** — a guaranteed 0 correct, with nothing to hope for or control about how people respond.

Step 3 — why this exact phrasing. The truth of “Will you answer no?” is simply a fact about the person’s own response, so each answer has a definite truth value and is definitely wrong: no paradox, nothing undefined.

Compare the tempting phrasing “*Will your answer to this question be wrong?*”

- Answering “true” (“yes, my answer will be wrong”) is a genuine *liar paradox*: it is correct exactly when it is wrong, so it has no consistent grade.
- Answering “false” (“no, my answer will not be wrong”) is *consistent either way* — it can be graded as correct.

So that version does *not* guarantee the minimum: anyone who answers “false” can be counted correct, and you cannot stop people from doing that. It reaches 0 only in the unrealistic case that *everyone* answers “true.” The “Will you answer no?” version closes both loopholes.

Ask “*Will you answer ‘no’ to this question?*” (equivalently, “*True or false: you will answer ‘false’ to this question?*”). Every possible response is wrong, so the number of people answering correctly is guaranteed to be 0 — the smallest possible.

The Key Idea — Self-Reference That Negates Itself

- **Make the truth depend on the answer.** A question about the respondent’s own answer has a correct answer that changes with what they say. That is the only way to beat the $\sim 50\%$ that random guessing gives on a factual question.
- **Make it negate the answer.** “Will you answer *no*?” is built so that each response makes the *opposite* response the true one. Every answer refutes itself.
- **Avoid genuine paradoxes.** A statement like “this sentence is false” (or “will your answer be wrong?”) has no consistent truth value on some branch, so it can’t be graded reliably. A good self-referential question has a *definite* truth value for every possible answer — here, always “wrong.”
- **Check every branch.** For any self-referential puzzle, list each possible response and ask “given that this was the answer, what is actually true?” — exactly as in Step 2.

Concept Summary and Applications

- **Self-reference** lets the correctness of an answer depend on the answer itself; designed as a negation, it forces every response to be wrong.
- **Liar-type statements** (“this is false”) are paradoxes with no consistent truth value; well-posed puzzles avoid them.
- **Method:** enumerate the possible answers and evaluate the truth of the question under each one.
- **Applications:** knights-and-knaves and liar puzzles, “what question would you ask the guard?” riddles, the halting problem and Gödel-style arguments in logic and computer science, and interview questions that test careful reasoning over recall.

42 De Bruijn Sequences

A **de Bruijn sequence** of order n over an alphabet of k symbols is a *cyclic* sequence in which every possible string of length n appears as a block of consecutive symbols **exactly once**. It is the shortest possible sequence that “contains every combination.” For binary strings ($k = 2$) of length $n = 5$ there are $2^5 = 32$ possible blocks, so the cyclic sequence has length 32.

Representative Problem — A Length-36 Binary Sequence (Q185)

Problem: Build a sequence of 36 digits, each 0 or 1, that starts with 00000 and in which every block of 5 consecutive digits is *different* from every other. (For example, 00000100001... fails, because the block 00001 appears twice.) How would you construct it?

Step 1 — count the blocks. A sequence of length 36 has

$$36 - 5 + 1 = 32$$

blocks of 5 consecutive digits (starting at positions 1, 2, ..., 32). There are exactly $2^5 = 32$ different binary strings of length 5. So “all 32 blocks distinct” means **every 5-bit string appears exactly once**. That is exactly a de Bruijn sequence of order 5: a cyclic sequence of length 32, written out linearly by repeating its first 4 digits at the end so that the blocks wrapping around are visible:

$$32 + 4 = 36.$$

Step 2 — turn it into a graph problem. Make a directed graph whose *vertices* are the 16 binary strings of length 4. From a vertex $a_1a_2a_3a_4$, draw two arrows: to $a_2a_3a_40$ and to $a_2a_3a_41$ (drop the first digit, append a new one). Each arrow corresponds to the 5-bit string $a_1a_2a_3a_4b$ — so

$$\text{vertices} = 4\text{-bit strings}, \quad \text{arrows} = 5\text{-bit strings}.$$

Walking along arrows and writing down the appended digit builds the sequence one digit at a time; the 5-bit block just completed is the arrow just used. A sequence using every 5-bit block exactly once is therefore a walk using *every arrow exactly once* — an **Eulerian cycle**. Every vertex has 2 arrows in and 2 arrows out (and the graph is connected), so by Euler’s theorem such a cycle always exists.

Step 3 — one explicit answer. A valid cyclic de Bruijn sequence of order 5 is

$$00000100011001010011101011011111 \quad (\text{length } 32).$$

Appending its first four digits 0000 gives a length-36 answer starting with 00000:

$$\boxed{000001000110010100111010110111110000}.$$

Its 32 blocks of length 5 are exactly the 32 binary strings, each once.

Constructing It by Hand — the Greedy “Prefer-1” Rule

You do not need to draw the graph. A simple greedy rule (Martin’s algorithm) produces a valid sequence directly:

1. Start with 00000.
2. Look at the *last 4 digits*. Try appending a **1**: if the new 5-digit block has not appeared before, keep it.
3. Otherwise append a **0** (if that block is new).
4. Stop when neither choice gives a new block — this happens after exactly 36 digits.

last 4 digits	append	new block
0000	1	00001
0001	1	00011
0011	1	00111
0111	1	01111
1111	1	11111
1111	0	11110 (11111 already used)
1110	1	11101
		⋮

Carrying this to the end gives

000001111101110011010110001010010000,

which also starts with 00000 and has all 32 blocks distinct. (De Bruijn sequences are *not* unique; there are $2^{11} = 2048$ different cyclic binary ones of order 5.)

Why the Greedy “Prefer-1” Rule Always Works

It is not obvious that such a short-sighted rule never paints itself into a corner. It does not, provided you **start from all zeros** (from other starting seeds it usually stalls early: from 11111 it covers only 6 of the 32 blocks). This is **Martin’s theorem** (1934). The proof uses the graph from Step 2 of the representative problem.

Set-up in graph language. Vertices are the 4-bit strings; the arrow vb leaves vertex v and appends bit b . Every vertex has exactly 2 arrows out ($v0$ and $v1$) and 2 arrows in. The rule is a walk that starts at vertex 0000 (the seed 00000 counts as having already used the loop arrow 00000) and, at each vertex v , takes arrow $v1$ if unused, else $v0$ if unused, else stops. Each arrow is a 5-bit block, and the walk never reuses an arrow — so the blocks are automatically distinct. What must be shown is that the walk does not stop until *all* 32 arrows are used.

Fact 1 — the walk can only stop at the start vertex 0000. Suppose the walk is at some vertex $u \neq 0000$. It got there by using an arrow *into* u , and every earlier visit to u also entered and then left. So the number of arrows used into u is one more than the number used out of u . Since u has only 2 arrows in, it has used at most 1 arrow out — so an unused arrow out of u remains, and the walk continues. Hence it can only get stuck back at 0000.

Fact 2 — at the end, every vertex is “balanced.” Each visit to a vertex enters once and leaves once, and the walk starts and ends at 0000. So when it stops, every vertex has used *the same number* of arrows in as arrows out.

Fact 3 — “prefer-1” means a used 0-arrow implies a used 1-arrow. The first time the walk leaves any vertex v , the arrow $v1$ is still unused (only a departure from v can use it), so the rule takes $v1$. Therefore, if $v0$ was ever used, $v1$ was used before it. Equivalently: *if v has any unused arrow out, then $v0$ is unused.*

Putting it together. Suppose, for contradiction, that when the walk stops some vertex $v = c_1c_2c_3c_4$ still has an unused arrow out.

1. By Fact 3, the arrow $v0 = c_1c_2c_3c_40$ is unused.
2. That arrow points *into* the vertex $w = c_2c_3c_40$ (drop the first digit, append 0). So w has an unused arrow in, and by Fact 2 (balance) it also has an unused arrow out.
3. Repeat with w : its 0-arrow is unused, so c_3c_400 has an unused arrow out; then c_4000 ; then 0000 .

After four steps every original digit has been pushed out and replaced by 0: the vertex 0000 has an unused arrow out. But by Fact 1 the walk stopped *at* 0000 , which means both of its arrows out were used — a contradiction. So no arrow is left unused: the walk uses all $2^5 = 32$ blocks, each exactly once, and the output has $32 + 4 = 36$ digits.

The same argument works for any length n (push in 0s $n - 1$ times) and for any alphabet of k symbols (start from all copies of the smallest symbol and always prefer the largest symbol that gives a new block).

Any binary de Bruijn sequence of order 5, written out with its first 4 digits repeated at the end, works. For example

000001000110010100111010110111110000

(from an Eulerian cycle) or 000001111101110011010110001010010000 (greedy “prefer-1”). Each has $36 - 5 + 1 = 32$ distinct blocks of length 5 — all 2^5 binary strings once.

Concept Summary and Applications

- **Counting first:** a sequence of length L has $L - n + 1$ windows of length n ; with k^n possible strings, the shortest sequence containing all of them has length $k^n + n - 1$ (cyclically, k^n).
- **Graph view:** vertices = $(n-1)$ -strings, arrows = n -strings; a de Bruijn sequence is an Eulerian cycle, which exists because every vertex has equal in- and out-degree.
- **Two constructions:** an Eulerian cycle in the de Bruijn graph, or the greedy “prefer-1” rule from all zeros (Martin’s theorem guarantees it never gets stuck early).
- **Applications:** cracking combination/keypad locks with the fewest presses, position encoding (reading any n consecutive marks tells you where you are), test-pattern generation, and DNA sequence assembly.

43 Recovering Five Numbers from Their Pairwise Sums

Representative Problem — Five Pumpkins (Q284)

Problem: There are 5 pumpkins. You are told the total weight of every pair of pumpkins — all $\binom{5}{2} = 10$ pairwise sums — but not which pair each sum came from. Determine the weight of each pumpkin.

The crux: the sums are unlabelled. If you knew which pair produced each sum, this would be an easy system of linear equations. You don't — you only have a list of ten numbers. (A formula like “pumpkin $i = \frac{S_i - T/4}{3}$ ” where S_i is the total of the four sums containing pumpkin i is true but useless, because you cannot tell which four sums those are.) The trick is to **sort** the sums and read the information hiding in the extremes.

Setup. Name the weights in increasing order, $a \leq b \leq c \leq d \leq e$, and sort the ten pairwise sums: $s_1 \leq s_2 \leq \dots \leq s_{10}$.

Step 1 — the grand total, by counting. Each pumpkin is paired with each of the other 4 pumpkins, so it appears in exactly 4 of the 10 sums. Adding all ten sums therefore counts every weight 4 times:

$$s_1 + s_2 + \dots + s_{10} = 4(a + b + c + d + e) \implies T := a + b + c + d + e = \frac{1}{4} \sum_{i=1}^{10} s_i.$$

Step 2 — four sums are identified for free. The extremes can only come from specific pairs:

- $s_1 = a + b$: the two lightest pumpkins give the smallest sum.
- $s_2 = a + c$: every other sum except $a + b$ is at least $a + c$ (for example $a + d \geq a + c$ and $b + c \geq a + c$), so $a + c$ is next smallest.
- $s_{10} = d + e$: the two heaviest give the largest sum.
- $s_9 = c + e$: symmetrically, every other sum except $d + e$ is at most $c + e$.

(The middle sums $s_3, \dots, s_8$ are ambiguous — e.g. $a + d$ and $b + c$ can come in either order — so we avoid them.)

Step 3 — solve, starting from the middle pumpkin. The sums $s_1 = a + b$ and $s_{10} = d + e$ together contain every pumpkin except c , so

$$c = T - (a + b) - (d + e) = T - s_1 - s_{10}.$$

Now peel off the rest using the identified sums:

$$a = s_2 - c, \quad b = s_1 - a, \quad e = s_9 - c, \quad d = s_{10} - e.$$

Worked example. Suppose the ten pairwise sums are $\{3, 4, 5, 5, 6, 6, 7, 7, 8, 9\}$.

- Total: $\sum s_i = 60$, so $T = \frac{60}{4} = 15$.
- Middle pumpkin: $c = 15 - s_1 - s_{10} = 15 - 3 - 9 = 3$.

- Then $a = s_2 - c = 4 - 3 = 1$, $b = s_1 - a = 3 - 1 = 2$, $e = s_9 - c = 8 - 3 = 5$, $d = s_{10} - e = 9 - 5 = 4$.

Check: the ten pairwise sums of $\{1, 2, 3, 4, 5\}$ are 3, 4, 5, 5, 6, 6, 7, 7, 8, 9 — exactly the given list.

Sort the ten sums $s_1 \leq \dots \leq s_{10}$. Compute the total $T = \frac{1}{4} \sum s_i$, then $c = T - s_1 - s_{10}$, $a = s_2 - c$, $b = s_1 - a$, $e = s_9 - c$, $d = s_{10} - e$. In the example the weights are 1, 2, 3, 4, 5.

Why It Works, and a Caveat

- **Counting gives the total.** With n numbers, each appears in $n - 1$ pairwise sums, so the total is $\frac{1}{n-1} \sum(\text{pairwise sums})$.
- **Sorting gives anchors.** The two smallest and two largest sums must be $a+b$, $a+c$, $c+e$, $d+e$ no matter what the weights are. Together with the total, these four equations pin down all five weights, with the middle weight c as the linchpin. (Checked by exact recovery on many random weight sets.)
- **Caveat — the answer is not always unique for other n .** For 5 numbers the list of pairwise sums determines the numbers. But for some sizes it does not: with 4 numbers, $\{1, 4, 6, 7\}$ and $\{2, 3, 5, 8\}$ both have pairwise sums $\{5, 7, 8, 10, 11, 13\}$. (A classical result: non-uniqueness can only happen when n is a power of 2.)

Concept Summary and Applications

- **Double counting:** summing all the given quantities and counting how often each unknown appears turns a messy list into one clean equation.
- **Extremes are informative:** when data are unlabelled, sorted extremes (the smallest and largest values) are often the only unambiguous pieces — anchor on them.
- **Solve from the linchpin:** find one unknown directly (here c), then unwind the rest.
- **Applications:** reconstruction puzzles (“weigh in pairs”), recovering positions from pairwise distances (the turnpike problem), and checksum-style reasoning in coding and data validation.

44 Secure Average Computation

A group wants a *collective* statistic (a total or an average) without anyone revealing their *individual* number. The trick is to hide each number behind random “masks” that cancel out only when everything is added together. This is the simplest example of **secure multi-party computation**.

Representative Problem — Eight Quants’ Average Salary (Q294)

Problem: Eight quants from different banks want to know the average salary of the group, but none wants to disclose their own salary to the others. Design a procedure that lets the group compute the average without revealing individual salaries.

Call the salaries $s_1, s_2, \dots, s_8$. The goal is

$$\frac{s_1 + s_2 + \dots + s_8}{8}$$

without anyone learning any single s_i . Since the average is just the total divided by 8, it is enough to compute the **total** privately.

Method 1 — the random-offset chain.

1. Quant 1 secretly picks a large random number R and tells no one.
2. She adds her salary and passes $R + s_1$ to Quant 2.
3. Each quant in turn adds their own salary and passes the running total on:

$$R \rightarrow R + s_1 \rightarrow R + s_1 + s_2 \rightarrow \dots \rightarrow R + \sum_{i=1}^8 s_i.$$

4. The final total returns to Quant 1, who subtracts her secret R to get $\sum_i s_i$, divides by 8, and announces the average.

Example (salaries in \$k): say $R = 5,317$ and the salaries are 200, 150, 300, 180, 250, 220, 170, 130 (total 1,600). Quant 2 receives 5,517, Quant 3 receives 5,667, ..., and Quant 1 finally receives 6,917; subtracting R gives 1,600, so the average is 200. Every number passed around looks like an arbitrary large number to its receiver.

Method 2 — circulating masks (symmetric). This version has no single “master” holding the secret offset.

1. **Sit in a circle**, labelled $1, 2, \dots, 8$ (so the neighbour after 8 is 1).
2. **Each quant picks a private random number** r_i .
3. **Pass it on privately:** quant i secretly sends r_i to quant $i + 1$. Now each quant knows their salary s_i , their own r_i , and the r_{i-1} received from the previous person.
4. **Announce a masked value publicly:**

$$a_i = s_i + r_i - r_{i-1} \quad (\text{indices mod } 8).$$

5. **Add up the announcements.**

$$\sum_{i=1}^8 a_i = \sum_{i=1}^8 s_i + \sum_{i=1}^8 r_i - \sum_{i=1}^8 r_{i-1}.$$

The last two sums contain the *same eight numbers* $r_1, \dots, r_8$ (just listed in a different order), so they cancel:

$$\sum_{i=1}^8 a_i = \sum_{i=1}^8 s_i, \quad \text{average} = \frac{1}{8} \sum_{i=1}^8 a_i.$$

Each announcement a_i is a salary plus one random number minus another — on its own it says nothing about s_i . Only the grand total is meaningful, because only there do the masks cancel.

Either method gives the exact total, hence the average, while hiding each salary from any single curious participant. Method 1: pass a secret random offset R around the circle, each person adding their salary; the originator subtracts R . Method 2: each quant privately passes a random r_i to the next and publicly announces $s_i + r_i - r_{i-1}$; summing the announcements cancels all masks.

Why the Methods Preserve Privacy

- **Everything a person sees is masked.** In Method 1, everyone except Quant 1 sees only $R + (\text{some salaries})$ with R unknown to them, so the number they receive is consistent with *any* set of earlier salaries. Quant 1 knows R but only ever sees the final grand total, never a partial sum. In Method 2, unmasking a_i requires *both* r_i and r_{i-1} , and each of those is known to only one other person.
- **Masks cancel only in the total.** The random numbers are designed so that they disappear exactly when all contributions are combined, and nowhere else.
- **Making the hiding exact.** “Large random number” hides salaries well but not perfectly (a huge received value slightly hints at large salaries). The clean fix is to do all arithmetic modulo a big number M (larger than any possible total) and choose R (or each r_i) uniformly from $0, 1, \dots, M - 1$. Then every value a person sees is *uniformly random*, independent of the salaries — perfect secrecy against a single observer.

The Limits

- **Two neighbours can collude.** In Method 1, the person before quant i knows the total they sent, $R + s_1 + \dots + s_{i-1}$, and the person after knows the total they received, $R + s_1 + \dots + s_i$; the difference is exactly s_i . In Method 2, quant $i - 1$ chose r_{i-1} and quant $i + 1$ received r_i , so together they know both masks and compute $s_i = a_i - r_i + r_{i-1}$. Both methods are safe only against a *single* curious participant.
- **Honest-but-curious only.** The methods assume everyone follows the steps but may try to learn more. A cheater who adds a wrong number corrupts the answer, and nobody can detect it.
- **The answer itself leaks something.** Once the average is announced, each quant can compute the *sum of everyone else's* salaries (total minus their own). With only two people, the average reveals the other person's salary completely — no protocol can prevent what the output itself reveals.

- **Collusion needs a stronger scheme** — additive secret sharing, below.

The Stronger Fix — Additive Secret Sharing

With n participants (here $n = 8$) and salaries $s_1, \dots, s_n$, fix a modulus M larger than any possible total (e.g. $M = 10^9$ dollars). All arithmetic is done “mod M ” — keep only the remainder after dividing by M , like a clock that wraps around.

Step 1 — each quant splits their salary into n random shares. Quant i picks $n - 1$ numbers $x_{i,1}, \dots, x_{i,n-1}$ uniformly at random from $\{0, 1, \dots, M - 1\}$, and sets the last share so that all n shares add up to the salary:

$$x_{i,n} = s_i - \sum_{j=1}^{n-1} x_{i,j} \pmod{M}, \quad \text{so} \quad \sum_{j=1}^n x_{i,j} \equiv s_i \pmod{M}.$$

Step 2 — hand out the shares. Quant i privately sends share $x_{i,j}$ to quant j (and keeps $x_{i,i}$). After this, quant j holds one share from everybody: the column $x_{1,j}, x_{2,j}, \dots, x_{n,j}$.

Step 3 — each quant announces the sum of what they received.

$$y_j = \sum_{i=1}^n x_{i,j} \pmod{M}.$$

Step 4 — add the announcements. Summing over j and swapping the order of summation,

$$\sum_{j=1}^n y_j \equiv \sum_{j=1}^n \sum_{i=1}^n x_{i,j} = \sum_{i=1}^n \underbrace{\sum_{j=1}^n x_{i,j}}_{\equiv s_i} \equiv \sum_{i=1}^n s_i \pmod{M}.$$

Because M exceeds the true total, the remainder *is* the total, and the average is $\frac{1}{n} \sum_j y_j$. Picture an $n \times n$ grid of shares: row i adds up to salary s_i , column j is what quant j receives and announces; the grand total of the grid is the same whether you add rows or columns.

Small example ($n = 3$, $M = 10,000$, salaries 200, 150, 300).

	to 1	to 2	to 3	row sum mod M
Quant 1 ($s_1 = 200$)	4213	7650	8337	$20200 \equiv 200$
Quant 2 ($s_2 = 150$)	9005	1234	9911	$20150 \equiv 150$
Quant 3 ($s_3 = 300$)	5500	6789	8011	$20300 \equiv 300$
announced y_j	8718	5673	6259	

(Each quant chose the first two shares at random; the third makes the row add up to the salary, e.g. $200 - 4213 - 7650 = -11663 \equiv 8337$.) The announcements add to $8718 + 5673 + 6259 = 20650 \equiv 650 = 200 + 150 + 300$, so the average is $650/3 \approx 216.7$ — yet every number anyone saw looks random.

Why it resists collusion. Any $n - 1$ of the n shares of a salary are independent and uniformly random; only *all* n together determine it. So a coalition learns something about quant i 's salary only by collecting all of i 's shares — including the one i keeps, which leaks only through i 's announcement y_i , and extracting it from y_i requires knowing every other share in column i . In short, a coalition of colluders learns nothing beyond the sum of the *non-colluders*' salaries — which

the published total reveals anyway. Only when all $n - 1$ others collude is s_i exposed, and at that point the average alone already gives it away.

The price. Method 1 sends n messages; secret sharing sends $n(n - 1)$ private shares (56 for eight quants). It still assumes people follow the rules — it protects against curious coalitions, not against a cheater who hands out wrong shares.

Concept Summary and Applications

- **Compute the total, not the parts:** an average only needs the sum, and a sum can be built from masked pieces.
- **Random masks that cancel:** add random numbers that sum to zero across the group (or are removed by their owner), so individual values are hidden and the total survives.
- **Know the threat model:** single curious party vs. colluding groups vs. cheaters; simple masking handles the first, secret sharing the second, verification the third.
- **Applications:** salary and pay-gap surveys, secure voting and polling, privacy-preserving aggregation of data (e.g. federated learning), and auctions or benchmarks among competitors.

45 Information-Theoretic Guessing: Spending One Bit Well

A yes/no answer carries at most **one bit** of information. These puzzles ask how to spend that single bit. The best question depends entirely on how you are scored, so the first step is always to write the payoff down precisely.

Representative Problem — Guess 100 Coins with One Question (Q372)

Problem: A coin is flipped 100 times. You may ask one yes/no question about the sequence of results, and then you must guess the entire sequence. You earn \$1 for each coin you guess correctly and lose \$1 for each coin you guess wrong. Find a good strategy and compute its expected return.

Step 1 — write the payoff in terms of the number correct. The hidden sequence is $X = (X_1, \dots, X_{100})$, all 2^{100} sequences equally likely. Since $(\# \text{correct}) + (\# \text{wrong}) = 100$,

$$\text{return} = (\# \text{correct}) - (\# \text{wrong}) = 2(\# \text{correct}) - 100.$$

Where the factor 2 comes from: the payoff is a $+1/-1$ spread, so turning one wrong guess into a right one both gains a dollar and avoids losing one — a swing of \$2. With no information every coin is a fair 50/50 guess, so you expect 50 correct and a return of \$0. The game is about using one bit to push the expected number correct above 50.

Step 2 — the strategy. Let H be the number of heads and $T = 100 - H$ the number of tails. Ask:

“Are there at least 50 heads?”

Guess **all heads** if the answer is yes, **all tails** if no. You then get exactly the majority colour right, so $\# \text{correct} = \max(H, T)$. Because $\max(H, T) = 50 + |H - 50|$,

$$\text{return} = 2 \max(H, T) - 100 = 2 |H - 50| \geq 0.$$

The one bit tells you which side is the majority, and you collect twice how far the flips strayed from a perfect 50–50 split. You can never lose money.

***The trap — the parity question is worthless here.** “Is the number of heads even?” sounds clever, but knowing the total parity leaves every individual coin still exactly 50/50 (fixing one coin and demanding a given overall parity only constrains the other 99). So no coin can be guessed better than even odds and the expected return stays \$0. The bit must bias the coins themselves — which the majority question does.*

Step 3 — the distribution of H (binomial). Each flip is heads with probability $\frac{1}{2}$, independently. A particular sequence with exactly k heads has probability $(\frac{1}{2})^{100}$, and the number of such sequences is the number of ways to choose which k of the 100 positions are heads, $\binom{100}{k}$. Hence

$$P(H = k) = \binom{100}{k} \left(\frac{1}{2}\right)^{100} = \frac{\binom{100}{k}}{2^{100}}, \quad k = 0, 1, \dots, 100,$$

i.e. $H \sim \text{Binomial}(100, \frac{1}{2})$. Writing H as a sum of 100 independent coin indicators (each with mean $\frac{1}{2}$ and variance $\frac{1}{4}$) gives mean $\mu = 50$ and variance $\sigma^2 = 100 \cdot \frac{1}{4} = 25$, so $\sigma = 5$. The expected return is

$$\mathbb{E}[\text{return}] = 2 \mathbb{E}|H - 50| = \frac{2}{2^{100}} \sum_{k=0}^{100} \binom{100}{k} |k - 50|.$$

Step 4 — a closed form for the sum. Write $100 = 2N$ with $N = 50$. We show

$$\sum_{k=0}^{2N} \binom{2N}{k} |k - N| = N \binom{2N}{N}.$$

Use symmetry. Since $\binom{2N}{k} = \binom{2N}{2N-k}$, the terms for k and $2N - k$ are equal, and the $k = N$ term is 0. So the sum is twice the upper half:

$$S := \sum_{k=0}^{2N} \binom{2N}{k} |k - N| = 2 \sum_{k=N+1}^{2N} \binom{2N}{k} (k - N), \quad \frac{S}{2} = \sum_{k=N+1}^{2N} k \binom{2N}{k} - N \sum_{k=N+1}^{2N} \binom{2N}{k}.$$

First piece. The “absorption” identity $k \binom{2N}{k} = 2N \binom{2N-1}{k-1}$ (choosing a team of k with a captain) gives

$$\sum_{k=N+1}^{2N} k \binom{2N}{k} = 2N \sum_{j=N}^{2N-1} \binom{2N-1}{j}.$$

Row $2N - 1$ is odd, so its coefficients pair up as mirror images with no middle term, and the upper half is exactly half the total 2^{2N-1} : $\sum_{j=N}^{2N-1} \binom{2N-1}{j} = 2^{2N-2}$. So this piece is $2N \cdot 2^{2N-2} = N 2^{2N-1}$.

Second piece. By symmetry the sum above the middle is half of everything except the middle term: $\sum_{k=N+1}^{2N} \binom{2N}{k} = \frac{1}{2}(2^{2N} - \binom{2N}{N})$, so

$$N \sum_{k=N+1}^{2N} \binom{2N}{k} = N 2^{2N-1} - \frac{N}{2} \binom{2N}{N}.$$

Subtract. The $N 2^{2N-1}$ terms cancel:

$$\frac{S}{2} = \frac{N}{2} \binom{2N}{N} \implies S = N \binom{2N}{N}.$$

Therefore

$$\mathbb{E}|H - N| = \frac{N \binom{2N}{N}}{2^{2N}}, \quad \mathbb{E}[\text{return}] = \frac{2N \binom{2N}{N}}{2^{2N}} = \frac{100 \binom{100}{50}}{2^{100}} \approx \$7.96.$$

(Sanity check at $N = 1$, two coins: $\frac{1 \cdot \binom{2}{1}}{4} = \frac{1}{2}$, and indeed $|H - 1|$ is 1 with probability $\frac{1}{2}$.)

Step 5 — the normal approximation and the $\sqrt{2/\pi}$ factor. H is a sum of 100 independent fair bits, so by the central limit theorem it is approximately normal, $H \approx \mathcal{N}(50, 5^2)$. Let $Z = H - 50 \approx \mathcal{N}(0, \sigma^2)$ with density $\varphi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2\sigma^2}$.

Why the integral runs from 0 to ∞ . By definition $\mathbb{E}|Z| = \int_{-\infty}^{\infty} |z| \varphi(z) dz$. The absolute value is awkward, but the density is *symmetric* ($\varphi(-z) = \varphi(z)$) and $|z|$ is symmetric too, so the left half ($z < 0$) contributes exactly the same as the right half ($z > 0$). On the right half $|z| = z$, so

$$\mathbb{E}|Z| = 2 \int_0^{\infty} z \varphi(z) dz = 2 \int_0^{\infty} \frac{z}{\sigma\sqrt{2\pi}} e^{-z^2/2\sigma^2} dz.$$

Substitute $u = \frac{z^2}{2\sigma^2}$, so $du = \frac{z}{\sigma^2} dz$, i.e. $z dz = \sigma^2 du$:

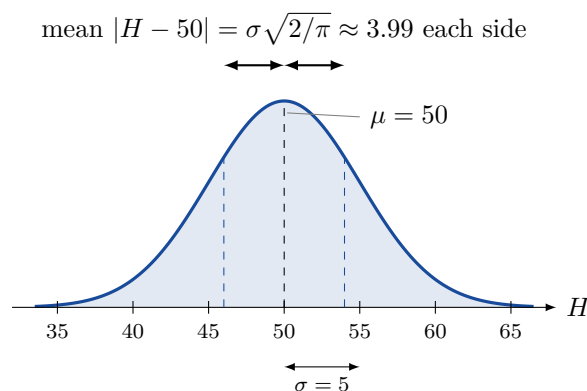
$$\mathbb{E}|Z| = \frac{2}{\sigma\sqrt{2\pi}} \sigma^2 \int_0^{\infty} e^{-u} du = \frac{2\sigma}{\sqrt{2\pi}} = \sigma \sqrt{\frac{2}{\pi}}.$$

The universal ratio. For *any* normal distribution, the mean absolute deviation is a fixed fraction of the standard deviation:

$$\frac{\mathbb{E}|X - \mu|}{\sigma} = \sqrt{\frac{2}{\pi}} \approx 0.798.$$

So “average distance from the mean” $\approx 0.8 \times$ “standard deviation” — a handy mental-math constant.

Here $\mathbb{E}|H - 50| \approx 5\sqrt{2/\pi} \approx 3.99$, so the return is about $2 \times 3.99 \approx \$8$ — matching the exact \$7.96 (the tiny gap, 3.979 exact vs. 3.989 normal, is only the discreteness of H). Stirling’s formula $\binom{2N}{N}/2^{2N} \approx 1/\sqrt{\pi N}$ shows the exact value $N\binom{2N}{N}/2^{2N} \approx \sqrt{N/\pi}$ is the same expression as $\sigma\sqrt{2/\pi}$ with $\sigma = \sqrt{N/2}$.



The mean absolute deviation (arrows) sits at $\sigma\sqrt{2/\pi} \approx 3.99$, a bit inside one standard deviation $\sigma = 5$. The expected return is twice this.

Ask “Are there at least 50 heads?” and guess every coin as the majority colour. The expected return is

$$\frac{100\binom{100}{50}}{2^{100}} = 2\mathbb{E}|H - 50| \approx \$7.96$$

(about 53.98 coins correct on average). *Mental shortcut:* $\sigma = 5$ and $\mathbb{E}|H - 50| \approx 0.8\sigma = 4$, so the return is about \$8.

Contrast — When Only an Exact Match Counts (Q293, Q355)

Change the scoring: now you win only if you guess *all* 100 coins exactly right. A yes/no question splits the 2^{100} equally likely sequences into a “yes” group and a “no” group, and after the answer you can guess just *one* sequence. So at most two sequences (one per answer) can ever win:

$$P(\text{win}) \leq \frac{2}{2^{100}} = \frac{1}{2^{99}},$$

and *any* non-trivial question — for example “Is the first flip heads?” — achieves it. One bit can at most **double** your chance of an exact hit. Under this scoring the majority question is no better

than any other; under per-coin scoring it is by far the best. Same puzzle, different payoff, different answer.

Concept Summary and Applications

- **Write the payoff first:** “exact match” and “count of correct pieces” call for completely different questions.
- **One bit at most doubles an exact-match probability;** for per-item scoring, use the bit to *bias many items at once* (the majority question), not to learn a global property that leaves each item 50/50 (parity).
- **Binomial mean absolute deviation:** for $2N$ fair coins, $\mathbb{E}|H - N| = N \binom{2N}{N} / 2^{2N} \approx \sqrt{N/\pi}$.
- **Normal rule of thumb:** $\mathbb{E}|X - \mu| = \sigma \sqrt{2/\pi} \approx 0.8 \sigma$ for any normal.
- **Applications:** designing informative questions/experiments, 20-questions and search strategies, compression and coding (bits of information), and quick estimates of average absolute error.

46 Rope-Burning Timer Puzzles

Representative Problem — Measure 45 Minutes with Two Ropes (Q302)

Problem: You have two ropes. Each takes exactly 1 hour to burn from one end to the other, but they burn at *non-uniform* rates along their length (some stretches burn fast, others slowly). Using only the ropes and a lighter, how do you measure exactly 45 minutes?

Why the obvious idea fails. You cannot cut a rope in half and expect 30 minutes: because the burn rate varies, half the *length* can take anywhere from almost 0 to almost 60 minutes. The only thing you know for sure is *total* burn time. So the solution must use only that fact.

The key trick — light both ends. A rope lit at *both* ends burns out in exactly **half** its remaining burn time, however uneven it is.

Why. Measure positions along the rope by “burn time from the left end”: let $T(x)$ be the time a single flame from the left takes to reach point x , so T runs from 0 at the left end to 60 at the right end. A flame lit at the right end reaches x after $60 - T(x)$ minutes (it has to burn everything the left flame hasn’t). The two flames meet at the point where they arrive at the same moment:

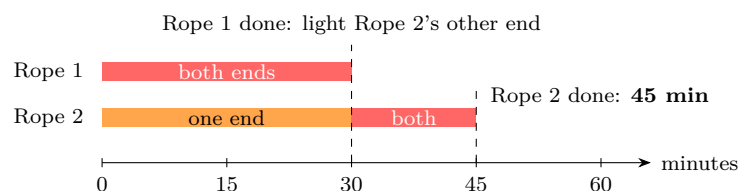
$$T(x) = 60 - T(x) \implies T(x) = 30.$$

So they meet after exactly 30 minutes — at an unknown *place* (not necessarily the midpoint), but at a known *time*. The same argument applies to any piece of rope: if the remaining piece would take R minutes from one end, lighting both ends finishes it in $R/2$.

The solution.

1. At time 0, light **Rope 1 at both ends** and **Rope 2 at one end**.
2. Rope 1 burns out after exactly 30 minutes (by the trick above).
3. At that instant, Rope 2 has burned for 30 minutes from one end. You don’t know *how much length* is left, but you do know the rest would take exactly $60 - 30 = 30$ more minutes from one end. Now **light the other end of Rope 2**.
4. Burning from both ends, the remaining 30 minutes of rope finishes in $30/2 = 15$ minutes.

When Rope 2 burns out, $30 + 15 = 45$ minutes have passed.



Light Rope 1 at both ends and Rope 2 at one end. When Rope 1 burns out (30 min), light Rope 2’s other end. Rope 2 burns out 15 minutes later: exactly 45 minutes in total.

The General Principle

- **Only burn *time* is reliable, never length.** Non-uniform ropes make length useless; reason purely about remaining burn time.
- **Both ends halves the remaining time.** A piece with R minutes left, lit at both ends, lasts $R/2$ — the two flames always meet at the time midpoint.
- **Chain the events.** One rope burning out is a clock tick you can use to light the next end. With two ropes you can measure, for example, 15 min (the last stretch above, from minute 30 to 45), 30, 45, 60, and 90 (Rope 1 from one end, then Rope 2 from both ends).
- **More ropes, finer times.** Each extra rope lets you halve a remaining time again: with three ropes you can reach $52\frac{1}{2}$ minutes ($30 + 15 + 7\frac{1}{2}$). Light Rope 3 at one end at time 0 too; at minute 45 it has 15 minutes left, so lighting its other end then finishes it $7\frac{1}{2}$ minutes later.

Concept Summary and Applications

- **Find the invariant:** when details (the burn profile) are unknown, work only with quantities that are fixed regardless (total burn time).
- **Symmetry trick:** two fronts moving toward each other meet at the midpoint of *time*, not of distance.
- **Applications:** lateral-thinking interview puzzles, measuring with unreliable instruments, and more generally designing procedures that are robust to unknown non-uniformity.

47 Weighing with a Balance Scale

A two-pan balance does not show a number — it only tells you *which side is heavier*, or that they balance. That makes it a **three-outcome** device (left heavy / balance / right heavy) and, when you may put objects on both pans, a **three-choice** device for each known weight (this pan / other pan / not used). Both problems below come down to counting with powers of 3.

Representative Problem — Which Weights Must Be Known? (Q296)

Problem: You have 40 balls weighing 1 g, 2 g, ..., 40 g (one of each), but you don't know which ball is which. You also have a fair two-pan balance. You are told the weights of a few of the balls in advance, and you will use *those known balls as reference weights*: a ball is “measured” when you find an arrangement in which it balances a combination of known balls (known balls may go on either pan). What is the minimum number of balls whose weights you need to know in advance so that every other ball can be measured?

Step 1 — what one weighing can express. Put the unknown ball on the left pan. Each known ball a_i can go on the *right* pan (it counts $+a_i$), on the *left* pan with the unknown (it counts $-a_i$), or be left off (0). The scale balances when

$$m = \sum_{i=1}^k \varepsilon_i a_i, \quad \varepsilon_i \in \{-1, 0, +1\}.$$

So with k known balls there are at most 3^k sign patterns $(\varepsilon_1, \dots, \varepsilon_k)$.

Step 2 — why only $\frac{3^k-1}{2}$ positive weights. Two adjustments:

- The all-zero pattern (every known ball set aside) nets 0 and measures nothing: $3^k - 1$ patterns remain.
- The rest come in **mirror pairs**: flipping every sign (swapping the two pans) turns a pattern netting $+v$ into one netting $-v$. They describe the same weighing, so each positive weight is counted twice.

Hence at most

$$\frac{3^k - 1}{2}$$

distinct positive weights can be measured. *Example* ($k = 2$, known balls 1 and 3): the 9 patterns give $0, \pm 1, \pm 2, \pm 3, \pm 4$ (e.g. $2 = 3 - 1$: unknown + 1 g balances 3 g), i.e. exactly $\frac{9-1}{2} = 4$ weights $\{1, 2, 3, 4\}$.

ball 1	ball 3	net
0	0	0
+	0	+1
−	0	−1
0	+	+3
0	−	−3
+	+	+4
−	−	−4
−	+	+2
+	−	−2

Step 3 — the lower bound. We need all 40 weights, so

$$\frac{3^k - 1}{2} \geq 40 : \quad k = 3 \Rightarrow \frac{27-1}{2} = 13 < 40, \quad k = 4 \Rightarrow \frac{81-1}{2} = 40.$$

At least 4 known balls are necessary.

Step 4 — four suffice: 1, 3, 9, 27 (**balanced ternary**). Every integer from 1 to 40 can be written as

$$m = \varepsilon_0 \cdot 1 + \varepsilon_1 \cdot 3 + \varepsilon_2 \cdot 9 + \varepsilon_3 \cdot 27, \quad \varepsilon_i \in \{-1, 0, 1\}$$

(“balanced ternary” — base 3 with digits $-1, 0, 1$). For example

$$2 = 3 - 1, \quad 5 = 9 - 3 - 1, \quad 8 = 9 - 1, \quad 20 = 27 - 9 + 3 - 1, \quad 40 = 27 + 9 + 3 + 1.$$

So knowing the 1, 3, 9 and 27 g balls lets you measure every ball.

The minimum is 4: know the balls of 1, 3, 9, 27 g. With k reference weights at most $\frac{3^k-1}{2}$ values are measurable, which reaches 40 only at $k = 4$, and 1, 3, 9, 27 achieve every value by balanced ternary.

The answer depends on the rules. If a ball, once measured, may be reused as a new reference, far fewer are needed: knowing 1 g and 3 g lets you find 2 and 4, then $\{1, 2, 3, 4\}$ reaches everything up to 10, and so on — 2 known balls suffice. And if you are told the set is exactly $\{1, \dots, 40\}$, you could simply sort the balls by pairwise comparisons and need to know nothing. State your interpretation; the standard (Bachet) answer is 4.

Additional Problem — 12 Coins, One Counterfeit

Problem: You have 12 coins. One is counterfeit and is either *heavier* or *lighter* than the others — you don’t know which. Using a balance scale only 3 times, find the fake coin *and* whether it is heavy or light.

Why 3 weighings can be enough (counting). Each weighing has 3 outcomes, so 3 weighings produce at most $3^3 = 27$ different outcome sequences. There are $12 \times 2 = 24$ possibilities to tell apart (which coin, and heavy or light). Since $24 \leq 27$, three weighings are *not ruled out*. (Counting only gives a necessary condition — 14 coins, with 28 possibilities, is impossible — so a concrete scheme is still needed.) The trick is to design weighings whose outcomes split the remaining possibilities as *evenly* as possible; each weighing should leave at most 9 possibilities, then at most 3, then 1.

Why the first weighing is 4 vs 4. Put k coins on each pan. If it tips, the fake is one of the $2k$ coins on the scale, each with a known direction: $2k$ possibilities. If it balances, it is one of the $12 - 2k$ off-scale coins, heavy or light: $2(12 - 2k)$ possibilities. Both must be ≤ 9 , so $2k \leq 9$ and $24 - 4k \leq 9$, giving $k = 4$ exactly (leaving 8 and 8).

A concrete scheme (label the coins 1–12).

Weighing 1: $\{1, 2, 3, 4\}$ vs $\{5, 6, 7, 8\}$.

- **Balances:** coins 1–8 are genuine; the fake is in $\{9, 10, 11, 12\}$, direction unknown (8 possibilities).

- **Unbalanced**, say the *left* pan is heavier (the right-heavy case is the mirror image): either one of $\{1, 2, 3, 4\}$ is *heavy* or one of $\{5, 6, 7, 8\}$ is *light* (8 possibilities). Coins 9–12 are now known to be genuine.

Balance case (fake in $\{9, 10, 11, 12\}$).

- *Weighing 2*: $\{9, 10, 11\}$ vs $\{1, 2, 3\}$ (known genuine).
- If it balances, the fake is 12; *weighing 3*: 12 vs a genuine coin shows whether it is heavy or light.
- If $\{9, 10, 11\}$ is heavier, one of them is *heavy*; if lighter, one of them is *light* — 3 suspects with a known direction. *Weighing 3*: 9 vs 10; the one tipping the expected way is the fake, and if they balance it is 11.

Imbalance case (left was heavy: 1, 2, 3, 4 maybe-heavy; 5, 6, 7, 8 maybe-light).

- *Weighing 2*: $\{1, 2, 5\}$ vs $\{3, 4, 6\}$, setting 7, 8 aside. This deliberately *mixes* the groups, so every remaining possibility pushes the scale a predictable way.
- **Left heavy**: the only consistent causes are a heavy 1 or 2 (on the left) or a light 6 (on the right). A light 5 would make the left *lighter*, and a heavy 3 or 4 would make the right heavier. Suspects $\{1H, 2H, 6L\}$. *Weighing 3*: 1 vs 2 — the heavier is the fake; if they balance, 6 is light.
- **Right heavy**: causes are a heavy 3 or 4 or a light 5. *Weighing 3*: 3 vs 4 — the heavier is the fake; if they balance, 5 is light.
- **Balances**: none of 1–6 is fake, so it is 7 or 8, both maybe-light. *Weighing 3*: 7 vs a genuine coin — if 7 is lighter it is the fake; otherwise 8 is the light fake.

Weigh $\{1, 2, 3, 4\}$ vs $\{5, 6, 7, 8\}$, then follow the branches above: every one of the 24 possibilities (coin and direction) is identified in exactly 3 weighings. (Checked by running all 24 cases.)

How far this goes. With 3 weighings the most coins you can handle while also naming the direction is 12; with one extra coin known to be genuine you can handle 13 ($26 \leq 27$ possibilities); 14 coins ($28 > 27$) is impossible. In general, w weighings handle $\frac{3^w-3}{2}$ coins, or $\frac{3^w-1}{2}$ with a spare genuine coin.

What a “spare genuine coin” is, and why it helps. It is an extra coin you are told is real, given in addition to the suspects. It can never be the fake; its only use is as a known-good filler on a pan. A balance compares equal numbers of coins, so without a spare every weighing is k suspects vs k suspects. With 13 suspects that fails already at the first weighing, which must leave at most 9 possibilities for the last two weighings:

- 4 vs 4: if it tips, 8 possibilities remain; if it balances, the 5 unweighed coins remain, each heavy or light — 10 possibilities (too many).
- 5 vs 5: if it tips, 10 possibilities remain (too many).

With a spare, weigh 5 suspects vs 4 suspects + the genuine coin. If it tips, 9 possibilities remain (the 9 suspects on the scale, each with a known direction); if it balances, the 4 unweighed coins give 8. Both are at most 9, so the remaining two weighings can finish — which is why the spare raises the limit from 12 to 13 coins.

Concept Summary and Applications

- **A balance is ternary:** each weighing has 3 outcomes, and each reference weight has 3 placements. Count with powers of 3.
- **Information bound:** w weighings distinguish at most 3^w cases; k reference weights measure at most $\frac{3^k - 1}{2}$ positive values.
- **Balanced ternary** (digits $-1, 0, 1$) shows $1, 3, 9, 27, \dots$ achieve the bound.
- **Design for even splits:** choose each weighing so that the three outcomes leave roughly equal numbers of possibilities; mixing suspects across pans is what makes the counterfeit puzzle work.
- **Applications:** optimal search and group testing, coin/weight puzzles, balanced-ternary number systems, and designing tests with few measurements.

48 The Airplane (Train) Seating Problem

Representative Problem — Will the Last Passenger Get Their Seat? (Q384)

Problem: A train (or plane) has 100 seats and is fully booked; passenger i is assigned seat i . Passengers board in order $1, 2, \dots, 100$. The first passenger has lost their ticket and sits in a *random* seat. Every later passenger sits in their own seat if it is free; otherwise they pick one of the remaining empty seats at random. What is the probability that the last passenger ends up in their own seat?

Warm-up with small cases.

- $n = 2$: passenger 1 takes seat 1 or seat 2 with probability $\frac{1}{2}$ each, so the last passenger gets their seat with probability $\frac{1}{2}$.
- $n = 3$: passenger 1 picks seat 1 (prob. $\frac{1}{3}$): everyone is fine. Seat 3 (prob. $\frac{1}{3}$): passenger 3 loses. Seat 2 (prob. $\frac{1}{3}$): passenger 2 must choose between seats 1 and 3 at random, so passenger 3 is fine half of that time. Total: $\frac{1}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

Both give $\frac{1}{2}$. The number of seats seems not to matter — here is why.

Step 1 — only two seats matter: seat 1 and seat 100. Whenever someone has to choose at random (the first passenger, or anyone who finds their seat taken), the choice falls into one of three kinds:

- **Seat 1** (the first passenger's own seat): this “closes the loop.” Every later passenger now finds their own seat free, so passenger 100 gets seat 100.
- **Seat 100**: passenger 100 has lost their seat for good.
- **Some other seat k** (between 2 and 99): nothing is decided yet. Passengers before k sit normally; passenger k then finds their seat taken and becomes the next random chooser.

So the process keeps passing the “random chooser” role down the line until someone picks seat 1 or seat 100. (It must happen by the end: every seat $2, \dots, 99$ is taken by its owner or earlier, so when passenger 100 boards, the one empty seat is either 1 or 100.)

Step 2 — symmetry. At every random choice, seats 1 and 100 are both still empty and neither belongs to the chooser, so they are equally likely to be picked. The first time the process lands on one of them, it is seat 1 or seat 100 with probability $\frac{1}{2}$ each. Therefore

$$P(\text{last passenger gets their own seat}) = \frac{1}{2}.$$

Step 3 — a check by recursion (for any n). Let p_n be the answer with n seats. Passenger 1 picks seat 1 (prob. $\frac{1}{n}$, last passenger succeeds), seat n (prob. $\frac{1}{n}$, fails), or seat k with $2 \leq k \leq n-1$ (prob. $\frac{1}{n}$ each). In the last case passengers $2, \dots, k-1$ sit normally and passenger k faces the same problem with the $n-k+1$ seats $\{1, k+1, \dots, n\}$ (seat 1 now plays the role of k 's own seat). So

$$p_n = \frac{1}{n} \cdot 1 + \frac{1}{n} \cdot 0 + \sum_{k=2}^{n-1} \frac{1}{n} p_{n-k+1}.$$

With $p_2 = \frac{1}{2}$ and assuming $p_m = \frac{1}{2}$ for all smaller m ,

$$p_n = \frac{1}{n} + \frac{n-2}{n} \cdot \frac{1}{2} = \frac{2+n-2}{2n} = \frac{1}{2}.$$

The last passenger sits in their own seat with probability $\boxed{\frac{1}{2}}$ — for any number of seats $n \geq 2$ (the “100” is a red herring).

A Useful Extension

The same reasoning gives the chance for *every* passenger. Fix passenger k ($2 \leq k \leq n$) and look at the $n - k + 2$ seats $S = \{1, k, k + 1, \dots, n\}$. Consider the first time any random chooser picks a seat in S — by symmetry, each seat of S is equally likely to be that first pick.

- If it is seat 1, the loop closes and passenger k gets seat k .
- If it is some seat $j > k$, the displaced person is seated and nobody is displaced again until passenger j boards — after passenger k — so k gets seat k .
- Only if it is seat k itself is passenger k displaced.

So passenger k loses their seat with probability $\frac{1}{n-k+2}$, i.e.

$$P(\text{passenger } k \text{ gets their own seat}) = \frac{n - k + 1}{n - k + 2}.$$

For $k = n$ this is $\frac{1}{2}$; for $k = 2$ it is $\frac{n-1}{n}$ (passenger 2 loses only if passenger 1 took seat 2). With $n = 10$, a simulation gives 0.899, 0.857, 0.667, 0.501 for passengers 2, 5, 9, 10, matching $\frac{9}{10}, \frac{6}{7}, \frac{2}{3}, \frac{1}{2}$.

Concept Summary and Applications

- **Reduce to what matters:** most of the process is irrelevant; the outcome is decided the first time a random chooser picks one of two special seats.
- **Symmetry beats computation:** those two seats are always treated identically, so each wins with probability $\frac{1}{2}$.
- **Check small cases and recurse:** $n = 2, 3$ suggest the answer; the recursion $p_n = \frac{1}{n} + \frac{n-2}{n}p$ confirms it for all n .
- **Applications:** random-displacement and “chain of evictions” problems, hashing collisions, and a classic interview test of spotting symmetry instead of brute-force enumeration.